

10_layer_simulation_stack.yaml

Source: ukg_yaml_v2/10_layer_simulation_stack.yaml

Content follows on next page.

10_layer_simulation_stack.yaml

```
00001      # =====
00002      # 10_layer_simulation_stack.yaml
00003      # Generated: 2025-12-21T22:20:42.682885
00004      # Format: Tree / numbered node style + YAML
00005      # =====
00006      # ASCII TREE (high-level)
00007      # simulation_stack
00008 # ■■ L1 Context Init & Parsing
00009 # ■■ L2 Retrieval & Traversal
00010 # ■■ L3 Research Agents
00011 # ■■ L4 Stakeholder POV
00012 # ■■ L5 Quad-Persona Debate
00013 # ■■ L6 Quant Validation
00014 # ■■ L7 Scenario Simulation
00015 # ■■ L8 Consistency Verification
00016 # ■■ L9 Strategic Alignment
00017 # ■■ L10 Emergence & Assembly
00018 #
00019      # -----
00020 simulation_stack:
00021   "1) metadata":
00022     name: "UKG 10-Layer Simulation Stack"
00023     version: "2.0"
00024     description: "Recursive reasoning pipeline with layered verification and synthesis."
00025     default_layers_for_complexity:
00026       simple: [1,2,10]
00027       moderate: [1,2,5,8,10]
00028       complex: [1,2,3,4,5,6,7,8,9,10]
00029
00030   "2) layers":
00031     "2.1) L1":
00032       name: "Context Initialization and Query Parsing"
00033       goals:
00034         - "normalize input"
00035         - "resolve ambiguity"
00036         - "map to 17-axis coordinates"
00037       invoke_ka: ["KA-004", "KA-005", "KA-113"]
00038       inputs: ["raw_query", "user_context"]
00039       outputs: ["parsed_query", "axis_coordinates", "stakes_profile", "complexity_tier"]
00040
00041     "2.2) L2":
00042       name: "Knowledge Retrieval and Graph Traversal"
00043       goals:
00044         - "build local working subgraph"
00045         - "retrieve facts, rules, constraints, sources"
00046       invoke_ka: ["KA-020", "KA-030", "KA-031"]
00047       inputs: ["axis_coordinates", "graph_index"]
00048       outputs: ["local_working_subgraph", "evidence_pack"]
00049
00050     "2.3) L3":
00051       name: "Simulated Research Agents and Deep Investigation"
00052       goals:
00053         - "spawn targeted research agents"
00054         - "deep dive into sub-questions"
00055       invoke_ka: ["KA-012", "KA-018"]
00056       invoke_registry: "ukg_knowledge_agents_registry.yaml"
00057       inputs: ["local_working_subgraph", "stakes_profile"]
00058       outputs: ["agent_findings", "expanded_evidence_pack"]
00059
00060     "2.4) L4":
00061       name: "Point of View Integration and Stakeholder Analysis"
00062       goals:
00063         - "capture affected stakeholders"
00064         - "requirements + impact analysis"
00065       invoke_ka: ["KA-027", "KA-022"]
00066       inputs: ["agent_findings", "axis_coordinates"]
00067       outputs: ["stakeholder_matrix", "constraints_matrix"]
```

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00068
00069 "2.5) L5":
00070   name: "Multi-Agent Orchestration and Collaborative Debate"
00071   goals:
00072     - "independent persona drafts"
00073     - "cross-critique"
00074     - "consensus synthesis"
00075   invoke_ka: ["KA-026", "KA-012"]
00076   personas: ["P8_Knowledge", "P9_Sector", "P10_Regulatory", "P11_Compliance"]
00077   inputs: ["expanded_evidence_pack", "stakeholder_matrix"]
00078   outputs: ["draft_solution", "persona_positions", "debate_log"]
00079
00080 "2.6) L6":
00081   name: "Pattern Analysis and Quantitative Validation"
00082   goals:
00083     - "statistical validation"
00084     - "risk modeling"
00085     - "cost/benefit"
00086   invoke_ka: ["KA-011", "KA-022", "KA-102"]
00087   inputs: ["draft_solution", "available_data"]
00088   outputs: ["quant_validation_report", "risk_scores", "updated_confidence_factors"]
00089
00090 "2.7) L7":
00091   name: "Long-Range Scenario Simulation and Strategic Foresight"
00092   goals:
00093     - "30/60/90-day windows"
00094     - "multi-scenario robustness"
00095     - "regulatory/tech drift sensitivity"
00096   invoke_ka: ["KA-018", "KA-101", "KA-022"]
00097   inputs: ["draft_solution", "assumptions"]
00098   outputs: ["scenario_runs", "robustness_notes", "contingency_plan"]
00099
00100 "2.8) L8":
00101   name: "Recursive Cross-Verification and Consistency Validation"
00102   goals:
00103     - "detect contradictions"
00104     - "constraint satisfaction"
00105     - "evidence consistency"
00106   invoke_ka: ["KA-003", "KA-008", "KA-014"]
00107   inputs: ["draft_solution", "expanded_evidence_pack", "scenario_runs"]
00108   outputs: ["consistency_report", "fix_requests", "revised_solution"]
00109
00110 "2.9) L9":
00111   name: "Deep Integration and Strategic Alignment"
00112   goals:
00113     - "align with enterprise strategy"
00114     - "integration with existing systems"
00115     - "portfolio coherence"
00116   invoke_ka: ["KA-022", "KA-012"]
00117   inputs: ["revised_solution", "enterprise_context"]
00118   outputs: ["aligned_solution", "implementation_roadmap"]
00119
00120 "2.10) L10":
00121   name: "Emergence Detection and Final Assembly"
00122   goals:
00123     - "final assembly"
00124     - "emergent side-effect checks"
00125     - "confidence computation"
00126   invoke_ka: ["KA-014", "KA-102"]
00127   inputs: ["aligned_solution", "audit_bundle_inputs"]
00128   outputs: ["preliminary_answer", "confidence_vector", "audit_bundle_inputs"]
00129
00130 "3) routing":
00131   router_ka: "KA-113"
00132   rules:
00133     - if: "complexity_tier == 'simple'"
00134       run_layers: [1,2,10]
```

```
00135     - if: "complexity_tier == 'moderate'"
00136       run_layers: [1,2,5,8,10]
00137     - if: "complexity_tier == 'complex'"
00138       run_layers: [1,2,3,4,5,6,7,8,9,10]
```

12_step_refinement_engine.yaml

Source: ukg_yaml_v2/12_step_refinement_engine.yaml
Content follows on next page.

12_step_refinement_engine.yaml

```
00001      # =====
00002      # 12_step_refinement_engine.yaml
00003      # Generated: 2025-12-21T22:20:42.683452
00004      # Format: Tree / numbered node style + YAML
00005      # =====
00006      # ASCII TREE (high-level)
00007      # refinement_engine
00008 # ■■ 1 AoT
00009 # ■■ 2 ToT
00010 # ■■ 3 Gap Analysis
00011 # ■■ 4 Data Validation
00012 # ■■ 5 NLP Enhancement
00013 # ■■ 6 Consistency Validation
00014 # ■■ 7 Cross-Reference Validation
00015 # ■■ 8 Logic Check
00016 # ■■ 9 Ethical/Bias Audit
00017 # ■■ 10 Regulatory Compliance
00018 # ■■ 11 Security Validation
00019 # ■■ 12 Final Synthesis
00020 #
00021 # -----
00022 refinement_engine:
00023   "1) metadata":
00024     name: "UKG 12-Step Refinement Engine"
00025     version: "2.0"
00026     description: "Quality assurance + validation workflow targeting mission-grade \
confidence thresholds."
00027     max_cycles: 3
00028     default_confidence_thresholds:
00029       high_stakes: 0.995
00030       medium_stakes: 0.95
00031       low_stakes: 0.90
00032
00033   "2) steps":
00034     "2.1) step_1":
00035       name: "Algorithm of Thought (AoT) – Structured Decomposition"
00036       invoke_ka: ["KA-001"]
00037       outputs: ["aot_dag", "premises", "inference_steps", "claim_index"]
00038
00039     "2.2) step_2":
00040       name: "Tree of Thought (ToT) – Alternative Exploration"
00041       invoke_ka: ["KA-002"]
00042       outputs: ["tot_branches", "branch_scores", "selected_branch"]
00043
00044     "2.3) step_3":
00045       name: "Gap Analysis – Completeness Verification"
00046       invoke_ka: ["KA-003"]
00047       outputs: ["gap_list", "coverage_report", "unresolved_questions"]
00048
00049     "2.4) step_4":
00050       name: "Data Validation – Evidence Quality Assurance"
00051       invoke_ka: ["KA-040", "KA-041", "KA-042"]
00052       outputs: ["evidence_quality_scores", "source_corroboration", "currency_check"]
00053
00054     "2.5) step_5":
00055       name: "NLP Enhancement – Clarity + Precision"
00056       invoke_ka: ["KA-050"]
00057       outputs: ["edited_solution_text", "terminology_map", "audience_fit_score"]
00058
00059     "2.6) step_6":
00060       name: "Data Consistency Validation"
00061       invoke_ka: ["KA-008"]
00062       outputs: ["conflict_list", "resolved_conflicts", "consistency_pass"]
00063
00064     "2.7) step_7":
00065       name: "Cross-Reference Validation"
00066       invoke_ka: ["KA-052"]
```

```
00067     outputs: ["citation_map", "citation_support_scores"]
00068
00069     "2.8) step_8":
00070       name: "Logic Check – Formal Reasoning Validation"
00071       invoke_ka: ["KA-053"]
00072       outputs: ["logic_validity_report", "fallacy_flags"]
00073
00074     "2.9) step_9":
00075       name: "Ethical and Bias Audit"
00076       invoke_ka: ["KA-010", "KA-027"]
00077       outputs: ["bias_report", "ethical_tradeoffs", "mitigations"]
00078
00079     "2.10) step_10":
00080       name: "Regulatory Compliance Check"
00081       invoke_ka: ["KA-066", "KA-067"]
00082       outputs: ["regulatory_pass", "regulatory_notes", "required_controls"]
00083
00084     "2.11) step_11":
00085       name: "Security Validation"
00086       invoke_ka: ["KA-061", "KA-107"]
00087       outputs: ["security_pass", "threat_model_notes"]
00088
00089     "2.12) step_12":
00090       name: "Final Logical and Compliance Synthesis"
00091       invoke_ka: ["KA-014", "KA-026", "KA-012"]
00092       outputs: ["final_answer", "final_confidence", "audit_bundle"]
00093
00094     "3) convergence":
00095       compute_confidence:
00096         formula_id: "F-CONF-01"
00097         gate_formula_id: "F-CONF-02"
00098       fail_actions:
00099         - if: "final_confidence < confidence_threshold AND cycles_remaining > 0"
00100           action: "rerun_refinement_with_gap_feedback"
00101         - if: "final_confidence < confidence_threshold AND cycles_remaining == 0"
00102           action: "manual_review_alert"
```

17_axis_coordinate_schema.yaml

Source: ukg_yaml_v2/17_axis_coordinate_schema.yaml
Content follows on next page.

17_axis_coordinate_schema.yaml

```
00001      # =====
00002      # 17_axis_coordinate_schema.yaml
00003      # Generated: 2025-12-21T22:20:42.682384
00004      # Format: Tree / numbered node style + YAML
00005      # =====
00006      # ASCII TREE (high-level)
00007      # axes
00008      # ■■ A) canonical_ukg_axes (Axis 1..17)
00009      # ■■ B) alternate_axis_view (Temporal..Learning)
00010      # ■■ C) coordinate_tuple_encoding
00011      # ■■ D) parsing_rules
00012      # ■■ E) examples
00013      #
00014      # -----
00015      axes:
00016        "A) canonical_ukg_axes":
00017          description: "Primary 17-axis model (Axis 1..17) used by UKG v2.0 technical \
architecture."
00018          "A.1) group_1_knowledge_organization (1..7)":
00019            - axis_id: 1
00020              name: "Pillar Level System"
00021              code: "PL"
00022              description: "107 foundational pillars (PL0001-PL0107) spanning human knowledge."
00023              encoding: "PL####"
00024            - axis_id: 2
00025              name: "Sector of Industry"
00026              code: "IND"
00027              description: "Industry context using NAICS/SIC/ISIC/NACE/UNSPSC/GICS codes + \
crosswalks."
00028              encoding: "scheme:code"
00029            - axis_id: 3
00030              name: "Honeycomb System"
00031              code: "HC"
00032              description: "Cross-disciplinary semantic bridges (non-hierarchical mesh)."
00033              encoding: "edge_bundle_id"
00034            - axis_id: 4
00035              name: "Branch System"
00036              code: "BR"
00037              description: "Hierarchical subdomains within pillars/sectors (3-7 levels typical)."
00038              encoding: "dotted_decimal"
00039            - axis_id: 5
00040              name: "Node System"
00041              code: "ND"
00042              description: "Interdisciplinary convergence nodes (e.g., Bioinformatics)."
00043              encoding: "node_id"
00044            - axis_id: 6
00045              name: "Octopus Node"
00046              code: "OCT"
00047              description: "Meta-regulatory context across jurisdictions/standards."
00048              encoding: "reg_mesh_id"
00049            - axis_id: 7
00050              name: "Spiderweb Node"
00051              code: "SPW"
00052              description: "Complex compliance constraint web; harmonization/conflict \
resolution."
00053              encoding: "constraint_graph_id"
00054          "A.2) group_2_persona_axes (8..11)":
00055            - axis_id: 8
00056              name: "Knowledge Role Mapping"
00057              code: "P8"
00058              description: "Scholar persona (domain expert) instantiation template."
00059              encoding: "persona_profile_id"
00060            - axis_id: 9
00061              name: "Sector Expert Mapping"
00062              code: "P9"
00063              description: "Practitioner persona (industry operator) instantiation template."
```

17_axis_coordinate_schema.yaml

```
00065     encoding: "persona_profile_id"
00066   - axis_id: 10
00067     name: "Octopus Regulatory Expert"
00068     code: "P10"
00069     description: "Regulatory persona (external laws/standards).\"
00070     encoding: "persona_profile_id"
00071   - axis_id: 11
00072     name: "Spiderweb Compliance Expert"
00073     code: "P11"
00074     description: "Compliance persona (internal governance/ethics/policy mesh).\"
00075     encoding: "persona_profile_id"
00076
00077   "A.3) group_3_spatiotemporal (12..13)":
00078     - axis_id: 12
00079       name: "Location Context"
00080       code: "LOC"
00081       description: "Geo + org + facility context.\"
00082       encoding: "ISO3166-1.ISO3166-2.MUNICIPALITY.ORG.FACILITY"
00083     - axis_id: 13
00084       name: "Temporal Context"
00085       code: "TIME"
00086       description: "Validity intervals + observation timestamps.\"
00087       encoding: "ISO_8601_interval"
00088
00089   "A.4) group_4_advanced_analytics (14..17)":
00090     - axis_id: 14
00091       name: "Certainty and Evidence"
00092       code: "CONF"
00093       description: "Probabilistic confidence + evidence strength; calibrated.\"
00094       encoding: "C:EVID_Q:SOURCE_COUNT"
00095     - axis_id: 15
00096       name: "Scale and Scope"
00097       code: "SCALE"
00098       description: "Granularity levels L1..L5.\"
00099       encoding: "L1_ATOMIC|L2_COMPONENT|L3_SYSTEM|L4_ENTERPRISE|L5_ECOSYSTEM"
00100     - axis_id: 16
00101       name: "Modality and Format Context"
00102       code: "MOD"
00103       description: "TEXT/NUMERIC/VISUAL/CODE/STRUCTURED/MULTIMODAL.\"
00104       encoding: "tag"
00105     - axis_id: 17
00106       name: "Continuous Learning"
00107       code: "LEARN"
00108       description: "Outcome tracking, calibration, graph updates with safety gates.\"
00109       encoding: "learning_cycle_id"
00110
00111   "B) alternate_axis_view":
00112     description: "Secondary axis naming used in some documents; mapped to canonical axes \
for compatibility.\"
00113     mapping:
00114       Temporal: 13
00115       Spatial: 12
00116       Semantic: 3
00117       Procedural: 4
00118       Analytical: 15
00119       Normative: 6
00120       "Risk/Impact": 7
00121       Performance: 15
00122       Expertise: 8
00123       Ethical: 11
00124       Regulatory: 10
00125       Compliance: 11
00126       Simulation: 15
00127       AI_Generated: 5
00128       "Cultural/Social": 2
00129       "Innovation/Frontier": 3
00130       "Learning/Continuous Improvement": 17
```

17_axis_coordinate_schema.yaml

```
00131
00132 coordinate_tuple_encoding:
00133   "C) tuple":
00134     format: "[Pillar.Branch.Node].[Sector.Industry].[Location].[Temporal].[Certainty].[Scale].[Modality]"
00135     notes:
00136       - "Axes not placed in the tuple are encoded as attributes/tags (e.g., \
00137         Honeycomb/Octopus/Spiderweb).\"
00138       - "Truncation enables hierarchical generalization (e.g., PL0013.02 → all Clinical \
00139         Medicine).\"
00138     "D) field_encodings":
00139       pillar_branch: "PL####.##.##.## (variable depth)"
00140       sector:
00141         examples:
00142           - "NAICS:622110"
00143           - "ISIC:8610"
00144           - "GICS:35102010"
00145       location:
00146         example: "US.CA.SAN_FRANCISCO.UCSF.PARNASSUS_CAMPUS"
00147       temporal:
00148         example: "2018-05-25/9999-12-31"
00149       certainty:
00150         example: "0.95:HIGH:12"
00151       scale:
00152         example: "L4_ENTERPRISE"
00153       modality:
00154         example: "TEXT"
00155
00156 parsing_rules:
00157   "E) coordinate_parsing":
00158     input_formats_supported:
00159       - "free_text_query"
00160       - "explicit_coordinate_tuple"
00161       - "axis_name:value list (17 entries)"
00162     router:
00163       invoke_ka: ["KA-004", "KA-005", "KA-113"]
00164     outputs:
00165       - "axis_coordinates"
00166       - "entity_resolution"
00167       - "query_type"
00168       - "stakes_profile"
00169       - "confidence_target"
00170
00171 examples:
00172   "E.1) example_medical_gdpr":
00173     query: "Implement GDPR-compliant patient data analytics in a hospital"
00174     coordinates:
00175       pillar_branch: "PL0013.02.05"
00176       sector: ["NAICS:622110"]
00177       location: "EU.*"
00178       temporal: "2018-05-25/9999-12-31"
00179       certainty_target: "0.995"
00180       scale: "L4_ENTERPRISE"
00181       modality: "TEXT|STRUCTURED"
00182     tags:
00183       honeycomb_links: ["PL0003", "PL0007"]
00184       octopus_reg_mesh: ["GDPR", "ISO27001"]
00185       spiderweb_constraints: ["HIPAA_if_US_data", "internal_policy_mesh"]
```

ka_registry.yaml

Source: ukg_yaml_v2/ka_registry.yaml
Content follows on next page.

```

00001      # =====
00002      # ka_registry.yaml
00003      # Generated: 2025-12-21T22:20:42.708649
00004      # Format: Tree / numbered node style + YAML
00005      # =====
00006      # ASCII TREE (high-level)
00007      # ka_registry
00008      # ■■ 1 metadata
00009      # ■■ 2 catalog (KA-001..KA-114)
00010      # ■■ 3 lookup_indexes
00011      #
00012      # -----
00013      ka_registry:
00014      1) metadata:
00015          name: UKG Knowledge Algorithm Registry
00016          version: '2.0'
00017          count: 114
00018          source: UKG_KA_1_to_114_Enterprise_Registry_with_Libraries_and_Formulas.xlsx / \
      KA_Catalog
00019      2) catalog:
00020          KA-001:
00021              KA_Name: Algorithm of Thought
00022              Short_Name: AoT
00023              Purpose: Decompose query into ordered tasks and dependencies
00024              Category: Reasoning
00025              Primary_Layers:
00026                  - L1
00027                  - L2
00028              Allowed_Layers:
00029                  - L1
00030                  - L2
00031              Inputs: Raw query; policy profile; initial context
00032              Outputs: Task graph (DAG); subtask list
00033              Reads_Memory: 'Yes'
00034              Writes_Memory: 'No'
00035              Can_Invoke_Chaos: 'No'
00036              Can_Invoke_External_Research: 'No'
00037              Can_Trigger_Recursion: 'Yes'
00038              Can_Veto: 'No'
00039              Risk_Class: Low
00040              Confidence_Impact: Indirect ↑
00041              Entropy_Signal: 'No'
00042              Default_Params: null
00043              Dependencies:
00044                  - KA-4
00045              Produces_Artifacts: 'Yes'
00046              Audit_Events:
00047                  - 'Yes'
00048              Version: 1.0.0
00049              Owner: UKG/USKD Core
00050              Status: Active
00051              Notes: Planning primitive; feeds ToT, gap analysis.
00052              Implementation_Mode: IN_MODEL
00053              Primary_Libraries: null
00054              Primary_Library_Versions: null
00055              Fallback_Libraries: null
00056              Fallback_Library_Versions: null
00057              Runtime_Env: in-model
00058              Test_Harness: pytest + reasoning-evals
00059              Has_Math: 'No'
00060              Math_Component_Type: null
00061              Math_Object_IDs: null
00062              Math_Scope: null
00063              Variables_Used: null
00064              Units_Normalization: null
00065              Assumptions_Constraints: null
00066          KA-002:

```

```
00067     KA_Name: Tree of Thought
00068     Short_Name: ToT
00069     Purpose: Generate and score parallel reasoning branches
00070     Category: Reasoning
00071     Primary_Layers:
00072     - L3
00073     - L7
00074     Allowed_Layers:
00075     - L3
00076     - L7
00077     Inputs: Task graph; context state
00078     Outputs: Branch set; branch scores; candidate solutions
00079     Reads_Memory: 'Yes'
00080     Writes_Memory: 'No'
00081     Can_Invoke_Chaos: 'Yes'
00082     Can_Invoke_External_Research: 'No'
00083     Can_Trigger_Recursion: 'Yes'
00084     Can_Veto: 'No'
00085     Risk_Class: Medium
00086     Confidence_Impact: Indirect ↑
00087     Entropy_Signal: Contributes
00088     Default_Params: null
00089     Dependencies:
00090     - KA-1
00091     - KA-29
00092     Produces_Artifacts: 'Yes'
00093     Audit_Events:
00094     - 'Yes'
00095     Version: 1.0.0
00096     Owner: UKG/USKD Core
00097     Status: Active
00098     Notes: Primary exploration engine; supports bounded chaos.
00099     Implementation_Mode: HYBRID
00100     Primary_Libraries:
00101     - numpy
00102     - scipy
00103     - networkx
00104     - pydantic
00105     Primary_Library_Versions: numpy>=1.26,<3; scipy>=1.11,<2; networkx>=3.2,<4; \
pydantic>=2.6,<3
00106     Fallback_Libraries:
00107     - pandas
00108     Fallback_Library_Versions: pandas>=2.2,<3
00109     Runtime_Env: python + in-model
00110     Test_Harness: pytest + reasoning-evals
00111     Has_Math: 'No'
00112     Math_Component_Type: null
00113     Math_Object_IDs: null
00114     Math_Scope: null
00115     Variables_Used: null
00116     Units_Normalization: null
00117     Assumptions_Constraints: null
00118     KA-003:
00119     KA_Name: Gap Analysis
00120     Short_Name: GAP
00121     Purpose: Detect missing/weak knowledge, contradictions, ambiguity
00122     Category: Analysis
00123     Primary_Layers:
00124     - L2
00125     Allowed_Layers:
00126     - L2
00127     Inputs: Context state; evidence set
00128     Outputs: Gap vector; missing-entity list; uncertainty map
00129     Reads_Memory: 'Yes'
00130     Writes_Memory: 'No'
00131     Can_Invoke_Chaos: 'No'
00132     Can_Invoke_External_Research: 'No'
```

```
00133     Can_Trigger_Recursion: 'Yes'
00134     Can_Veto: 'No'
00135     Risk_Class: Low
00136     Confidence_Impact: Neutral
00137     Entropy_Signal: Contributes
00138     Default_Params: null
00139     Dependencies:
00140     - KA-29
00141     Produces_Artifacts: 'Yes'
00142     Audit_Events:
00143     - 'Yes'
00144     Version: 1.0.0
00145     Owner: UKG/USKD Core
00146     Status: Active
00147     Notes: Main recursion trigger when gaps exceed threshold.
00148     Implementation_Mode: PYTHON_LIB
00149     Primary_Libraries:
00150     - pydantic
00151     - numpy
00152     - pandas
00153     - rapidfuzz
00154     - opentelemetry
00155     Primary_Library_Versions: pydantic>=2.6,<3; numpy>=1.26,<3; pandas>=2.2,<3; \
    rapidfuzz>=3.6,<4; opentelemetry-api>=1.26,<2;
00156     opentelemetry-sdk>=1.26,<2
00157     Fallback_Libraries:
00158     - scikit-learn
00159     Fallback_Library_Versions: scikit-learn>=1.4,<2
00160     Runtime_Env: python
00161     Test_Harness: pytest + evidence fixtures
00162     Has_Math: 'No'
00163     Math_Component_Type: null
00164     Math_Object_IDs: null
00165     Math_Scope: null
00166     Variables_Used: null
00167     Units_Normalization: null
00168     Assumptions_Constraints: null
00169 KA-004:
00170     KA_Name: Input Validation & Normalization
00171     Short_Name: VALID
00172     Purpose: Sanitize, normalize, and validate query and metadata
00173     Category: Safety
00174     Primary_Layers:
00175     - L1
00176     Allowed_Layers:
00177     - L1
00178     Inputs: User input; auth context; policy
00179     Outputs: Normalized query; safe tokens; reject reasons
00180     Reads_Memory: 'No'
00181     Writes_Memory: 'No'
00182     Can_Invoke_Chaos: 'No'
00183     Can_Invoke_External_Research: 'No'
00184     Can_Trigger_Recursion: 'No'
00185     Can_Veto: 'Yes'
00186     Risk_Class: Critical
00187     Confidence_Impact: Neutral
00188     Entropy_Signal: 'No'
00189     Default_Params: null
00190     Dependencies: null
00191     Produces_Artifacts: 'Yes'
00192     Audit_Events:
00193     - 'Yes'
00194     Version: 1.0.0
00195     Owner: UKG/USKD Core
00196     Status: Active
00197     Notes: First hard gate; blocks unsafe/invalid inputs.
00198     Implementation_Mode: PYTHON_LIB
```

```
00199     Primary_Libraries:
00200     - pydantic
00201     - opentelemetry
00202     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
00203     Fallback_Libraries:
00204     - asynpg
00205     Fallback_Library_Versions: asynpg>=0.29,<1
00206     Runtime_Env: python
00207     Test_Harness: pytest + compliance fixtures
00208     Has_Math: 'No'
00209     Math_Component_Type: null
00210     Math_Object_IDs: null
00211     Math_Scope: null
00212     Variables_Used: null
00213     Units_Normalization: null
00214     Assumptions_Constraints: null
00215 KA-005:
00216     KA_Name: Query Classification
00217     Short_Name: QCLASS
00218     Purpose: Classify domain, complexity, stakes, and ambiguity
00219     Category: Routing
00220     Primary_Layers:
00221     - L1
00222     Allowed_Layers:
00223     - L1
00224     Inputs: Normalized query; user/org policy
00225     Outputs: Domain tags; complexity tier; risk hints
00226     Reads_Memory: 'No'
00227     Writes_Memory: 'No'
00228     Can_Invoke_Chaos: 'No'
00229     Can_Invoke_External_Research: 'No'
00230     Can_Trigger_Recursion: 'No'
00231     Can_Veto: 'No'
00232     Risk_Class: Medium
00233     Confidence_Impact: Neutral
00234     Entropy_Signal: 'No'
00235     Default_Params: null
00236     Dependencies:
00237     - KA-4
00238     Produces_Artifacts: 'Yes'
00239     Audit_Events:
00240     - 'Yes'
00241     Version: 1.0.0
00242     Owner: UKG/USKD Core
00243     Status: Active
00244     Notes: Feeds KA-31/KA-113/KA-59.
00245     Implementation_Mode: HYBRID
00246     Primary_Libraries:
00247     - pydantic
00248     - pyyaml
00249     - orjson
00250     - opentelemetry
00251     - prometheus-client
00252     - networkx
00253     Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \
    opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
00254     prometheus-client>=0.20,<1; networkx>=3.2,<4
00255     Fallback_Libraries:
00256     - redis
00257     Fallback_Library_Versions: redis>=5.0,<6
00258     Runtime_Env: python + in-model
00259     Test_Harness: pytest + golden-trace tests
00260     Has_Math: 'No'
00261     Math_Component_Type: null
00262     Math_Object_IDs: null
00263     Math_Scope: null
```



```
00264     Variables_Used: null
00265     Units_Normalization: null
00266     Assumptions_Constraints: null
00267 KA-006:
00268     KA_Name: Deep Planning
00269     Short_Name: PLAN
00270     Purpose: Select reasoning strategy, depth, and verification approach
00271     Category: Reasoning
00272     Primary_Layers:
00273     - L2
00274     - L7
00275     Allowed_Layers:
00276     - L2
00277     - L7
00278     Inputs: Gap vector; domain tags; constraints
00279     Outputs: Strategy map; verification plan; depth targets
00280     Reads_Memory: 'Yes'
00281     Writes_Memory: 'No'
00282     Can_Invoke_Chaos: 'No'
00283     Can_Invoke_External_Research: 'No'
00284     Can_Trigger_Recursion: 'Yes'
00285     Can_Veto: 'No'
00286     Risk_Class: Medium
00287     Confidence_Impact: Indirect ↑
00288     Entropy_Signal: 'No'
00289     Default_Params: null
00290     Dependencies:
00291     - KA-3
00292     - KA-5
00293     Produces_Artifacts: 'Yes'
00294     Audit_Events:
00295     - 'Yes'
00296     Version: 1.0.0
00297     Owner: UKG/USKD Core
00298     Status: Active
00299     Notes: null
00300     Implementation_Mode: IN_MODEL
00301     Primary_Libraries: null
00302     Primary_Library_Versions: null
00303     Fallback_Libraries: null
00304     Fallback_Library_Versions: null
00305     Runtime_Env: in-model
00306     Test_Harness: pytest + reasoning-evals
00307     Has_Math: 'No'
00308     Math_Component_Type: null
00309     Math_Object_IDs: null
00310     Math_Scope: null
00311     Variables_Used: null
00312     Units_Normalization: null
00313     Assumptions_Constraints: null
00314 KA-007:
00315     KA_Name: Recursive Reasoning Control
00316     Short_Name: RECURSE
00317     Purpose: Control recursion depth/breadth; prevent runaway loops
00318     Category: Control
00319     Primary_Layers:
00320     - L7
00321     Allowed_Layers:
00322     - L7
00323     Inputs: Entropy/confidence; depth budget; branch stats
00324     Outputs: Recurse/stop decisions; fork limits
00325     Reads_Memory: 'Yes'
00326     Writes_Memory: 'No'
00327     Can_Invoke_Chaos: 'Yes'
00328     Can_Invoke_External_Research: 'No'
00329     Can_Trigger_Recursion: 'Yes'
00330     Can_Veto: 'No'
```

```
00331     Risk_Class: Critical
00332     Confidence_Impact: Neutral
00333     Entropy_Signal: Consumes
00334     Default_Params: null
00335     Dependencies:
00336     - KA-14
00337     - KA-102
00338     - KA-81
00339     Produces_Artifacts: 'Yes'
00340     Audit_Events:
00341     - 'Yes'
00342     Version: 1.0.0
00343     Owner: UKG/USKD Core
00344     Status: Active
00345     Notes: Key containment control for FROST fractal sims.
00346     Implementation_Mode: HYBRID
00347     Primary_Libraries:
00348     - pydantic
00349     - pyyaml
00350     - orjson
00351     - opentelemetry
00352     - prometheus-client
00353     - networkx
00354     Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \
    opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
00355     prometheus-client>=0.20,<1; networkx>=3.2,<4
00356     Fallback_Libraries:
00357     - redis
00358     Fallback_Library_Versions: redis>=5.0,<6
00359     Runtime_Env: python + in-model
00360     Test_Harness: pytest + golden-trace tests
00361     Has_Math: 'No'
00362     Math_Component_Type: null
00363     Math_Object_IDs: null
00364     Math_Scope: null
00365     Variables_Used: null
00366     Units_Normalization: null
00367     Assumptions_Constraints: null
00368 KA-008:
00369     KA_Name: Self-Critique & Reflection
00370     Short_Name: CRITIQUE
00371     Purpose: Attack draft output for flaws and assumptions
00372     Category: QA
00373     Primary_Layers:
00374     - L9
00375     Allowed_Layers:
00376     - L9
00377     Inputs: Draft answer object; reasoning trace
00378     Outputs: Critique list; correction tasks
00379     Reads_Memory: 'Yes'
00380     Writes_Memory: 'No'
00381     Can_Invoke_Chaos: 'No'
00382     Can_Invoke_External_Research: 'No'
00383     Can_Trigger_Recursion: 'Yes'
00384     Can_Veto: 'No'
00385     Risk_Class: Low
00386     Confidence_Impact: Indirect ↑
00387     Entropy_Signal: 'No'
00388     Default_Params: null
00389     Dependencies: null
00390     Produces_Artifacts: 'Yes'
00391     Audit_Events:
00392     - 'Yes'
00393     Version: 1.0.0
00394     Owner: UKG/USKD Core
00395     Status: Active
00396     Notes: null
```

```
00397      Implementation_Mode: IN_MODEL
00398      Primary_Libraries: null
00399      Primary_Library_Versions: null
00400      Fallback_Libraries: null
00401      Fallback_Library_Versions: null
00402      Runtime_Env: in-model
00403      Test_Harness: pytest + reasoning-evals
00404      Has_Math: 'No'
00405      Math_Component_Type: null
00406      Math_Object_IDs: null
00407      Math_Scope: null
00408      Variables_Used: null
00409      Units_Normalization: null
00410      Assumptions_Constraints: null
00411  KA-009:
00412      KA_Name: Evidence Validation
00413      Short_Name: EVID
00414      Purpose: Validate evidence consistency and claims support
00415      Category: Truth
00416      Primary_Layers:
00417      - L6
00418      Allowed_Layers:
00419      - L6
00420      Inputs: Evidence matrix; claims list
00421      Outputs: Validated evidence; rejected evidence; rationale
00422      Reads_Memory: 'Yes'
00423      Writes_Memory: 'No'
00424      Can_Invoke_Chaos: 'No'
00425      Can_Invoke_External_Research: 'No'
00426      Can_Trigger_Recursion: 'No'
00427      Can_Veto: 'No'
00428      Risk_Class: High
00429      Confidence_Impact: Direct ↑
00430      Entropy_Signal: 'No'
00431      Default_Params: null
00432      Dependencies:
00433      - KA-18
00434      - KA-62
00435      Produces_Artifacts: 'Yes'
00436      Audit_Events:
00437      - 'Yes'
00438      Version: 1.0.0
00439      Owner: UKG/USKD Core
00440      Status: Active
00441      Notes: null
00442      Implementation_Mode: PYTHON_LIB
00443      Primary_Libraries:
00444      - pydantic
00445      - numpy
00446      - pandas
00447      - rapidfuzz
00448      - opentelemetry
00449      Primary_Library_Versions: pydantic>=2.6,<3; numpy>=1.26,<3; pandas>=2.2,<3; \
    rapidfuzz>=3.6,<4; opentelemetry-api>=1.26,<2;
00450      opentelemetry-sdk>=1.26,<2
00451      Fallback_Libraries:
00452      - scikit-learn
00453      Fallback_Library_Versions: scikit-learn>=1.4,<2
00454      Runtime_Env: python
00455      Test_Harness: pytest + evidence fixtures
00456      Has_Math: 'No'
00457      Math_Component_Type: null
00458      Math_Object_IDs: null
00459      Math_Scope: null
00460      Variables_Used: null
00461      Units_Normalization: null
00462      Assumptions_Constraints: null
```

```
00463     KA-010:
00464         KA_Name: Bias Detection
00465         Short_Name: BIAS-DETECT
00466         Purpose: Detect bias signals in reasoning/output
00467         Category: Ethics
00468         Primary_Layers:
00469             - L8
00470             - L9
00471         Allowed_Layers:
00472             - L8
00473             - L9
00474         Inputs: Reasoning trace; draft output
00475         Outputs: Bias flags; bias vectors
00476         Reads_Memory: 'Yes'
00477         Writes_Memory: 'No'
00478         Can_Invoke_Chaos: 'No'
00479         Can_Invoke_External_Research: 'No'
00480         Can_Trigger_Recursion: 'No'
00481         Can_Veto: 'No'
00482         Risk_Class: High
00483         Confidence_Impact: Neutral
00484         Entropy_Signal: 'No'
00485         Default_Params: null
00486         Dependencies:
00487             - KA-45
00488         Produces_Artifacts: 'Yes'
00489         Audit_Events:
00490             - 'Yes'
00491         Version: 1.0.0
00492         Owner: UKG/USKD Core
00493         Status: Active
00494         Notes: null
00495         Implementation_Mode: PYTHON_LIB
00496         Primary_Libraries:
00497             - pydantic
00498             - opentelemetry
00499         Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
00500             opentelemetry-sdk>=1.26,<2
00501         Fallback_Libraries:
00502             - asynccpg
00503         Fallback_Library_Versions: asynccpg>=0.29,<1
00504         Runtime_Env: python
00505         Test_Harness: pytest + compliance fixtures
00506         Has_Math: 'No'
00507         Math_Component_Type: null
00508         Math_Object_IDs: null
00509         Math_Scope: null
00510         Variables_Used: null
00511         Units_Normalization: null
00512         Assumptions_Constraints: null
00512     KA-011:
00513         KA_Name: Analytical Modeling
00514         Short_Name: MODEL
00515         Purpose: Perform math/statistical/structural modeling
00516         Category: Reasoning
00517         Primary_Layers:
00518             - L3
00519         Allowed_Layers:
00520             - L3
00521         Inputs: Data model; constraints
00522         Outputs: Derived results; model fit; assumptions
00523         Reads_Memory: 'Yes'
00524         Writes_Memory: 'No'
00525         Can_Invoke_Chaos: 'No'
00526         Can_Invoke_External_Research: 'No'
00527         Can_Trigger_Recursion: 'No'
00528         Can_Veto: 'No'
```

```
00529     Risk_Class: Medium
00530     Confidence_Impact: Indirect ↑
00531     Entropy_Signal: 'No'
00532     Default_Params: null
00533     Dependencies: null
00534     Produces_Artifacts: 'Yes'
00535     Audit_Events:
00536     - 'Yes'
00537     Version: 1.0.0
00538     Owner: UKG/USKD Core
00539     Status: Active
00540     Notes: null
00541     Implementation_Mode: HYBRID
00542     Primary_Libraries:
00543     - numpy
00544     - scipy
00545     - networkx
00546     - pydantic
00547     Primary_Library_Versions: numpy>=1.26,<3; scipy>=1.11,<2; networkx>=3.2,<4; \
pydantic>=2.6,<3
00548     Fallback_Libraries:
00549     - pandas
00550     Fallback_Library_Versions: pandas>=2.2,<3
00551     Runtime_Env: python + in-model
00552     Test_Harness: pytest + reasoning-evals
00553     Has_Math: 'No'
00554     Math_Component_Type: null
00555     Math_Object_IDs: null
00556     Math_Scope: null
00557     Variables_Used: null
00558     Units_Normalization: null
00559     Assumptions_Constraints: null
00560 KA-012:
00561     KA_Name: Persona Simulation
00562     Short_Name: PERSONA
00563     Purpose: Run expert personas (knowledge/sector/regulatory/compliance)
00564     Category: Persona
00565     Primary_Layers:
00566     - L4
00567     Allowed_Layers:
00568     - L4
00569     Inputs: Persona specs; context state
00570     Outputs: Persona findings; objections; constraints
00571     Reads_Memory: 'Yes'
00572     Writes_Memory: 'No'
00573     Can_Invoke_Chaos: 'Yes'
00574     Can_Invoke_External_Research: 'No'
00575     Can_Trigger_Recursion: 'No'
00576     Can_Veto: 'No'
00577     Risk_Class: Medium
00578     Confidence_Impact: Indirect ↑
00579     Entropy_Signal: Contributes
00580     Default_Params: null
00581     Dependencies:
00582     - KA-29
00583     - KA-13
00584     Produces_Artifacts: 'Yes'
00585     Audit_Events:
00586     - 'Yes'
00587     Version: 1.0.0
00588     Owner: UKG/USKD Core
00589     Status: Active
00590     Notes: null
00591     Implementation_Mode: IN_MODEL
00592     Primary_Libraries: null
00593     Primary_Library_Versions: null
00594     Fallback_Libraries: null
```

```
00595     Fallback_Library_Versions: null
00596     Runtime_Env: in-model
00597     Test_Harness: pytest + reasoning-evals
00598     Has_Math: 'No'
00599     Math_Component_Type: null
00600     Math_Object_IDs: null
00601     Math_Scope: null
00602     Variables_Used: null
00603     Units_Normalization: null
00604     Assumptions_Constraints: null
00605 KA-013:
00606     KA_Name: Persona Weighting
00607     Short_Name: P-WEIGHT
00608     Purpose: Weight persona outputs by relevance/authority
00609     Category: Persona
00610     Primary_Layers:
00611     - L4
00612     Allowed_Layers:
00613     - L4
00614     Inputs: Persona outputs; domain tags
00615     Outputs: Weighted persona views; priority map
00616     Reads_Memory: 'Yes'
00617     Writes_Memory: 'No'
00618     Can_Invoke_Chaos: 'No'
00619     Can_Invoke_External_Research: 'No'
00620     Can_Trigger_Recursion: 'No'
00621     Can_Veto: 'No'
00622     Risk_Class: Low
00623     Confidence_Impact: Neutral
00624     Entropy_Signal: 'No'
00625     Default_Params: null
00626     Dependencies: null
00627     Produces_Artifacts: 'Yes'
00628     Audit_Events:
00629     - 'Yes'
00630     Version: 1.0.0
00631     Owner: UKG/USKD Core
00632     Status: Active
00633     Notes: null
00634     Implementation_Mode: IN_MODEL
00635     Primary_Libraries: null
00636     Primary_Library_Versions: null
00637     Fallback_Libraries: null
00638     Fallback_Library_Versions: null
00639     Runtime_Env: in-model
00640     Test_Harness: pytest + reasoning-evals
00641     Has_Math: 'No'
00642     Math_Component_Type: null
00643     Math_Object_IDs: null
00644     Math_Scope: null
00645     Variables_Used: null
00646     Units_Normalization: null
00647     Assumptions_Constraints: null
00648 KA-014:
00649     KA_Name: Confidence Scoring
00650     Short_Name: CONF
00651     Purpose: Compute confidence and thresholds; certify outputs
00652     Category: Truth
00653     Primary_Layers:
00654     - L6
00655     - L10
00656     Allowed_Layers:
00657     - L6
00658     - L10
00659     Inputs: Truth scores; evidence weights; conflict metrics
00660     Outputs: Confidence score; pass/fail; deltas
00661     Reads_Memory: 'Yes'
```

```

00662     Writes_Memory: 'No'
00663     Can_Invoke_Chaos: 'No'
00664     Can_Invoke_External_Research: 'No'
00665     Can_Trigger_Recursion: 'No'
00666     Can_Veto: 'No'
00667     Risk_Class: Critical
00668     Confidence_Impact: Direct ↑/↓
00669     Entropy_Signal: Produces
00670     Default_Params: null
00671     Dependencies:
00672     - KA-9
00673     - KA-26
00674     - KA-41
00675     - KA-102
00676     Produces_Artifacts: 'Yes'
00677     Audit_Events:
00678     - 'Yes'
00679     Version: 1.0.0
00680     Owner: UKG/USKD Core
00681     Status: Active
00682     Notes: null
00683     Implementation_Mode: PYTHON_LIB
00684     Primary_Libraries:
00685     - pydantic
00686     - numpy
00687     - pandas
00688     - rapidfuzz
00689     - opentelemetry
00690     Primary_Library_Versions: pydantic>=2.6,<3; numpy>=1.26,<3; pandas>=2.2,<3; \
    rapidfuzz>=3.6,<4; opentelemetry-api>=1.26,<2;
00691     opentelemetry-sdk>=1.26,<2
00692     Fallback_Libraries:
00693     - scikit-learn
00694     Fallback_Library_Versions: scikit-learn>=1.4,<2
00695     Runtime_Env: python
00696     Test_Harness: pytest + evidence fixtures
00697     Has_Math: 'Yes'
00698     Math_Component_Type: Mixed
00699     Math_Object_IDs:
00700     - F-CONF-01
00701     - F-CONF-02
00702     Math_Scope: TruthEngine/KA
00703     Variables_Used:
00704     - C
00705     - H
00706     - w_i
00707     - e_i
00708     Units_Normalization: C∈[0,1]
00709     Assumptions_Constraints: Requires validated evidence weights
00710     KA-015:
00711     KA_Name: Temporal Reasoning
00712     Short_Name: TIME
00713     Purpose: Reason across time; timelines; validity windows
00714     Category: Reasoning
00715     Primary_Layers:
00716     - L3
00717     Allowed_Layers:
00718     - L3
00719     Inputs: Time-tagged facts; events
00720     Outputs: Temporal conclusions; timeline; validity range
00721     Reads_Memory: 'Yes'
00722     Writes_Memory: 'No'
00723     Can_Invoke_Chaos: 'No'
00724     Can_Invoke_External_Research: 'No'
00725     Can_Trigger_Recursion: 'No'
00726     Can_Veto: 'No'
00727     Risk_Class: Medium

```

```
00728     Confidence_Impact: Indirect ↑
00729     Entropy_Signal: 'No'
00730     Default_Params: null
00731     Dependencies:
00732     - KA-52
00733     Produces_Artifacts: 'Yes'
00734     Audit_Events:
00735     - 'Yes'
00736     Version: 1.0.0
00737     Owner: UKG/USKD Core
00738     Status: Active
00739     Notes: null
00740     Implementation_Mode: IN_MODEL
00741     Primary_Libraries: null
00742     Primary_Library_Versions: null
00743     Fallback_Libraries: null
00744     Fallback_Library_Versions: null
00745     Runtime_Env: in-model
00746     Test_Harness: pytest + reasoning-evals
00747     Has_Math: 'No'
00748     Math_Component_Type: null
00749     Math_Object_IDs: null
00750     Math_Scope: null
00751     Variables_Used: null
00752     Units_Normalization: null
00753     Assumptions_Constraints: null
00754 KA-016:
00755     KA_Name: Regulatory Mapping
00756     Short_Name: REGMAP
00757     Purpose: Map regs/policies to obligations and constraints
00758     Category: Compliance
00759     Primary_Layers:
00760     - L4
00761     - L8
00762     Allowed_Layers:
00763     - L4
00764     - L8
00765     Inputs: Jurisdiction; policy corpus; scenario
00766     Outputs: Reg matrix; obligations; citations map
00767     Reads_Memory: 'Yes'
00768     Writes_Memory: 'No'
00769     Can_Invoke_Chaos: 'No'
00770     Can_Invoke_External_Research: 'No'
00771     Can_Trigger_Recursion: 'No'
00772     Can_Veto: 'No'
00773     Risk_Class: Critical
00774     Confidence_Impact: Indirect ↑
00775     Entropy_Signal: 'No'
00776     Default_Params: null
00777     Dependencies:
00778     - KA-17
00779     - KA-18
00780     Produces_Artifacts: 'Yes'
00781     Audit_Events:
00782     - 'Yes'
00783     Version: 1.0.0
00784     Owner: UKG/USKD Core
00785     Status: Active
00786     Notes: null
00787     Implementation_Mode: PYTHON_LIB
00788     Primary_Libraries:
00789     - pydantic
00790     - opentelemetry
00791     - pyyaml
00792     - httpx
00793     - tenacity
00794     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
```



```
  opentelemetry-sdk>=1.26,<2; PyYAML>=6.0,<7;
00795     httpx>=0.27,<1; tenacity>=8.2,<9
00796     Fallback_Libraries:
00797     - asynccpg
00798     Fallback_Library_Versions: asynccpg>=0.29,<1
00799     Runtime_Env: python
00800     Test_Harness: pytest + compliance fixtures
00801     Has_Math: 'No'
00802     Math_Component_Type: null
00803     Math_Object_IDs: null
00804     Math_Scope: null
00805     Variables_Used: null
00806     Units_Normalization: null
00807     Assumptions_Constraints: null
00808 KA-017:
00809     KA_Name: Spatial/Jurisdiction Mapping
00810     Short_Name: GEO
00811     Purpose: Resolve geography/jurisdiction applicability
00812     Category: Context
00813     Primary_Layers:
00814     - L2
00815     Allowed_Layers:
00816     - L2
00817     Inputs: Location metadata; entity scope
00818     Outputs: Jurisdiction tags; applicability constraints
00819     Reads_Memory: 'Yes'
00820     Writes_Memory: 'No'
00821     Can_Invoke_Chaos: 'No'
00822     Can_Invoke_External_Research: 'No'
00823     Can_Trigger_Recursion: 'No'
00824     Can_Veto: 'No'
00825     Risk_Class: High
00826     Confidence_Impact: Indirect ↑
00827     Entropy_Signal: 'No'
00828     Default_Params: null
00829     Dependencies: null
00830     Produces_Artifacts: 'Yes'
00831     Audit_Events:
00832     - 'Yes'
00833     Version: 1.0.0
00834     Owner: UKG/USKD Core
00835     Status: Active
00836     Notes: null
00837     Implementation_Mode: IN_MODEL
00838     Primary_Libraries: null
00839     Primary_Library_Versions: null
00840     Fallback_Libraries: null
00841     Fallback_Library_Versions: null
00842     Runtime_Env: in-model
00843     Test_Harness: pytest + eval harness
00844     Has_Math: 'No'
00845     Math_Component_Type: null
00846     Math_Object_IDs: null
00847     Math_Scope: null
00848     Variables_Used: null
00849     Units_Normalization: null
00850     Assumptions_Constraints: null
00851 KA-018:
00852     KA_Name: Source Provenance
00853     Short_Name: PROV
00854     Purpose: Track source origin, authority, and trust weight
00855     Category: Truth
00856     Primary_Layers:
00857     - L6
00858     Allowed_Layers:
00859     - L6
00860     Inputs: Source metadata; doc hashes
```

```
00861      Outputs: Provenance graph; authority scores
00862      Reads_Memory: 'No'
00863      Writes_Memory: 'No'
00864      Can_Invoke_Chaos: 'No'
00865      Can_Invoke_External_Research: 'No'
00866      Can_Trigger_Recursion: 'No'
00867      Can_Veto: 'No'
00868      Risk_Class: High
00869      Confidence_Impact: Indirect ↑
00870      Entropy_Signal: 'No'
00871      Default_Params: null
00872      Dependencies: null
00873      Produces_Artifacts: 'Yes'
00874      Audit_Events:
00875      - 'Yes'
00876      Version: 1.0.0
00877      Owner: UKG/USKD Core
00878      Status: Active
00879      Notes: null
00880      Implementation_Mode: PYTHON_LIB
00881      Primary_Libraries:
00882      - pydantic
00883      - numpy
00884      - pandas
00885      - rapidfuzz
00886      - opentelemetry
00887      Primary_Library_Versions: pydantic>=2.6,<3; numpy>=1.26,<3; pandas>=2.2,<3; \
    rapidfuzz>=3.6,<4; opentelemetry-api>=1.26,<2;
00888      opentelemetry-sdk>=1.26,<2
00889      Fallback_Libraries:
00890      - scikit-learn
00891      Fallback_Library_Versions: scikit-learn>=1.4,<2
00892      Runtime_Env: python
00893      Test_Harness: pytest + evidence fixtures
00894      Has_Math: 'No'
00895      Math_Component_Type: null
00896      Math_Object_IDs: null
00897      Math_Scope: null
00898      Variables_Used: null
00899      Units_Normalization: null
00900      Assumptions_Constraints: null
00901      KA-019:
00902      KA_Name: Knowledge Synthesis
00903      Short_Name: SYNTH
00904      Purpose: Merge validated knowledge into unified state (no prose)
00905      Category: Synthesis
00906      Primary_Layers:
00907      - L9
00908      Allowed_Layers:
00909      - L9
00910      Inputs: Validated evidence; resolved state
00911      Outputs: Unified knowledge object; key points
00912      Reads_Memory: 'Yes'
00913      Writes_Memory: 'No'
00914      Can_Invoke_Chaos: 'No'
00915      Can_Invoke_External_Research: 'No'
00916      Can_Trigger_Recursion: 'No'
00917      Can_Veto: 'No'
00918      Risk_Class: Medium
00919      Confidence_Impact: Indirect ↑
00920      Entropy_Signal: 'No'
00921      Default_Params: null
00922      Dependencies:
00923      - KA-30
00924      - KA-9
00925      Produces_Artifacts: 'Yes'
00926      Audit_Events:
```

```
00927     - 'Yes'
00928     Version: 1.0.0
00929     Owner: UKG/USKD Core
00930     Status: Active
00931     Notes: null
00932     Implementation_Mode: IN_MODEL
00933     Primary_Libraries: null
00934     Primary_Library_Versions: null
00935     Fallback_Libraries: null
00936     Fallback_Library_Versions: null
00937     Runtime_Env: in-model
00938     Test_Harness: pytest + reasoning-evals
00939     Has_Math: 'No'
00940     Math_Component_Type: null
00941     Math_Object_IDs: null
00942     Math_Scope: null
00943     Variables_Used: null
00944     Units_Normalization: null
00945     Assumptions_Constraints: null
00946     KA-020:
00947         KA_Name: Loopback Trigger
00948         Short_Name: LOOP
00949         Purpose: Force loopback to earlier layers when thresholds fail
00950         Category: Control
00951         Primary_Layers:
00952             - L10
00953         Allowed_Layers:
00954             - L10
00955         Inputs: Confidence/entropy fail; missing obligations
00956         Outputs: Loopback directive; target layer
00957         Reads_Memory: 'Yes'
00958         Writes_Memory: 'No'
00959         Can_Invoke_Chaos: 'No'
00960         Can_Invoke_External_Research: 'No'
00961         Can_Trigger_Recursion: 'Yes'
00962         Can_Veto: 'No'
00963         Risk_Class: High
00964         Confidence_Impact: Neutral
00965         Entropy_Signal: Consumes
00966         Default_Params: null
00967         Dependencies:
00968             - KA-14
00969             - KA-102
00970         Produces_Artifacts: 'Yes'
00971         Audit_Events:
00972             - 'Yes'
00973         Version: 1.0.0
00974         Owner: UKG/USKD Core
00975         Status: Active
00976         Notes: null
00977         Implementation_Mode: HYBRID
00978         Primary_Libraries:
00979             - pydantic
00980             - pyyaml
00981             - orjson
00982             - opentelemetry
00983             - prometheus-client
00984             - networkx
00985         Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \
00986             opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
00987             prometheus-client>=0.20,<1; networkx>=3.2,<4
00988         Fallback_Libraries:
00989             - redis
00990         Fallback_Library_Versions: redis>=5.0,<6
00991         Runtime_Env: python + in-model
00992         Test_Harness: pytest + golden-trace tests
00993         Has_Math: 'No'
```

```
00993     Math_Component_Type: null
00994     Math_Object_IDs: null
00995     Math_Scope: null
00996     Variables_Used: null
00997     Units_Normalization: null
00998     Assumptions_Constraints: null
00999 KA-021:
01000     KA_Name: Emergence Detection
01001     Short_Name: EMERG
01002     Purpose: Detect novel patterns/insights; flag unexpected behavior
01003     Category: Analysis
01004     Primary_Layers:
01005     - L7
01006     - L10
01007     Allowed_Layers:
01008     - L7
01009     - L10
01010     Inputs: Reasoning graph; agent behaviors
01011     Outputs: Emergence flags; novelty score
01012     Reads_Memory: 'Yes'
01013     Writes_Memory: 'No'
01014     Can_Invoke_Chaos: 'Yes'
01015     Can_Invoke_External_Research: 'No'
01016     Can_Trigger_Recursion: 'No'
01017     Can_Veto: 'No'
01018     Risk_Class: High
01019     Confidence_Impact: Neutral
01020     Entropy_Signal: Produces
01021     Default_Params: null
01022     Dependencies: null
01023     Produces_Artifacts: 'Yes'
01024     Audit_Events:
01025     - 'Yes'
01026     Version: 1.0.0
01027     Owner: UKG/USKD Core
01028     Status: Active
01029     Notes: null
01030     Implementation_Mode: PYTHON_LIB
01031     Primary_Libraries:
01032     - pydantic
01033     - numpy
01034     - pandas
01035     - rapidfuzz
01036     - opentelemetry
01037     Primary_Library_Versions: pydantic>=2.6,<3; numpy>=1.26,<3; pandas>=2.2,<3; \
    rapidfuzz>=3.6,<4; opentelemetry-api>=1.26,<2;
01038     opentelemetry-sdk>=1.26,<2
01039     Fallback_Libraries:
01040     - scikit-learn
01041     Fallback_Library_Versions: scikit-learn>=1.4,<2
01042     Runtime_Env: python
01043     Test_Harness: pytest + evidence fixtures
01044     Has_Math: 'No'
01045     Math_Component_Type: null
01046     Math_Object_IDs: null
01047     Math_Scope: null
01048     Variables_Used: null
01049     Units_Normalization: null
01050     Assumptions_Constraints: null
01051 KA-022:
01052     KA_Name: Risk Assessment
01053     Short_Name: RISK
01054     Purpose: Compute operational/mission risk of recommendations
01055     Category: Safety
01056     Primary_Layers:
01057     - L8
01058     Allowed_Layers:
```

```
01059     - L8
01060     Inputs: Draft plan; domain constraints
01061     Outputs: Risk profile; mitigations
01062     Reads_Memory: 'Yes'
01063     Writes_Memory: 'No'
01064     Can_Invoke_Chaos: 'No'
01065     Can_Invoke_External_Research: 'No'
01066     Can_Trigger_Recursion: 'No'
01067     Can_Veto: 'No'
01068     Risk_Class: Critical
01069     Confidence_Impact: Neutral
01070     Entropy_Signal: 'No'
01071     Default_Params: null
01072     Dependencies: null
01073     Produces_Artifacts: 'Yes'
01074     Audit_Events:
01075     - 'Yes'
01076     Version: 1.0.0
01077     Owner: UKG/USKD Core
01078     Status: Active
01079     Notes: null
01080     Implementation_Mode: PYTHON_LIB
01081     Primary_Libraries:
01082     - pydantic
01083     - opentelemetry
01084     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
01085     Fallback_Libraries:
01086     - asynpg
01087     Fallback_Library_Versions: asynpg>=0.29,<1
01088     Runtime_Env: python
01089     Test_Harness: pytest + compliance fixtures
01090     Has_Math: 'No'
01091     Math_Component_Type: null
01092     Math_Object_IDs: null
01093     Math_Scope: null
01094     Variables_Used: null
01095     Units_Normalization: null
01096     Assumptions_Constraints: null
01097 KA-023:
01098     KA_Name: Belief Decay
01099     Short_Name: DECAY
01100     Purpose: Decay stale beliefs; reduce reliance on outdated knowledge
01101     Category: Lifecycle
01102     Primary_Layers:
01103     - L10
01104     Allowed_Layers:
01105     - L10
01106     Inputs: Knowledge timestamps; usage
01107     Outputs: Adjusted belief weights; decay log
01108     Reads_Memory: 'Yes'
01109     Writes_Memory: 'Yes'
01110     Can_Invoke_Chaos: 'No'
01111     Can_Invoke_External_Research: 'No'
01112     Can_Trigger_Recursion: 'No'
01113     Can_Veto: 'No'
01114     Risk_Class: Medium
01115     Confidence_Impact: Indirect ↑/↓
01116     Entropy_Signal: 'No'
01117     Default_Params: null
01118     Dependencies: null
01119     Produces_Artifacts: 'Yes'
01120     Audit_Events:
01121     - 'Yes'
01122     Version: 1.0.0
01123     Owner: UKG/USKD Core
01124     Status: Active
```

```
01125     Notes: null
01126     Implementation_Mode: PYTHON_LIB
01127     Primary_Libraries:
01128     - pydantic
01129     - opentelemetry
01130     - sqlalchemy
01131     - psycpg
01132     - alembic
01133     - redis
01134     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
01135     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
01136     Fallback_Libraries:
01137     - asyncpg
01138     Fallback_Library_Versions: asyncpg>=0.29,<1
01139     Runtime_Env: python
01140     Test_Harness: pytest + compliance fixtures
01141     Has_Math: 'No'
01142     Math_Component_Type: null
01143     Math_Object_IDs: null
01144     Math_Scope: null
01145     Variables_Used: null
01146     Units_Normalization: null
01147     Assumptions_Constraints: null
01148     KA-024:
01149     KA_Name: Trust Gate
01150     Short_Name: TRUST-GATE
01151     Purpose: Approve/veto outputs based on trust/risk/policy
01152     Category: Safety
01153     Primary_Layers:
01154     - L8
01155     Allowed_Layers:
01156     - L8
01157     Inputs: Risk profile; trust score; policy
01158     Outputs: Pass/fail; block reason
01159     Reads_Memory: 'No'
01160     Writes_Memory: 'No'
01161     Can_Invoke_Chaos: 'No'
01162     Can_Invoke_External_Research: 'No'
01163     Can_Trigger_Recursion: 'No'
01164     Can_Veto: 'Yes'
01165     Risk_Class: Critical
01166     Confidence_Impact: Neutral
01167     Entropy_Signal: 'No'
01168     Default_Params: null
01169     Dependencies:
01170     - KA-62
01171     - KA-22
01172     - KA-107
01173     Produces_Artifacts: 'Yes'
01174     Audit_Events:
01175     - 'Yes'
01176     Version: 1.0.0
01177     Owner: UKG/USKD Core
01178     Status: Active
01179     Notes: null
01180     Implementation_Mode: PYTHON_LIB
01181     Primary_Libraries:
01182     - pydantic
01183     - opentelemetry
01184     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
01185     Fallback_Libraries:
01186     - asyncpg
01187     Fallback_Library_Versions: asyncpg>=0.29,<1
01188     Runtime_Env: python
01189     Test_Harness: pytest + compliance fixtures
```

```
01190     Has_Math: 'No'
01191     Math_Component_Type: null
01192     Math_Object_IDs: null
01193     Math_Scope: null
01194     Variables_Used: null
01195     Units_Normalization: null
01196     Assumptions_Constraints: null
01197   KA-025:
01198     KA_Name: Dependency Mapping
01199     Short_Name: DEP
01200     Purpose: Track logical/claim dependencies for audit
01201     Category: Analysis
01202     Primary_Layers:
01203     - L3
01204     Allowed_Layers:
01205     - L3
01206     Inputs: Task graph; claims
01207     Outputs: Dependency tree; impact graph
01208     Reads_Memory: 'Yes'
01209     Writes_Memory: 'No'
01210     Can_Invoke_Chaos: 'No'
01211     Can_Invoke_External_Research: 'No'
01212     Can_Trigger_Recursion: 'No'
01213     Can_Veto: 'No'
01214     Risk_Class: Low
01215     Confidence_Impact: Neutral
01216     Entropy_Signal: 'No'
01217     Default_Params: null
01218     Dependencies: null
01219     Produces_Artifacts: 'Yes'
01220     Audit_Events:
01221     - 'Yes'
01222     Version: 1.0.0
01223     Owner: UKG/USKD Core
01224     Status: Active
01225     Notes: null
01226     Implementation_Mode: PYTHON_LIB
01227     Primary_Libraries:
01228     - pydantic
01229     - numpy
01230     - pandas
01231     - rapidfuzz
01232     - opentelemetry
01233     Primary_Library_Versions: pydantic>=2.6,<3; numpy>=1.26,<3; pandas>=2.2,<3; \
    rapidfuzz>=3.6,<4; opentelemetry-api>=1.26,<2;
01234     opentelemetry-sdk>=1.26,<2
01235     Fallback_Libraries:
01236     - scikit-learn
01237     Fallback_Library_Versions: scikit-learn>=1.4,<2
01238     Runtime_Env: python
01239     Test_Harness: pytest + evidence fixtures
01240     Has_Math: 'No'
01241     Math_Component_Type: null
01242     Math_Object_IDs: null
01243     Math_Scope: null
01244     Variables_Used: null
01245     Units_Normalization: null
01246     Assumptions_Constraints: null
01247   KA-026:
01248     KA_Name: Contradiction Detection
01249     Short_Name: CONTRA
01250     Purpose: Detect conflicts across branches/personas/evidence
01251     Category: Truth
01252     Primary_Layers:
01253     - L6
01254     Allowed_Layers:
01255     - L6
```

```
01256      Inputs: Persona outputs; evidence; branches
01257      Outputs: Conflict set; severity scores
01258      Reads_Memory: 'Yes'
01259      Writes_Memory: 'No'
01260      Can_Invoke_Chaos: 'Yes'
01261      Can_Invoke_External_Research: 'No'
01262      Can_Trigger_Recursion: 'No'
01263      Can_Veto: 'No'
01264      Risk_Class: High
01265      Confidence_Impact: Direct ↓
01266      Entropy_Signal: Produces
01267      Default_Params: null
01268      Dependencies:
01269      - KA-12
01270      - KA-9
01271      Produces_Artifacts: 'Yes'
01272      Audit_Events:
01273      - 'Yes'
01274      Version: 1.0.0
01275      Owner: UKG/USKD Core
01276      Status: Active
01277      Notes: null
01278      Implementation_Mode: PYTHON_LIB
01279      Primary_Libraries:
01280      - pydantic
01281      - numpy
01282      - pandas
01283      - rapidfuzz
01284      - opentelemetry
01285      Primary_Library_Versions: pydantic>=2.6,<3; numpy>=1.26,<3; pandas>=2.2,<3; \
    rapidfuzz>=3.6,<4; opentelemetry-api>=1.26,<2;
01286      opentelemetry-sdk>=1.26,<2
01287      Fallback_Libraries:
01288      - scikit-learn
01289      Fallback_Library_Versions: scikit-learn>=1.4,<2
01290      Runtime_Env: python
01291      Test_Harness: pytest + evidence fixtures
01292      Has_Math: 'No'
01293      Math_Component_Type: null
01294      Math_Object_IDs: null
01295      Math_Scope: null
01296      Variables_Used: null
01297      Units_Normalization: null
01298      Assumptions_Constraints: null
01299      KA-027:
01300      KA_Name: Ethical Impact Analysis
01301      Short_Name: ETHICS
01302      Purpose: Evaluate ethical implications; harm assessment
01303      Category: Ethics
01304      Primary_Layers:
01305      - L8
01306      Allowed_Layers:
01307      - L8
01308      Inputs: Draft output; domain context
01309      Outputs: Ethics score; harm flags; mitigation guidance
01310      Reads_Memory: 'Yes'
01311      Writes_Memory: 'No'
01312      Can_Invoke_Chaos: 'No'
01313      Can_Invoke_External_Research: 'No'
01314      Can_Trigger_Recursion: 'No'
01315      Can_Veto: 'No'
01316      Risk_Class: Critical
01317      Confidence_Impact: Neutral
01318      Entropy_Signal: 'No'
01319      Default_Params: null
01320      Dependencies: null
01321      Produces_Artifacts: 'Yes'
```



```

01322     Audit_Events:
01323     - 'Yes'
01324     Version: 1.0.0
01325     Owner: UKG/USKD Core
01326     Status: Active
01327     Notes: null
01328     Implementation_Mode: PYTHON_LIB
01329     Primary_Libraries:
01330     - pydantic
01331     - opentelemetry
01332     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
01333     Fallback_Libraries:
01334     - asynpg
01335     Fallback_Library_Versions: asynpg>=0.29,<1
01336     Runtime_Env: python
01337     Test_Harness: pytest + compliance fixtures
01338     Has_Math: 'No'
01339     Math_Component_Type: null
01340     Math_Object_IDs: null
01341     Math_Scope: null
01342     Variables_Used: null
01343     Units_Normalization: null
01344     Assumptions_Constraints: null
01345     KA-028:
01346     KA_Name: Point-of-View Expansion
01347     Short_Name: POV
01348     Purpose: Add alternate viewpoints beyond base personas
01349     Category: Persona
01350     Primary_Layers:
01351     - L4
01352     Allowed_Layers:
01353     - L4
01354     Inputs: Context state; debate state
01355     Outputs: Additional POV set; objections
01356     Reads_Memory: 'Yes'
01357     Writes_Memory: 'No'
01358     Can_Invoke_Chaos: 'Yes'
01359     Can_Invoke_External_Research: 'No'
01360     Can_Trigger_Recursion: 'No'
01361     Can_Veto: 'No'
01362     Risk_Class: Medium
01363     Confidence_Impact: Indirect ↑
01364     Entropy_Signal: Contributes
01365     Default_Params: null
01366     Dependencies: null
01367     Produces_Artifacts: 'Yes'
01368     Audit_Events:
01369     - 'Yes'
01370     Version: 1.0.0
01371     Owner: UKG/USKD Core
01372     Status: Active
01373     Notes: null
01374     Implementation_Mode: IN_MODEL
01375     Primary_Libraries: null
01376     Primary_Library_Versions: null
01377     Fallback_Libraries: null
01378     Fallback_Library_Versions: null
01379     Runtime_Env: in-model
01380     Test_Harness: pytest + reasoning-evals
01381     Has_Math: 'No'
01382     Math_Component_Type: null
01383     Math_Object_IDs: null
01384     Math_Scope: null
01385     Variables_Used: null
01386     Units_Normalization: null
01387     Assumptions_Constraints: null

```

```
01388     KA-029:
01389         KA_Name: Knowledge Expansion
01390         Short_Name: EXPAND
01391         Purpose: Traverse UKG/USKD cross-links to expand context
01392         Category: Context
01393         Primary_Layers:
01394         - L2
01395         - L3
01396         Allowed_Layers:
01397         - L2
01398         - L3
01399         Inputs: Axis coordinates; entity seeds
01400         Outputs: Expanded node set; link paths
01401         Reads_Memory: 'Yes'
01402         Writes_Memory: 'No'
01403         Can_Invoke_Chaos: 'No'
01404         Can_Invoke_External_Research: 'No'
01405         Can_Trigger_Recursion: 'No'
01406         Can_Veto: 'No'
01407         Risk_Class: Medium
01408         Confidence_Impact: Indirect ↑
01409         Entropy_Signal: 'No'
01410         Default_Params: null
01411         Dependencies: null
01412         Produces_Artifacts: 'Yes'
01413         Audit_Events:
01414         - 'Yes'
01415         Version: 1.0.0
01416         Owner: UKG/USKD Core
01417         Status: Active
01418         Notes: null
01419         Implementation_Mode: PYTHON_LIB
01420         Primary_Libraries:
01421         - networkx
01422         Primary_Library_Versions: networkx>=3.2,<4
01423         Fallback_Libraries:
01424         - neo4j
01425         - chromadb
01426         Fallback_Library_Versions: neo4j>=5.18,<6; chromadb>=0.5,<1
01427         Runtime_Env: python
01428         Test_Harness: pytest + graph fixtures
01429         Has_Math: 'No'
01430         Math_Component_Type: null
01431         Math_Object_IDs: null
01432         Math_Scope: null
01433         Variables_Used: null
01434         Units_Normalization: null
01435         Assumptions_Constraints: null
01436     KA-030:
01437         KA_Name: Conflict Resolution
01438         Short_Name: RESOLVE
01439         Purpose: Arbitrate and resolve conflicts; produce coherent state
01440         Category: Coordination
01441         Primary_Layers:
01442         - L5
01443         Allowed_Layers:
01444         - L5
01445         Inputs: Conflict set; priority rules
01446         Outputs: Resolved state; decision log
01447         Reads_Memory: 'Yes'
01448         Writes_Memory: 'No'
01449         Can_Invoke_Chaos: 'No'
01450         Can_Invoke_External_Research: 'No'
01451         Can_Trigger_Recursion: 'No'
01452         Can_Veto: 'No'
01453         Risk_Class: High
01454         Confidence_Impact: Indirect ↑
```

```
01455     Entropy_Signal: Consumes
01456     Default_Params: null
01457     Dependencies:
01458     - KA-26
01459     - KA-13
01460     Produces_Artifacts: 'Yes'
01461     Audit_Events:
01462     - 'Yes'
01463     Version: 1.0.0
01464     Owner: UKG/USKD Core
01465     Status: Active
01466     Notes: null
01467     Implementation_Mode: IN_MODEL
01468     Primary_Libraries: null
01469     Primary_Library_Versions: null
01470     Fallback_Libraries: null
01471     Fallback_Library_Versions: null
01472     Runtime_Env: in-model
01473     Test_Harness: pytest + eval harness
01474     Has_Math: 'No'
01475     Math_Component_Type: null
01476     Math_Object_IDs: null
01477     Math_Scope: null
01478     Variables_Used: null
01479     Units_Normalization: null
01480     Assumptions_Constraints: null
01481 KA-031:
01482     KA_Name: Algorithm Selection Engine
01483     Short_Name: ASE
01484     Purpose: Select KA pipeline given query, policies, and budget
01485     Category: Routing
01486     Primary_Layers:
01487     - L1
01488     Allowed_Layers:
01489     - L1
01490     Inputs: Query class; complexity tier; policy; budget
01491     Outputs: Ordered KA pipeline; allowed sets
01492     Reads_Memory: 'No'
01493     Writes_Memory: 'No'
01494     Can_Invoke_Chaos: 'No'
01495     Can_Invoke_External_Research: 'No'
01496     Can_Trigger_Recursion: 'No'
01497     Can_Veto: 'No'
01498     Risk_Class: High
01499     Confidence_Impact: Neutral
01500     Entropy_Signal: 'No'
01501     Default_Params: null
01502     Dependencies:
01503     - KA-5
01504     - KA-113
01505     - KA-81
01506     Produces_Artifacts: 'Yes'
01507     Audit_Events:
01508     - 'Yes'
01509     Version: 1.0.0
01510     Owner: UKG/USKD Core
01511     Status: Active
01512     Notes: null
01513     Implementation_Mode: HYBRID
01514     Primary_Libraries:
01515     - pydantic
01516     - pyyaml
01517     - orjson
01518     - opentelemetry
01519     - prometheus-client
01520     - networkx
01521     Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \
```

```

    opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
01522     prometheus-client>=0.20,<1; networkx>=3.2,<4
01523     Fallback_Libraries:
01524     - redis
01525     Fallback_Library_Versions: redis>=5.0,<6
01526     Runtime_Env: python + in-model
01527     Test_Harness: pytest + golden-trace tests
01528     Has_Math: 'No'
01529     Math_Component_Type: null
01530     Math_Object_IDs: null
01531     Math_Scope: null
01532     Variables_Used: null
01533     Units_Normalization: null
01534     Assumptions_Constraints: null
01535 KA-032:
01536     KA_Name: Simulation Orchestration Controller
01537     Short_Name: ORCH
01538     Purpose: Sequence KAs; manage layer transitions; enforce exit rules
01539     Category: Control
01540     Primary_Layers:
01541     - L1
01542     - L2
01543     - L3
01544     - L4
01545     - L5
01546     - L6
01547     - L7
01548     - L8
01549     - L9
01550     - L10
01551     Allowed_Layers:
01552     - L1-L10
01553     Inputs: KA pipeline; simulation state
01554     Outputs: Execution schedule; checkpoints
01555     Reads_Memory: 'Yes'
01556     Writes_Memory: 'Yes'
01557     Can_Invoke_Chaos: 'No'
01558     Can_Invoke_External_Research: 'No'
01559     Can_Trigger_Recursion: 'No'
01560     Can_Veto: 'No'
01561     Risk_Class: Critical
01562     Confidence_Impact: Neutral
01563     Entropy_Signal: 'No'
01564     Default_Params: null
01565     Dependencies:
01566     - KA-31
01567     - KA-107
01568     Produces_Artifacts: 'Yes'
01569     Audit_Events:
01570     - 'Yes'
01571     Version: 1.0.0
01572     Owner: UKG/USKD Core
01573     Status: Active
01574     Notes: null
01575     Implementation_Mode: HYBRID
01576     Primary_Libraries:
01577     - pydantic
01578     - pyyaml
01579     - orjson
01580     - opentelemetry
01581     - prometheus-client
01582     - networkx
01583     - celery
01584     - langgraph
01585     - llama-index
01586     - redis
01587     Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \

```

```
    opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
01588      prometheus-client>=0.20,<1; networkx>=3.2,<4; celery>=5.3,<6; langgraph>=0.2,<1; \
    llama-index>=0.10,<1; redis>=5.0,<6
01589      Fallback_Libraries:
01590        - redis
01591      Fallback_Library_Versions: redis>=5.0,<6
01592      Runtime_Env: python + in-model
01593      Test_Harness: pytest + integration tests
01594      Has_Math: 'No'
01595      Math_Component_Type: null
01596      Math_Object_IDs: null
01597      Math_Scope: null
01598      Variables_Used: null
01599      Units_Normalization: null
01600      Assumptions_Constraints: null
01601  KA-033:
01602    KA_Name: Reserved (Expansion Slot)
01603    Short_Name: RESERVED
01604    Purpose: Reserved ID for future algorithm
01605    Category: Reserved
01606    Primary_Layers: null
01607    Allowed_Layers: null
01608    Inputs: null
01609    Outputs: null
01610    Reads_Memory: 'No'
01611    Writes_Memory: 'No'
01612    Can_Invoke_Chaos: 'No'
01613    Can_Invoke_External_Research: 'No'
01614    Can_Trigger_Recursion: 'No'
01615    Can_Veto: 'No'
01616    Risk_Class: Low
01617    Confidence_Impact: Neutral
01618    Entropy_Signal: 'No'
01619    Default_Params: null
01620    Dependencies: null
01621    Produces_Artifacts: 'No'
01622    Audit_Events:
01623      - 'No'
01624    Version: 1.0.0
01625    Owner: UKG/USKD Core
01626    Status: Reserved
01627    Notes: null
01628    Implementation_Mode: IN_MODEL
01629    Primary_Libraries: null
01630    Primary_Library_Versions: null
01631    Fallback_Libraries: null
01632    Fallback_Library_Versions: null
01633    Runtime_Env: in-model
01634    Test_Harness: pytest + eval harness
01635    Has_Math: 'No'
01636    Math_Component_Type: null
01637    Math_Object_IDs: null
01638    Math_Scope: null
01639    Variables_Used: null
01640    Units_Normalization: null
01641    Assumptions_Constraints: null
01642  KA-034:
01643    KA_Name: Adversarial Reasoning
01644    Short_Name: ADV
01645    Purpose: Simulate adversarial constraints/attacks on assumptions
01646    Category: Safety
01647    Primary_Layers:
01648      - L3
01649      - L7
01650    Allowed_Layers:
01651      - L3
01652      - L7
```

```
01653      Inputs: Scenario; threat model
01654      Outputs: Robust alternatives; failure points
01655      Reads_Memory: 'Yes'
01656      Writes_Memory: 'No'
01657      Can_Invoke_Chaos: 'Yes'
01658      Can_Invoke_External_Research: 'No'
01659      Can_Trigger_Recursion: 'No'
01660      Can_Veto: 'No'
01661      Risk_Class: High
01662      Confidence_Impact: Indirect ↑
01663      Entropy_Signal: Contributes
01664      Default_Params: null
01665      Dependencies: null
01666      Produces_Artifacts: 'Yes'
01667      Audit_Events:
01668      - 'Yes'
01669      Version: 1.0.0
01670      Owner: UKG/USKD Core
01671      Status: Active
01672      Notes: null
01673      Implementation_Mode: PYTHON_LIB
01674      Primary_Libraries:
01675      - pydantic
01676      - opentelemetry
01677      Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
01678      Fallback_Libraries:
01679      - asyncpg
01680      Fallback_Library_Versions: asyncpg>=0.29,<1
01681      Runtime_Env: python
01682      Test_Harness: pytest + compliance fixtures
01683      Has_Math: 'No'
01684      Math_Component_Type: null
01685      Math_Object_IDs: null
01686      Math_Scope: null
01687      Variables_Used: null
01688      Units_Normalization: null
01689      Assumptions_Constraints: null
01690      KA-035:
01691      KA_Name: Bayesian Gap Imputation
01692      Short_Name: IMPUTE
01693      Purpose: Probabilistically fill missing data with uncertainty bounds
01694      Category: Reasoning
01695      Primary_Layers:
01696      - L2
01697      - L7
01698      Allowed_Layers:
01699      - L2
01700      - L7
01701      Inputs: Gap vector; priors
01702      Outputs: Imputed values; uncertainty intervals
01703      Reads_Memory: 'Yes'
01704      Writes_Memory: 'No'
01705      Can_Invoke_Chaos: 'No'
01706      Can_Invoke_External_Research: 'No'
01707      Can_Trigger_Recursion: 'No'
01708      Can_Veto: 'No'
01709      Risk_Class: High
01710      Confidence_Impact: Direct ↓/↑
01711      Entropy_Signal: Produces
01712      Default_Params: null
01713      Dependencies: null
01714      Produces_Artifacts: 'Yes'
01715      Audit_Events:
01716      - 'Yes'
01717      Version: 1.0.0
01718      Owner: UKG/USKD Core
```

```
01719     Status: Active
01720     Notes: null
01721     Implementation_Mode: IN_MODEL
01722     Primary_Libraries: null
01723     Primary_Library_Versions: null
01724     Fallback_Libraries: null
01725     Fallback_Library_Versions: null
01726     Runtime_Env: in-model
01727     Test_Harness: pytest + reasoning-evals
01728     Has_Math: 'Yes'
01729     Math_Component_Type: Equation
01730     Math_Object_IDs:
01731     - F-IMPUTE-01
01732     Math_Scope: KA
01733     Variables_Used:
01734     - x
01735     -  $\mu$ 
01736     -  $\sigma$ 
01737     - D
01738     Units_Normalization:  $\sigma > 0$ 
01739     Assumptions_Constraints: Assumes reasonable priors
01740 KA-036:
01741     KA_Name: Pareto Optimization Engine
01742     Short_Name: PARETO
01743     Purpose: Optimize multi-objective trade-offs
01744     Category: Optimization
01745     Primary_Layers:
01746     - L7
01747     Allowed_Layers:
01748     - L7
01749     Inputs: Objectives; constraints; options
01750     Outputs: Pareto front; recommended set
01751     Reads_Memory: 'Yes'
01752     Writes_Memory: 'No'
01753     Can_Invoke_Chaos: 'No'
01754     Can_Invoke_External_Research: 'No'
01755     Can_Trigger_Recursion: 'No'
01756     Can_Veto: 'No'
01757     Risk_Class: Medium
01758     Confidence_Impact: Indirect  $\uparrow$ 
01759     Entropy_Signal: 'No'
01760     Default_Params: null
01761     Dependencies: null
01762     Produces_Artifacts: 'Yes'
01763     Audit_Events:
01764     - 'Yes'
01765     Version: 1.0.0
01766     Owner: UKG/USKD Core
01767     Status: Active
01768     Notes: null
01769     Implementation_Mode: HYBRID
01770     Primary_Libraries:
01771     - numpy
01772     - scipy
01773     - networkx
01774     - pydantic
01775     Primary_Library_Versions: numpy>=1.26,<3; scipy>=1.11,<2; networkx>=3.2,<4; \
pydantic>=2.6,<3
01776     Fallback_Libraries:
01777     - pandas
01778     Fallback_Library_Versions: pandas>=2.2,<3
01779     Runtime_Env: python + in-model
01780     Test_Harness: pytest + reasoning-evals
01781     Has_Math: 'Yes'
01782     Math_Component_Type: Algorithm
01783     Math_Object_IDs:
01784     - A-PARETO-01
```

```
01785     Math_Scope: KA
01786     Variables_Used:
01787     - f_i
01788     - S
01789     Units_Normalization: Depends on objective units
01790     Assumptions_Constraints: Normalize objectives if needed
01791 KA-037:
01792     KA_Name: Norm Emergence Detector
01793     Short_Name: NORM
01794     Purpose: Detect groupthink/unhealthy convergence in multi-agent outputs
01795     Category: Analysis
01796     Primary_Layers:
01797     - L5
01798     - L7
01799     Allowed_Layers:
01800     - L5
01801     - L7
01802     Inputs: Persona outputs; debate trace
01803     Outputs: Norm flags; convergence metrics
01804     Reads_Memory: 'Yes'
01805     Writes_Memory: 'No'
01806     Can_Invoke_Chaos: 'No'
01807     Can_Invoke_External_Research: 'No'
01808     Can_Trigger_Recursion: 'No'
01809     Can_Veto: 'No'
01810     Risk_Class: High
01811     Confidence_Impact: Neutral
01812     Entropy_Signal: Produces
01813     Default_Params: null
01814     Dependencies:
01815     - KA-12
01816     - KA-30
01817     Produces_Artifacts: 'Yes'
01818     Audit_Events:
01819     - 'Yes'
01820     Version: 1.0.0
01821     Owner: UKG/USKD Core
01822     Status: Active
01823     Notes: null
01824     Implementation_Mode: PYTHON_LIB
01825     Primary_Libraries:
01826     - pydantic
01827     - numpy
01828     - pandas
01829     - rapidfuzz
01830     - opentelemetry
01831     Primary_Library_Versions: pydantic>=2.6,<3; numpy>=1.26,<3; pandas>=2.2,<3; \
01832     rapidfuzz>=3.6,<4; opentelemetry-api>=1.26,<2;
01833     opentelemetry-sdk>=1.26,<2
01834     Fallback_Libraries:
01835     - scikit-learn
01836     Fallback_Library_Versions: scikit-learn>=1.4,<2
01837     Runtime_Env: python
01838     Test_Harness: pytest + evidence fixtures
01839     Has_Math: 'No'
01840     Math_Component_Type: null
01841     Math_Object_IDs: null
01842     Math_Scope: null
01843     Variables_Used: null
01844     Units_Normalization: null
01845     Assumptions_Constraints: null
01846 KA-038:
01847     KA_Name: Cross-Modal Synthesis
01848     Short_Name: XMOD
01849     Purpose: Fuse multi-modal evidence into unified representation
01850     Category: Reasoning
01851     Primary_Layers:
```



```
01851     - L3
01852     - L9
01853     Allowed_Layers:
01854     - L3
01855     - L9
01856     Inputs: Text+tables+images+code evidence
01857     Outputs: Unified evidence object
01858     Reads_Memory: 'Yes'
01859     Writes_Memory: 'No'
01860     Can_Invoke_Chaos: 'No'
01861     Can_Invoke_External_Research: 'No'
01862     Can_Trigger_Recursion: 'No'
01863     Can_Veto: 'No'
01864     Risk_Class: Medium
01865     Confidence_Impact: Indirect ↑
01866     Entropy_Signal: 'No'
01867     Default_Params: null
01868     Dependencies: null
01869     Produces_Artifacts: 'Yes'
01870     Audit_Events:
01871     - 'Yes'
01872     Version: 1.0.0
01873     Owner: UKG/USKD Core
01874     Status: Active
01875     Notes: null
01876     Implementation_Mode: IN_MODEL
01877     Primary_Libraries: null
01878     Primary_Library_Versions: null
01879     Fallback_Libraries: null
01880     Fallback_Library_Versions: null
01881     Runtime_Env: in-model
01882     Test_Harness: pytest + reasoning-evals
01883     Has_Math: 'No'
01884     Math_Component_Type: null
01885     Math_Object_IDs: null
01886     Math_Scope: null
01887     Variables_Used: null
01888     Units_Normalization: null
01889     Assumptions_Constraints: null
01890 KA-039:
01891     KA_Name: Ontology Drift Detection
01892     Short_Name: ODRIFT
01893     Purpose: Detect semantic drift in ontology definitions over time
01894     Category: Lifecycle
01895     Primary_Layers:
01896     - L6
01897     - L10
01898     Allowed_Layers:
01899     - L6
01900     - L10
01901     Inputs: Ontology snapshots
01902     Outputs: Drift alerts; impacted nodes
01903     Reads_Memory: 'Yes'
01904     Writes_Memory: 'Yes'
01905     Can_Invoke_Chaos: 'No'
01906     Can_Invoke_External_Research: 'No'
01907     Can_Trigger_Recursion: 'No'
01908     Can_Veto: 'No'
01909     Risk_Class: High
01910     Confidence_Impact: Indirect ↓
01911     Entropy_Signal: Produces
01912     Default_Params: null
01913     Dependencies: null
01914     Produces_Artifacts: 'Yes'
01915     Audit_Events:
01916     - 'Yes'
01917     Version: 1.0.0
```

```
01918      Owner: UKG/USKD Core
01919      Status: Active
01920      Notes: null
01921      Implementation_Mode: PYTHON_LIB
01922      Primary_Libraries:
01923      - pydantic
01924      - opentelemetry
01925      - sqlalchemy
01926      - psycpg
01927      - alembic
01928      - redis
01929      Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
01930      psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
01931      Fallback_Libraries:
01932      - asynpg
01933      Fallback_Library_Versions: asynpg>=0.29,<1
01934      Runtime_Env: python
01935      Test_Harness: pytest + compliance fixtures
01936      Has_Math: 'No'
01937      Math_Component_Type: null
01938      Math_Object_IDs: null
01939      Math_Scope: null
01940      Variables_Used: null
01941      Units_Normalization: null
01942      Assumptions_Constraints: null
01943      KA-040:
01944      KA_Name: Semantic Alignment Engine
01945      Short_Name: ALIGN
01946      Purpose: Align synonyms/overlaps; prevent fragmentation
01947      Category: Context
01948      Primary_Layers:
01949      - L3
01950      - L6
01951      Allowed_Layers:
01952      - L3
01953      - L6
01954      Inputs: Concept sets; embeddings
01955      Outputs: Alignment mappings; merge suggestions
01956      Reads_Memory: 'Yes'
01957      Writes_Memory: 'Yes'
01958      Can_Invoke_Chaos: 'No'
01959      Can_Invoke_External_Research: 'No'
01960      Can_Trigger_Recursion: 'No'
01961      Can_Veto: 'No'
01962      Risk_Class: Medium
01963      Confidence_Impact: Indirect ↑
01964      Entropy_Signal: 'No'
01965      Default_Params: null
01966      Dependencies: null
01967      Produces_Artifacts: 'Yes'
01968      Audit_Events:
01969      - 'Yes'
01970      Version: 1.0.0
01971      Owner: UKG/USKD Core
01972      Status: Active
01973      Notes: null
01974      Implementation_Mode: PYTHON_LIB
01975      Primary_Libraries:
01976      - networkx
01977      Primary_Library_Versions: networkx>=3.2,<4
01978      Fallback_Libraries:
01979      - neo4j
01980      - chromadb
01981      Fallback_Library_Versions: neo4j>=5.18,<6; chromadb>=0.5,<1
01982      Runtime_Env: python
01983      Test_Harness: pytest + graph fixtures
```

```
01984     Has_Math: 'No'
01985     Math_Component_Type: null
01986     Math_Object_IDs: null
01987     Math_Scope: null
01988     Variables_Used: null
01989     Units_Normalization: null
01990     Assumptions_Constraints: null
01991 KA-041:
01992     KA_Name: Concept Confidence Normalization
01993     Short_Name: CNORM
01994     Purpose: Normalize confidence scales across domains
01995     Category: Truth
01996     Primary_Layers:
01997     - L6
01998     Allowed_Layers:
01999     - L6
02000     Inputs: Confidence vectors
02001     Outputs: Normalized confidence
02002     Reads_Memory: 'Yes'
02003     Writes_Memory: 'No'
02004     Can_Invoke_Chaos: 'No'
02005     Can_Invoke_External_Research: 'No'
02006     Can_Trigger_Recursion: 'No'
02007     Can_Veto: 'No'
02008     Risk_Class: High
02009     Confidence_Impact: Direct ↑
02010     Entropy_Signal: Consumes
02011     Default_Params: null
02012     Dependencies: null
02013     Produces_Artifacts: 'Yes'
02014     Audit_Events:
02015     - 'Yes'
02016     Version: 1.0.0
02017     Owner: UKG/USKD Core
02018     Status: Active
02019     Notes: null
02020     Implementation_Mode: PYTHON_LIB
02021     Primary_Libraries:
02022     - pydantic
02023     - numpy
02024     - pandas
02025     - rapidfuzz
02026     - opentelemetry
02027     Primary_Library_Versions: pydantic>=2.6,<3; numpy>=1.26,<3; pandas>=2.2,<3; \
    rapidfuzz>=3.6,<4; opentelemetry-api>=1.26,<2;
02028     opentelemetry-sdk>=1.26,<2
02029     Fallback_Libraries:
02030     - scikit-learn
02031     Fallback_Library_Versions: scikit-learn>=1.4,<2
02032     Runtime_Env: python
02033     Test_Harness: pytest + evidence fixtures
02034     Has_Math: 'No'
02035     Math_Component_Type: null
02036     Math_Object_IDs: null
02037     Math_Scope: null
02038     Variables_Used: null
02039     Units_Normalization: null
02040     Assumptions_Constraints: null
02041 KA-042:
02042     KA_Name: Contradiction Propagation Analysis
02043     Short_Name: CPROP
02044     Purpose: Trace downstream impacts of contradictions
02045     Category: Analysis
02046     Primary_Layers:
02047     - L6
02048     Allowed_Layers:
02049     - L6
```

```
02050     Inputs: Conflict set; dependency graph
02051     Outputs: Affected paths; priority fixes
02052     Reads_Memory: 'Yes'
02053     Writes_Memory: 'No'
02054     Can_Invoke_Chaos: 'No'
02055     Can_Invoke_External_Research: 'No'
02056     Can_Trigger_Recursion: 'No'
02057     Can_Veto: 'No'
02058     Risk_Class: High
02059     Confidence_Impact: Indirect ↓
02060     Entropy_Signal: Produces
02061     Default_Params: null
02062     Dependencies:
02063     - KA-26
02064     - KA-25
02065     Produces_Artifacts: 'Yes'
02066     Audit_Events:
02067     - 'Yes'
02068     Version: 1.0.0
02069     Owner: UKG/USKD Core
02070     Status: Active
02071     Notes: null
02072     Implementation_Mode: PYTHON_LIB
02073     Primary_Libraries:
02074     - pydantic
02075     - numpy
02076     - pandas
02077     - rapidfuzz
02078     - opentelemetry
02079     Primary_Library_Versions: pydantic>=2.6,<3; numpy>=1.26,<3; pandas>=2.2,<3; \
    rapidfuzz>=3.6,<4; opentelemetry-api>=1.26,<2;
02080     opentelemetry-sdk>=1.26,<2
02081     Fallback_Libraries:
02082     - scikit-learn
02083     Fallback_Library_Versions: scikit-learn>=1.4,<2
02084     Runtime_Env: python
02085     Test_Harness: pytest + evidence fixtures
02086     Has_Math: 'No'
02087     Math_Component_Type: null
02088     Math_Object_IDs: null
02089     Math_Scope: null
02090     Variables_Used: null
02091     Units_Normalization: null
02092     Assumptions_Constraints: null
02093 KA-043:
02094     KA_Name: Knowledge Lineage Tracker
02095     Short_Name: LINEAGE
02096     Purpose: Track knowledge evolution across simulations/commits
02097     Category: Governance
02098     Primary_Layers:
02099     - L10
02100     Allowed_Layers:
02101     - L10
02102     Inputs: Commit logs; provenance
02103     Outputs: Lineage graph; change history
02104     Reads_Memory: 'Yes'
02105     Writes_Memory: 'Yes'
02106     Can_Invoke_Chaos: 'No'
02107     Can_Invoke_External_Research: 'No'
02108     Can_Trigger_Recursion: 'No'
02109     Can_Veto: 'No'
02110     Risk_Class: Medium
02111     Confidence_Impact: Neutral
02112     Entropy_Signal: 'No'
02113     Default_Params: null
02114     Dependencies: null
02115     Produces_Artifacts: 'Yes'
```

```
02116     Audit_Events:
02117     - 'Yes'
02118     Version: 1.0.0
02119     Owner: UKG/USKD Core
02120     Status: Active
02121     Notes: null
02122     Implementation_Mode: PYTHON_LIB
02123     Primary_Libraries:
02124     - pydantic
02125     - opentelemetry
02126     - sqlalchemy
02127     - psycpg
02128     - alembic
02129     - redis
02130     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
02131     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
02132     Fallback_Libraries:
02133     - asyncpg
02134     Fallback_Library_Versions: asyncpg>=0.29,<1
02135     Runtime_Env: python
02136     Test_Harness: pytest + compliance fixtures
02137     Has_Math: 'No'
02138     Math_Component_Type: null
02139     Math_Object_IDs: null
02140     Math_Scope: null
02141     Variables_Used: null
02142     Units_Normalization: null
02143     Assumptions_Constraints: null
02144 KA-044:
02145     KA_Name: Creative Knowledge Composer
02146     Short_Name: COMPOSE
02147     Purpose: Generate novel combinations/hypotheses (bounded)
02148     Category: Creativity
02149     Primary_Layers:
02150     - L7
02151     Allowed_Layers:
02152     - L7
02153     Inputs: Knowledge graph; constraints
02154     Outputs: Novel hypotheses; test plans
02155     Reads_Memory: 'Yes'
02156     Writes_Memory: 'No'
02157     Can_Invoke_Chaos: 'Yes'
02158     Can_Invoke_External_Research: 'No'
02159     Can_Trigger_Recursion: 'No'
02160     Can_Veto: 'No'
02161     Risk_Class: High
02162     Confidence_Impact: Neutral
02163     Entropy_Signal: Produces
02164     Default_Params: null
02165     Dependencies: null
02166     Produces_Artifacts: 'Yes'
02167     Audit_Events:
02168     - 'Yes'
02169     Version: 1.0.0
02170     Owner: UKG/USKD Core
02171     Status: Active
02172     Notes: null
02173     Implementation_Mode: IN_MODEL
02174     Primary_Libraries: null
02175     Primary_Library_Versions: null
02176     Fallback_Libraries: null
02177     Fallback_Library_Versions: null
02178     Runtime_Env: in-model
02179     Test_Harness: pytest + reasoning-evals
02180     Has_Math: 'No'
02181     Math_Component_Type: null
```

```
02182     Math_Object_IDs: null
02183     Math_Scope: null
02184     Variables_Used: null
02185     Units_Normalization: null
02186     Assumptions_Constraints: null
02187 KA-045:
02188   KA_Name: Bias Pattern Analyzer
02189   Short_Name: BIAS-PAT
02190   Purpose: Detect systemic bias patterns (population-level)
02191   Category: Ethics
02192   Primary_Layers:
02193   - L8
02194   Allowed_Layers:
02195   - L8
02196   Inputs: Traces over time; outputs corpus
02197   Outputs: Bias patterns; risk alerts
02198   Reads_Memory: 'Yes'
02199   Writes_Memory: 'No'
02200   Can_Invoke_Chaos: 'No'
02201   Can_Invoke_External_Research: 'No'
02202   Can_Trigger_Recursion: 'No'
02203   Can_Veto: 'No'
02204   Risk_Class: High
02205   Confidence_Impact: Neutral
02206   Entropy_Signal: 'No'
02207   Default_Params: null
02208   Dependencies: null
02209   Produces_Artifacts: 'Yes'
02210   Audit_Events:
02211   - 'Yes'
02212   Version: 1.0.0
02213   Owner: UKG/USKD Core
02214   Status: Active
02215   Notes: null
02216   Implementation_Mode: PYTHON_LIB
02217   Primary_Libraries:
02218   - pydantic
02219   - opentelemetry
02220   Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
02221   Fallback_Libraries:
02222   - asynpg
02223   Fallback_Library_Versions: asynpg>=0.29,<1
02224   Runtime_Env: python
02225   Test_Harness: pytest + compliance fixtures
02226   Has_Math: 'No'
02227   Math_Component_Type: null
02228   Math_Object_IDs: null
02229   Math_Scope: null
02230   Variables_Used: null
02231   Units_Normalization: null
02232   Assumptions_Constraints: null
02233 KA-046:
02234   KA_Name: Hierarchical Memory Patcher
02235   Short_Name: MEM-PATCH
02236   Purpose: Apply validated updates to correct memory tier
02237   Category: Memory
02238   Primary_Layers:
02239   - L10
02240   Allowed_Layers:
02241   - L10
02242   Inputs: Validated knowledge; containment class
02243   Outputs: Memory patch operations
02244   Reads_Memory: 'Yes'
02245   Writes_Memory: 'Yes'
02246   Can_Invoke_Chaos: 'No'
02247   Can_Invoke_External_Research: 'No'
```

```
02248     Can_Trigger_Recursion: 'No'
02249     Can_Veto: 'No'
02250     Risk_Class: Critical
02251     Confidence_Impact: Indirect ↑
02252     Entropy_Signal: 'No'
02253     Default_Params: null
02254     Dependencies:
02255     - KA-109
02256     - KA-79
02257     Produces_Artifacts: 'Yes'
02258     Audit_Events:
02259     - 'Yes'
02260     Version: 1.0.0
02261     Owner: UKG/USKD Core
02262     Status: Active
02263     Notes: null
02264     Implementation_Mode: PYTHON_LIB
02265     Primary_Libraries:
02266     - pydantic
02267     - opentelemetry
02268     - sqlalchemy
02269     - psycpg
02270     - alembic
02271     - redis
02272     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
02273     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
02274     Fallback_Libraries:
02275     - asynpg
02276     Fallback_Library_Versions: asynpg>=0.29,<1
02277     Runtime_Env: python
02278     Test_Harness: pytest + compliance fixtures
02279     Has_Math: 'No'
02280     Math_Component_Type: null
02281     Math_Object_IDs: null
02282     Math_Scope: null
02283     Variables_Used: null
02284     Units_Normalization: null
02285     Assumptions_Constraints: null
02286 KA-047:
02287     KA_Name: Meta-Algorithm Selection
02288     Short_Name: META-ASE
02289     Purpose: Select/invent algorithms dynamically (bounded)
02290     Category: Meta
02291     Primary_Layers:
02292     - L7
02293     Allowed_Layers:
02294     - L7
02295     Inputs: Performance history; problem signature
02296     Outputs: Tuned pipeline; candidate new KA config
02297     Reads_Memory: 'Yes'
02298     Writes_Memory: 'No'
02299     Can_Invoke_Chaos: 'No'
02300     Can_Invoke_External_Research: 'No'
02301     Can_Trigger_Recursion: 'No'
02302     Can_Veto: 'No'
02303     Risk_Class: Critical
02304     Confidence_Impact: Neutral
02305     Entropy_Signal: Produces
02306     Default_Params: null
02307     Dependencies:
02308     - KA-63
02309     - KA-64
02310     - KA-107
02311     Produces_Artifacts: 'Yes'
02312     Audit_Events:
02313     - 'Yes'
```

```
02314     Version: 1.0.0
02315     Owner: UKG/USKD Core
02316     Status: Active
02317     Notes: Must be tightly governed; cannot override safety gates.
02318     Implementation_Mode: IN_MODEL
02319     Primary_Libraries:
02320     - celery
02321     - langgraph
02322     - llama-index
02323     - redis
02324     Primary_Library_Versions: celery>=5.3,<6; langgraph>=0.2,<1; llama-index>=0.10,<1; \
redis>=5.0,<6
02325     Fallback_Libraries: null
02326     Fallback_Library_Versions: null
02327     Runtime_Env: in-model
02328     Test_Harness: pytest + integration tests
02329     Has_Math: 'No'
02330     Math_Component_Type: null
02331     Math_Object_IDs: null
02332     Math_Scope: null
02333     Variables_Used: null
02334     Units_Normalization: null
02335     Assumptions_Constraints: null
02336 KA-048:
02337     KA_Name: Ontological Conflict Resolver
02338     Short_Name: ONTO-RES
02339     Purpose: Reconcile conflicting ontologies/concept systems
02340     Category: Truth
02341     Primary_Layers:
02342     - L6
02343     Allowed_Layers:
02344     - L6
02345     Inputs: Ontology graphs; conflict map
02346     Outputs: Resolved ontology map; translation layer
02347     Reads_Memory: 'Yes'
02348     Writes_Memory: 'Yes'
02349     Can_Invoke_Chaos: 'No'
02350     Can_Invoke_External_Research: 'No'
02351     Can_Trigger_Recursion: 'No'
02352     Can_Veto: 'No'
02353     Risk_Class: High
02354     Confidence_Impact: Indirect ↑
02355     Entropy_Signal: Consumes
02356     Default_Params: null
02357     Dependencies: null
02358     Produces_Artifacts: 'Yes'
02359     Audit_Events:
02360     - 'Yes'
02361     Version: 1.0.0
02362     Owner: UKG/USKD Core
02363     Status: Active
02364     Notes: null
02365     Implementation_Mode: PYTHON_LIB
02366     Primary_Libraries:
02367     - pydantic
02368     - numpy
02369     - pandas
02370     - rapidfuzz
02371     - opentelemetry
02372     - networkx
02373     Primary_Library_Versions: pydantic>=2.6,<3; numpy>=1.26,<3; pandas>=2.2,<3; \
rapidfuzz>=3.6,<4; opentelemetry-api>=1.26,<2;
02374     opentelemetry-sdk>=1.26,<2; networkx>=3.2,<4
02375     Fallback_Libraries:
02376     - scikit-learn
02377     - neo4j
02378     - chromadb
```



```
02379     Fallback_Library_Versions: scikit-learn>=1.4,<2; neo4j>=5.18,<6; chromadb>=0.5,<1
02380     Runtime_Env: python
02381     Test_Harness: pytest + graph fixtures
02382     Has_Math: 'No'
02383     Math_Component_Type: null
02384     Math_Object_IDs: null
02385     Math_Scope: null
02386     Variables_Used: null
02387     Units_Normalization: null
02388     Assumptions_Constraints: null
02389 KA-049:
02390     KA_Name: Knowledge Redundancy Detector
02391     Short_Name: DEDUP
02392     Purpose: Detect duplicate/near-duplicate knowledge
02393     Category: Lifecycle
02394     Primary_Layers:
02395     - L10
02396     Allowed_Layers:
02397     - L10
02398     Inputs: Knowledge nodes; similarity metrics
02399     Outputs: Redundancy scores; merge candidates
02400     Reads_Memory: 'Yes'
02401     Writes_Memory: 'No'
02402     Can_Invoke_Chaos: 'No'
02403     Can_Invoke_External_Research: 'No'
02404     Can_Trigger_Recursion: 'No'
02405     Can_Veto: 'No'
02406     Risk_Class: Medium
02407     Confidence_Impact: Neutral
02408     Entropy_Signal: 'No'
02409     Default_Params: null
02410     Dependencies: null
02411     Produces_Artifacts: 'Yes'
02412     Audit_Events:
02413     - 'Yes'
02414     Version: 1.0.0
02415     Owner: UKG/USKD Core
02416     Status: Active
02417     Notes: null
02418     Implementation_Mode: PYTHON_LIB
02419     Primary_Libraries:
02420     - pydantic
02421     - opentelemetry
02422     - sqlalchemy
02423     - psycpg
02424     - alembic
02425     - redis
02426     - networkx
02427     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
02428     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6; networkx>=3.2,<4
02429     Fallback_Libraries:
02430     - asyncpg
02431     - neo4j
02432     - chromadb
02433     Fallback_Library_Versions: asyncpg>=0.29,<1; neo4j>=5.18,<6; chromadb>=0.5,<1
02434     Runtime_Env: python
02435     Test_Harness: pytest + graph fixtures
02436     Has_Math: 'No'
02437     Math_Component_Type: null
02438     Math_Object_IDs: null
02439     Math_Scope: null
02440     Variables_Used: null
02441     Units_Normalization: null
02442     Assumptions_Constraints: null
02443 KA-050:
02444     KA_Name: Knowledge Integrity Validator
```

```
02445     Short_Name: KB-INTEGRITY
02446     Purpose: Validate internal consistency before persistence
02447     Category: Safety
02448     Primary_Layers:
02449       - L10
02450     Allowed_Layers:
02451       - L10
02452     Inputs: KB snapshot; constraints
02453     Outputs: Integrity report; quarantine list
02454     Reads_Memory: 'Yes'
02455     Writes_Memory: 'No'
02456     Can_Invoke_Chaos: 'No'
02457     Can_Invoke_External_Research: 'No'
02458     Can_Trigger_Recursion: 'No'
02459     Can_Veto: 'Yes'
02460     Risk_Class: Critical
02461     Confidence_Impact: Direct ↑/↓
02462     Entropy_Signal: Produces
02463     Default_Params: null
02464     Dependencies:
02465       - KA-94
02466       - KA-99
02467     Produces_Artifacts: 'Yes'
02468     Audit_Events:
02469       - 'Yes'
02470     Version: 1.0.0
02471     Owner: UKG/USKD Core
02472     Status: Active
02473     Notes: null
02474     Implementation_Mode: PYTHON_LIB
02475     Primary_Libraries:
02476       - pydantic
02477       - opentelemetry
02478     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
02479       opentelemetry-sdk>=1.26,<2
02480     Fallback_Libraries:
02481       - asynpg
02482     Fallback_Library_Versions: asynpg>=0.29,<1
02483     Runtime_Env: python
02484     Test_Harness: pytest + compliance fixtures
02485     Has_Math: 'No'
02486     Math_Component_Type: null
02487     Math_Object_IDs: null
02488     Math_Scope: null
02489     Variables_Used: null
02490     Units_Normalization: null
02491     Assumptions_Constraints: null
02491 KA-051:
02492     KA_Name: Self-Correcting Knowledge Distillation
02493     Short_Name: DISTILL
02494     Purpose: Distill high-confidence outcomes into reusable abstractions
02495     Category: Learning
02496     Primary_Layers:
02497       - L10
02498     Allowed_Layers:
02499       - L10
02500     Inputs: Validated traces; commitments
02501     Outputs: Distilled rules; corrected nodes
02502     Reads_Memory: 'Yes'
02503     Writes_Memory: 'Yes'
02504     Can_Invoke_Chaos: 'No'
02505     Can_Invoke_External_Research: 'No'
02506     Can_Trigger_Recursion: 'No'
02507     Can_Veto: 'No'
02508     Risk_Class: High
02509     Confidence_Impact: Indirect ↑
02510     Entropy_Signal: 'No'
```

```

02511     Default_Params: null
02512     Dependencies:
02513     - KA-46
02514     - KA-79
02515     Produces_Artifacts: 'Yes'
02516     Audit_Events:
02517     - 'Yes'
02518     Version: 1.0.0
02519     Owner: UKG/USKD Core
02520     Status: Active
02521     Notes: null
02522     Implementation_Mode: PYTHON_LIB
02523     Primary_Libraries:
02524     - pydantic
02525     - opentelemetry
02526     - sqlalchemy
02527     - psycpg
02528     - alembic
02529     - redis
02530     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
02531     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
02532     Fallback_Libraries:
02533     - asyncpg
02534     Fallback_Library_Versions: asyncpg>=0.29,<1
02535     Runtime_Env: python
02536     Test_Harness: pytest + compliance fixtures
02537     Has_Math: 'No'
02538     Math_Component_Type: null
02539     Math_Object_IDs: null
02540     Math_Scope: null
02541     Variables_Used: null
02542     Units_Normalization: null
02543     Assumptions_Constraints: null
02544     KA-052:
02545     KA_Name: Temporal Knowledge Evolution
02546     Short_Name: TKE
02547     Purpose: Maintain time-versioned knowledge; retire outdated facts
02548     Category: Lifecycle
02549     Primary_Layers:
02550     - L10
02551     Allowed_Layers:
02552     - L10
02553     Inputs: Time axis data; policy updates
02554     Outputs: Temporal versions; retirements
02555     Reads_Memory: 'Yes'
02556     Writes_Memory: 'Yes'
02557     Can_Invoke_Chaos: 'No'
02558     Can_Invoke_External_Research: 'No'
02559     Can_Trigger_Recursion: 'No'
02560     Can_Veto: 'No'
02561     Risk_Class: High
02562     Confidence_Impact: Indirect ↑/↓
02563     Entropy_Signal: 'No'
02564     Default_Params: null
02565     Dependencies:
02566     - KA-83
02567     Produces_Artifacts: 'Yes'
02568     Audit_Events:
02569     - 'Yes'
02570     Version: 1.0.0
02571     Owner: UKG/USKD Core
02572     Status: Active
02573     Notes: null
02574     Implementation_Mode: PYTHON_LIB
02575     Primary_Libraries:
02576     - pydantic

```

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```
02577     - opentelemetry
02578     - sqlalchemy
02579     - psycpg
02580     - alembic
02581     - redis
02582     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
02583     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
02584     Fallback_Libraries:
02585     - asyncpg
02586     Fallback_Library_Versions: asyncpg>=0.29,<1
02587     Runtime_Env: python
02588     Test_Harness: pytest + compliance fixtures
02589     Has_Math: 'Yes'
02590     Math_Component_Type: Equation
02591     Math_Object_IDs:
02592     - F-TEMP-01
02593     Math_Scope: System
02594     Variables_Used:
02595     - t
02596     - window
02597     Units_Normalization: ISO time
02598     Assumptions_Constraints: Requires timestamps
02599     KA-053:
02600     KA_Name: Dynamic Knowledge Compression
02601     Short_Name: COMPRESS
02602     Purpose: Compress/merge knowledge while preserving utility
02603     Category: Lifecycle
02604     Primary_Layers:
02605     - L10
02606     Allowed_Layers:
02607     - L10
02608     Inputs: Graph segments; usage metrics
02609     Outputs: Compressed nodes; mapping table
02610     Reads_Memory: 'Yes'
02611     Writes_Memory: 'Yes'
02612     Can_Invoke_Chaos: 'No'
02613     Can_Invoke_External_Research: 'No'
02614     Can_Trigger_Recursion: 'No'
02615     Can_Veto: 'No'
02616     Risk_Class: Medium
02617     Confidence_Impact: Neutral
02618     Entropy_Signal: 'No'
02619     Default_Params: null
02620     Dependencies:
02621     - KA-49
02622     - KA-77
02623     Produces_Artifacts: 'Yes'
02624     Audit_Events:
02625     - 'Yes'
02626     Version: 1.0.0
02627     Owner: UKG/USKD Core
02628     Status: Active
02629     Notes: null
02630     Implementation_Mode: PYTHON_LIB
02631     Primary_Libraries:
02632     - pydantic
02633     - opentelemetry
02634     - sqlalchemy
02635     - psycpg
02636     - alembic
02637     - redis
02638     - networkx
02639     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
02640     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6; networkx>=3.2,<4
02641     Fallback_Libraries:
```

```
02642     - asyncpg
02643     - neo4j
02644     - chromadb
02645     Fallback_Library_Versions: asyncpg>=0.29,<1; neo4j>=5.18,<6; chromadb>=0.5,<1
02646     Runtime_Env: python
02647     Test_Harness: pytest + graph fixtures
02648     Has_Math: 'Yes'
02649     Math_Component_Type: Algorithm
02650     Math_Object_IDs:
02651     - A-COMP-01
02652     Math_Scope: System
02653     Variables_Used:
02654     - utility
02655     - rate
02656     Units_Normalization: utility∈[0,1]
02657     Assumptions_Constraints: Preserve utility ≥ threshold
02658 KA-054:
02659     KA_Name: Cross-Lingual Knowledge Fusion
02660     Short_Name: X-LING
02661     Purpose: Align and merge concepts across languages
02662     Category: Context
02663     Primary_Layers:
02664     - L3
02665     - L10
02666     Allowed_Layers:
02667     - L3
02668     - L10
02669     Inputs: Multilingual sources; embeddings
02670     Outputs: Language-agnostic nodes; alignments
02671     Reads_Memory: 'Yes'
02672     Writes_Memory: 'Yes'
02673     Can_Invoke_Chaos: 'No'
02674     Can_Invoke_External_Research: 'No'
02675     Can_Trigger_Recursion: 'No'
02676     Can_Veto: 'No'
02677     Risk_Class: Medium
02678     Confidence_Impact: Indirect ↑
02679     Entropy_Signal: 'No'
02680     Default_Params: null
02681     Dependencies:
02682     - KA-40
02683     Produces_Artifacts: 'Yes'
02684     Audit_Events:
02685     - 'Yes'
02686     Version: 1.0.0
02687     Owner: UKG/USKD Core
02688     Status: Active
02689     Notes: null
02690     Implementation_Mode: IN_MODEL
02691     Primary_Libraries: null
02692     Primary_Library_Versions: null
02693     Fallback_Libraries: null
02694     Fallback_Library_Versions: null
02695     Runtime_Env: in-model
02696     Test_Harness: pytest + eval harness
02697     Has_Math: 'No'
02698     Math_Component_Type: null
02699     Math_Object_IDs: null
02700     Math_Scope: null
02701     Variables_Used: null
02702     Units_Normalization: null
02703     Assumptions_Constraints: null
02704 KA-055:
02705     KA_Name: Adaptive Multi-Modal Integration
02706     Short_Name: MM-INTEG
02707     Purpose: Resolve conflicts across modalities
02708     Category: Reasoning
```

```
02709     Primary_Layers:
02710     - L3
02711     - L9
02712     Allowed_Layers:
02713     - L3
02714     - L9
02715     Inputs: Modal evidence; trust weights
02716     Outputs: Fused evidence; disagreement notes
02717     Reads_Memory: 'Yes'
02718     Writes_Memory: 'No'
02719     Can_Invoke_Chaos: 'No'
02720     Can_Invoke_External_Research: 'No'
02721     Can_Trigger_Recursion: 'No'
02722     Can_Veto: 'No'
02723     Risk_Class: High
02724     Confidence_Impact: Indirect ↑
02725     Entropy_Signal: Produces
02726     Default_Params: null
02727     Dependencies:
02728     - KA-9
02729     - KA-18
02730     Produces_Artifacts: 'Yes'
02731     Audit_Events:
02732     - 'Yes'
02733     Version: 1.0.0
02734     Owner: UKG/USKD Core
02735     Status: Active
02736     Notes: null
02737     Implementation_Mode: IN_MODEL
02738     Primary_Libraries: null
02739     Primary_Library_Versions: null
02740     Fallback_Libraries: null
02741     Fallback_Library_Versions: null
02742     Runtime_Env: in-model
02743     Test_Harness: pytest + reasoning-evals
02744     Has_Math: 'No'
02745     Math_Component_Type: null
02746     Math_Object_IDs: null
02747     Math_Scope: null
02748     Variables_Used: null
02749     Units_Normalization: null
02750     Assumptions_Constraints: null
02751 KA-056:
02752     KA_Name: Narrative Explainability Engine
02753     Short_Name: EXPLAIN
02754     Purpose: Generate audience-safe explanation from trace
02755     Category: Synthesis
02756     Primary_Layers:
02757     - L9
02758     Allowed_Layers:
02759     - L9
02760     Inputs: Trace DAG; decision log
02761     Outputs: Explainability report; rationale map
02762     Reads_Memory: 'Yes'
02763     Writes_Memory: 'No'
02764     Can_Invoke_Chaos: 'No'
02765     Can_Invoke_External_Research: 'No'
02766     Can_Trigger_Recursion: 'No'
02767     Can_Veto: 'No'
02768     Risk_Class: Medium
02769     Confidence_Impact: Neutral
02770     Entropy_Signal: 'No'
02771     Default_Params: null
02772     Dependencies: null
02773     Produces_Artifacts: 'Yes'
02774     Audit_Events:
02775     - 'Yes'
```

```
02776     Version: 1.0.0
02777     Owner: UKG/USKD Core
02778     Status: Active
02779     Notes: null
02780     Implementation_Mode: IN_MODEL
02781     Primary_Libraries:
02782     - pydantic
02783     Primary_Library_Versions: pydantic>=2.6,<3
02784     Fallback_Libraries: null
02785     Fallback_Library_Versions: null
02786     Runtime_Env: in-model
02787     Test_Harness: pytest + reasoning-evals
02788     Has_Math: 'No'
02789     Math_Component_Type: null
02790     Math_Object_IDs: null
02791     Math_Scope: null
02792     Variables_Used: null
02793     Units_Normalization: null
02794     Assumptions_Constraints: null
02795 KA-057:
02796     KA_Name: Persona/Emotion Adaptation
02797     Short_Name: STYLE
02798     Purpose: Adapt tone/structure to stakeholder role
02799     Category: UX
02800     Primary_Layers:
02801     - L9
02802     Allowed_Layers:
02803     - L9
02804     Inputs: User profile; sentiment; output object
02805     Outputs: Styled response plan
02806     Reads_Memory: 'No'
02807     Writes_Memory: 'No'
02808     Can_Invoke_Chaos: 'No'
02809     Can_Invoke_External_Research: 'No'
02810     Can_Trigger_Recursion: 'No'
02811     Can_Veto: 'No'
02812     Risk_Class: Low
02813     Confidence_Impact: Neutral
02814     Entropy_Signal: 'No'
02815     Default_Params: null
02816     Dependencies: null
02817     Produces_Artifacts: 'Yes'
02818     Audit_Events:
02819     - 'Yes'
02820     Version: 1.0.0
02821     Owner: UKG/USKD Core
02822     Status: Active
02823     Notes: null
02824     Implementation_Mode: IN_MODEL
02825     Primary_Libraries: null
02826     Primary_Library_Versions: null
02827     Fallback_Libraries: null
02828     Fallback_Library_Versions: null
02829     Runtime_Env: in-model
02830     Test_Harness: pytest + reasoning-evals
02831     Has_Math: 'No'
02832     Math_Component_Type: null
02833     Math_Object_IDs: null
02834     Math_Scope: null
02835     Variables_Used: null
02836     Units_Normalization: null
02837     Assumptions_Constraints: null
02838 KA-058:
02839     KA_Name: Interactive Clarification & Learning
02840     Short_Name: CLARIFY
02841     Purpose: Decide when to ask user follow-ups
02842     Category: UX
```

```
02843     Primary_Layers:
02844     - L1
02845     - L9
02846     Allowed_Layers:
02847     - L1
02848     - L9
02849     Inputs: Ambiguity metrics; competing intents
02850     Outputs: Clarifying questions; options
02851     Reads_Memory: 'Yes'
02852     Writes_Memory: 'No'
02853     Can_Invoke_Chaos: 'No'
02854     Can_Invoke_External_Research: 'No'
02855     Can_Trigger_Recursion: 'No'
02856     Can_Veto: 'No'
02857     Risk_Class: Low
02858     Confidence_Impact: Indirect ↑
02859     Entropy_Signal: Consumes
02860     Default_Params: null
02861     Dependencies:
02862     - KA-73
02863     - KA-102
02864     Produces_Artifacts: 'Yes'
02865     Audit_Events:
02866     - 'Yes'
02867     Version: 1.0.0
02868     Owner: UKG/USKD Core
02869     Status: Active
02870     Notes: null
02871     Implementation_Mode: IN_MODEL
02872     Primary_Libraries:
02873     - pydantic
02874     Primary_Library_Versions: pydantic>=2.6,<3
02875     Fallback_Libraries: null
02876     Fallback_Library_Versions: null
02877     Runtime_Env: in-model
02878     Test_Harness: pytest + reasoning-evals
02879     Has_Math: 'No'
02880     Math_Component_Type: null
02881     Math_Object_IDs: null
02882     Math_Scope: null
02883     Variables_Used: null
02884     Units_Normalization: null
02885     Assumptions_Constraints: null
02886 KA-059:
02887     KA_Name: Predictive Layer Preemption
02888     Short_Name: PREEMPT
02889     Purpose: Skip unneeded layers/KAs for easy queries; fast-path routing
02890     Category: Control
02891     Primary_Layers:
02892     - L1
02893     Allowed_Layers:
02894     - L1
02895     Inputs: Complexity tier; budget
02896     Outputs: Reduced path; fast-mode flag
02897     Reads_Memory: 'No'
02898     Writes_Memory: 'No'
02899     Can_Invoke_Chaos: 'No'
02900     Can_Invoke_External_Research: 'No'
02901     Can_Trigger_Recursion: 'No'
02902     Can_Veto: 'No'
02903     Risk_Class: Medium
02904     Confidence_Impact: Neutral
02905     Entropy_Signal: 'No'
02906     Default_Params: null
02907     Dependencies:
02908     - KA-113
02909     - KA-31
```



```
02910     Produces_Artifacts: 'Yes'
02911     Audit_Events:
02912     - 'Yes'
02913     Version: 1.0.0
02914     Owner: UKG/USKD Core
02915     Status: Active
02916     Notes: null
02917     Implementation_Mode: HYBRID
02918     Primary_Libraries:
02919     - pydantic
02920     - pyyaml
02921     - orjson
02922     - opentelemetry
02923     - prometheus-client
02924     - networkx
02925     Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \
    opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
02926     prometheus-client>=0.20,<1; networkx>=3.2,<4
02927     Fallback_Libraries:
02928     - redis
02929     Fallback_Library_Versions: redis>=5.0,<6
02930     Runtime_Env: python + in-model
02931     Test_Harness: pytest + golden-trace tests
02932     Has_Math: 'No'
02933     Math_Component_Type: null
02934     Math_Object_IDs: null
02935     Math_Scope: null
02936     Variables_Used: null
02937     Units_Normalization: null
02938     Assumptions_Constraints: null
02939 KA-060:
02940     KA_Name: Cognitive Load Balancer
02941     Short_Name: LOAD
02942     Purpose: Allocate effort to hardest subparts; prune low-yield branches
02943     Category: Control
02944     Primary_Layers:
02945     - L7
02946     Allowed_Layers:
02947     - L7
02948     Inputs: Branch stats; cost estimate
02949     Outputs: Resource allocation; pruned branches
02950     Reads_Memory: 'Yes'
02951     Writes_Memory: 'No'
02952     Can_Invoke_Chaos: 'Yes'
02953     Can_Invoke_External_Research: 'No'
02954     Can_Trigger_Recursion: 'Yes'
02955     Can_Veto: 'No'
02956     Risk_Class: High
02957     Confidence_Impact: Indirect ↑
02958     Entropy_Signal: Consumes
02959     Default_Params: null
02960     Dependencies:
02961     - KA-80
02962     - KA-81
02963     Produces_Artifacts: 'Yes'
02964     Audit_Events:
02965     - 'Yes'
02966     Version: 1.0.0
02967     Owner: UKG/USKD Core
02968     Status: Active
02969     Notes: null
02970     Implementation_Mode: HYBRID
02971     Primary_Libraries:
02972     - pydantic
02973     - pyyaml
02974     - orjson
02975     - opentelemetry
```

```
02976     - prometheus-client
02977     - networkx
02978     Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \
opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
02979         prometheus-client>=0.20,<1; networkx>=3.2,<4
02980     Fallback_Libraries:
02981     - redis
02982     Fallback_Library_Versions: redis>=5.0,<6
02983     Runtime_Env: python + in-model
02984     Test_Harness: pytest + golden-trace tests
02985     Has_Math: 'Yes'
02986     Math_Component_Type: Algorithm
02987     Math_Object_IDs:
02988     - A-LOAD-01
02989     Math_Scope: FROST/System
02990     Variables_Used:
02991     - b_i
02992     - c_i
02993     - τ
02994     Units_Normalization: budget units
02995     Assumptions_Constraints: Requires cost estimator
02996 KA-061:
02997     KA_Name: Adversarial Input Shield
02998     Short_Name: SHIELD
02999     Purpose: Detect/neutralize malicious inputs
03000     Category: Safety
03001     Primary_Layers:
03002     - L1
03003     Allowed_Layers:
03004     - L1
03005     Inputs: Raw query; metadata
03006     Outputs: Sanitized query; block decision
03007     Reads_Memory: 'No'
03008     Writes_Memory: 'No'
03009     Can_Invoke_Chaos: 'No'
03010     Can_Invoke_External_Research: 'No'
03011     Can_Trigger_Recursion: 'No'
03012     Can_Veto: 'Yes'
03013     Risk_Class: Critical
03014     Confidence_Impact: Neutral
03015     Entropy_Signal: 'No'
03016     Default_Params: null
03017     Dependencies:
03018     - KA-4
03019     Produces_Artifacts: 'Yes'
03020     Audit_Events:
03021     - 'Yes'
03022     Version: 1.0.0
03023     Owner: UKG/USKD Core
03024     Status: Active
03025     Notes: null
03026     Implementation_Mode: PYTHON_LIB
03027     Primary_Libraries:
03028     - pydantic
03029     - opentelemetry
03030     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
opentelemetry-sdk>=1.26,<2
03031     Fallback_Libraries:
03032     - asyncpg
03033     Fallback_Library_Versions: asyncpg>=0.29,<1
03034     Runtime_Env: python
03035     Test_Harness: pytest + compliance fixtures
03036     Has_Math: 'No'
03037     Math_Component_Type: null
03038     Math_Object_IDs: null
03039     Math_Scope: null
03040     Variables_Used: null
```

```
03041     Units_Normalization: null
03042     Assumptions_Constraints: null
03043   KA-062:
03044     KA_Name: Decentralized Trust Scoring
03045     Short_Name: D-TRUST
03046     Purpose: Compute trust score via provenance hashes/ledger concepts
03047     Category: Truth
03048     Primary_Layers:
03049       - L6
03050       - L10
03051     Allowed_Layers:
03052       - L6
03053       - L10
03054     Inputs: Evidence hashes; provenance graph
03055     Outputs: Trust score; trust report
03056     Reads_Memory: 'Yes'
03057     Writes_Memory: 'No'
03058     Can_Invoke_Chaos: 'No'
03059     Can_Invoke_External_Research: 'No'
03060     Can_Trigger_Recursion: 'No'
03061     Can_Veto: 'No'
03062     Risk_Class: Critical
03063     Confidence_Impact: Direct ↑/↓
03064     Entropy_Signal: 'No'
03065     Default_Params: null
03066     Dependencies:
03067       - KA-18
03068     Produces_Artifacts: 'Yes'
03069     Audit_Events:
03070       - 'Yes'
03071     Version: 1.0.0
03072     Owner: UKG/USKD Core
03073     Status: Active
03074     Notes: null
03075     Implementation_Mode: PYTHON_LIB
03076     Primary_Libraries:
03077       - pydantic
03078       - numpy
03079       - pandas
03080       - rapidfuzz
03081       - opentelemetry
03082     Primary_Library_Versions: pydantic>=2.6,<3; numpy>=1.26,<3; pandas>=2.2,<3; \
03083       rapidfuzz>=3.6,<4; opentelemetry-api>=1.26,<2;
03084       opentelemetry-sdk>=1.26,<2
03085     Fallback_Libraries:
03086       - scikit-learn
03087     Fallback_Library_Versions: scikit-learn>=1.4,<2
03088     Runtime_Env: python
03089     Test_Harness: pytest + evidence fixtures
03090     Has_Math: 'Yes'
03091     Math_Component_Type: Equation
03092     Math_Object_IDs:
03093       - F-TRUST-01
03094     Math_Scope: TruthEngine/System
03095     Variables_Used:
03096       - w_i
03097       - p_i
03098     Units_Normalization:  $T \in [0,1]$ 
03099     Assumptions_Constraints: Provenance weights required
03100   KA-063:
03101     KA_Name: Continuous Performance Learning
03102     Short_Name: PERF-LEARN
03103     Purpose: Learn which KAs/pipelines perform best over time
03104     Category: Learning
03105     Primary_Layers:
03106       - L10
03107     Allowed_Layers:
```

```
03107     - L10
03108     Inputs: Outcome metrics; feedback
03109     Outputs: Performance profiles; tuning suggestions
03110     Reads_Memory: 'Yes'
03111     Writes_Memory: 'Yes'
03112     Can_Invoke_Chaos: 'No'
03113     Can_Invoke_External_Research: 'No'
03114     Can_Trigger_Recursion: 'No'
03115     Can_Veto: 'No'
03116     Risk_Class: Medium
03117     Confidence_Impact: Neutral
03118     Entropy_Signal: 'No'
03119     Default_Params: null
03120     Dependencies: null
03121     Produces_Artifacts: 'Yes'
03122     Audit_Events:
03123     - 'Yes'
03124     Version: 1.0.0
03125     Owner: UKG/USKD Core
03126     Status: Active
03127     Notes: null
03128     Implementation_Mode: PYTHON_LIB
03129     Primary_Libraries:
03130     - pydantic
03131     - opentelemetry
03132     - sqlalchemy
03133     - psycpg
03134     - alembic
03135     - redis
03136     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
03137     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
03138     Fallback_Libraries:
03139     - asyncpg
03140     Fallback_Library_Versions: asyncpg>=0.29,<1
03141     Runtime_Env: python
03142     Test_Harness: pytest + compliance fixtures
03143     Has_Math: 'No'
03144     Math_Component_Type: null
03145     Math_Object_IDs: null
03146     Math_Scope: null
03147     Variables_Used: null
03148     Units_Normalization: null
03149     Assumptions_Constraints: null
03150 KA-064:
03151     KA_Name: Failure Pattern Detection
03152     Short_Name: FAIL-PAT
03153     Purpose: Detect recurring failure signatures
03154     Category: Learning
03155     Primary_Layers:
03156     - L10
03157     Allowed_Layers:
03158     - L10
03159     Inputs: Error logs; traces
03160     Outputs: Failure signatures; alerts
03161     Reads_Memory: 'Yes'
03162     Writes_Memory: 'Yes'
03163     Can_Invoke_Chaos: 'No'
03164     Can_Invoke_External_Research: 'No'
03165     Can_Trigger_Recursion: 'No'
03166     Can_Veto: 'No'
03167     Risk_Class: Medium
03168     Confidence_Impact: Neutral
03169     Entropy_Signal: 'No'
03170     Default_Params: null
03171     Dependencies: null
03172     Produces_Artifacts: 'Yes'
```

```

03173     Audit_Events:
03174     - 'Yes'
03175     Version: 1.0.0
03176     Owner: UKG/USKD Core
03177     Status: Active
03178     Notes: null
03179     Implementation_Mode: PYTHON_LIB
03180     Primary_Libraries:
03181     - pydantic
03182     - opentelemetry
03183     - sqlalchemy
03184     - psycpg
03185     - alembic
03186     - redis
03187     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
03188     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
03189     Fallback_Libraries:
03190     - asyncpg
03191     Fallback_Library_Versions: asyncpg>=0.29,<1
03192     Runtime_Env: python
03193     Test_Harness: pytest + compliance fixtures
03194     Has_Math: 'No'
03195     Math_Component_Type: null
03196     Math_Object_IDs: null
03197     Math_Scope: null
03198     Variables_Used: null
03199     Units_Normalization: null
03200     Assumptions_Constraints: null
03201 KA-065:
03202     KA_Name: Knowledge Regression Tester
03203     Short_Name: REGRESS
03204     Purpose: Ensure updates don't break prior knowledge
03205     Category: Safety
03206     Primary_Layers:
03207     - L10
03208     Allowed_Layers:
03209     - L10
03210     Inputs: KB snapshot; tests
03211     Outputs: Regression pass/fail; diffs
03212     Reads_Memory: 'Yes'
03213     Writes_Memory: 'No'
03214     Can_Invoke_Chaos: 'No'
03215     Can_Invoke_External_Research: 'No'
03216     Can_Trigger_Recursion: 'No'
03217     Can_Veto: 'Yes'
03218     Risk_Class: Critical
03219     Confidence_Impact: Direct ↑/↓
03220     Entropy_Signal: 'No'
03221     Default_Params: null
03222     Dependencies:
03223     - KA-99
03224     - KA-50
03225     Produces_Artifacts: 'Yes'
03226     Audit_Events:
03227     - 'Yes'
03228     Version: 1.0.0
03229     Owner: UKG/USKD Core
03230     Status: Active
03231     Notes: null
03232     Implementation_Mode: PYTHON_LIB
03233     Primary_Libraries:
03234     - pydantic
03235     - opentelemetry
03236     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
03237     Fallback_Libraries:

```

```
03238     - asyncpg
03239     Fallback_Library_Versions: asyncpg>=0.29,<1
03240     Runtime_Env: python
03241     Test_Harness: pytest + compliance fixtures
03242     Has_Math: 'No'
03243     Math_Component_Type: null
03244     Math_Object_IDs: null
03245     Math_Scope: null
03246     Variables_Used: null
03247     Units_Normalization: null
03248     Assumptions_Constraints: null
03249     KA-066:
03250     KA_Name: Causal Inference Engine
03251     Short_Name: CAUSAL
03252     Purpose: Infer cause-effect relationships
03253     Category: Reasoning
03254     Primary_Layers:
03255     - L3
03256     - L7
03257     Allowed_Layers:
03258     - L3
03259     - L7
03260     Inputs: Events; dependencies; interventions
03261     Outputs: Causal graph; causal claims
03262     Reads_Memory: 'Yes'
03263     Writes_Memory: 'No'
03264     Can_Invoke_Chaos: 'No'
03265     Can_Invoke_External_Research: 'No'
03266     Can_Trigger_Recursion: 'No'
03267     Can_Veto: 'No'
03268     Risk_Class: High
03269     Confidence_Impact: Indirect ↑
03270     Entropy_Signal: 'No'
03271     Default_Params: null
03272     Dependencies: null
03273     Produces_Artifacts: 'Yes'
03274     Audit_Events:
03275     - 'Yes'
03276     Version: 1.0.0
03277     Owner: UKG/USKD Core
03278     Status: Active
03279     Notes: null
03280     Implementation_Mode: HYBRID
03281     Primary_Libraries:
03282     - numpy
03283     - scipy
03284     - networkx
03285     - pydantic
03286     Primary_Library_Versions: numpy>=1.26,<3; scipy>=1.11,<2; networkx>=3.2,<4; \
pydantic>=2.6,<3
03287     Fallback_Libraries:
03288     - pandas
03289     Fallback_Library_Versions: pandas>=2.2,<3
03290     Runtime_Env: python + in-model
03291     Test_Harness: pytest + reasoning-evals
03292     Has_Math: 'No'
03293     Math_Component_Type: null
03294     Math_Object_IDs: null
03295     Math_Scope: null
03296     Variables_Used: null
03297     Units_Normalization: null
03298     Assumptions_Constraints: null
03299     KA-067:
03300     KA_Name: Analogical Reasoning Engine
03301     Short_Name: ANALOGY
03302     Purpose: Transfer structure/solutions across domains
03303     Category: Reasoning
```

```
03304     Primary_Layers:
03305     - L7
03306     Allowed_Layers:
03307     - L7
03308     Inputs: Source/target structures
03309     Outputs: Analogical mapping; candidate solution
03310     Reads_Memory: 'Yes'
03311     Writes_Memory: 'No'
03312     Can_Invoke_Chaos: 'Yes'
03313     Can_Invoke_External_Research: 'No'
03314     Can_Trigger_Recursion: 'No'
03315     Can_Veto: 'No'
03316     Risk_Class: Medium
03317     Confidence_Impact: Indirect ↑
03318     Entropy_Signal: Produces
03319     Default_Params: null
03320     Dependencies: null
03321     Produces_Artifacts: 'Yes'
03322     Audit_Events:
03323     - 'Yes'
03324     Version: 1.0.0
03325     Owner: UKG/USKD Core
03326     Status: Active
03327     Notes: null
03328     Implementation_Mode: IN_MODEL
03329     Primary_Libraries: null
03330     Primary_Library_Versions: null
03331     Fallback_Libraries: null
03332     Fallback_Library_Versions: null
03333     Runtime_Env: in-model
03334     Test_Harness: pytest + reasoning-evals
03335     Has_Math: 'No'
03336     Math_Component_Type: null
03337     Math_Object_IDs: null
03338     Math_Scope: null
03339     Variables_Used: null
03340     Units_Normalization: null
03341     Assumptions_Constraints: null
03342 KA-068:
03343     KA_Name: Domain Specialization Tuner
03344     Short_Name: DOMAIN-TUNE
03345     Purpose: Reweight domain-specific vs general reasoning
03346     Category: Control
03347     Primary_Layers:
03348     - L2
03349     - L4
03350     Allowed_Layers:
03351     - L2
03352     - L4
03353     Inputs: Domain context; persona set
03354     Outputs: Weights; routing hints
03355     Reads_Memory: 'Yes'
03356     Writes_Memory: 'No'
03357     Can_Invoke_Chaos: 'No'
03358     Can_Invoke_External_Research: 'No'
03359     Can_Trigger_Recursion: 'No'
03360     Can_Veto: 'No'
03361     Risk_Class: Medium
03362     Confidence_Impact: Neutral
03363     Entropy_Signal: 'No'
03364     Default_Params: null
03365     Dependencies: null
03366     Produces_Artifacts: 'Yes'
03367     Audit_Events:
03368     - 'Yes'
03369     Version: 1.0.0
03370     Owner: UKG/USKD Core
```

```
03371      Status: Active
03372      Notes: null
03373      Implementation_Mode: HYBRID
03374      Primary_Libraries:
03375      - pydantic
03376      - pyyaml
03377      - orjson
03378      - opentelemetry
03379      - prometheus-client
03380      - networkx
03381      Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \
    opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
03382      prometheus-client>=0.20,<1; networkx>=3.2,<4
03383      Fallback_Libraries:
03384      - redis
03385      Fallback_Library_Versions: redis>=5.0,<6
03386      Runtime_Env: python + in-model
03387      Test_Harness: pytest + golden-trace tests
03388      Has_Math: 'No'
03389      Math_Component_Type: null
03390      Math_Object_IDs: null
03391      Math_Scope: null
03392      Variables_Used: null
03393      Units_Normalization: null
03394      Assumptions_Constraints: null
03395      KA-069:
03396      KA_Name: Cultural Context Adapter
03397      Short_Name: CULTURE
03398      Purpose: Adapt examples/framing to cultural context
03399      Category: UX
03400      Primary_Layers:
03401      - L9
03402      Allowed_Layers:
03403      - L9
03404      Inputs: Locale; audience
03405      Outputs: Localized response framing
03406      Reads_Memory: 'No'
03407      Writes_Memory: 'No'
03408      Can_Invoke_Chaos: 'No'
03409      Can_Invoke_External_Research: 'No'
03410      Can_Trigger_Recursion: 'No'
03411      Can_Veto: 'No'
03412      Risk_Class: Low
03413      Confidence_Impact: Neutral
03414      Entropy_Signal: 'No'
03415      Default_Params: null
03416      Dependencies: null
03417      Produces_Artifacts: 'Yes'
03418      Audit_Events:
03419      - 'Yes'
03420      Version: 1.0.0
03421      Owner: UKG/USKD Core
03422      Status: Active
03423      Notes: null
03424      Implementation_Mode: IN_MODEL
03425      Primary_Libraries: null
03426      Primary_Library_Versions: null
03427      Fallback_Libraries: null
03428      Fallback_Library_Versions: null
03429      Runtime_Env: in-model
03430      Test_Harness: pytest + reasoning-evals
03431      Has_Math: 'No'
03432      Math_Component_Type: null
03433      Math_Object_IDs: null
03434      Math_Scope: null
03435      Variables_Used: null
03436      Units_Normalization: null
```



```
03437     Assumptions_Constraints: null
03438 KA-070:
03439     KA_Name: Counterfactual Scenario Simulator
03440     Short_Name: CF-SIM
03441     Purpose: Run what-if simulations; alternate premises
03442     Category: Reasoning
03443     Primary_Layers:
03444     - L7
03445     Allowed_Layers:
03446     - L7
03447     Inputs: Base model; interventions
03448     Outputs: Alternative outcomes; sensitivity
03449     Reads_Memory: 'Yes'
03450     Writes_Memory: 'No'
03451     Can_Invoke_Chaos: 'Yes'
03452     Can_Invoke_External_Research: 'No'
03453     Can_Trigger_Recursion: 'Yes'
03454     Can_Veto: 'No'
03455     Risk_Class: High
03456     Confidence_Impact: Neutral
03457     Entropy_Signal: Produces
03458     Default_Params: null
03459     Dependencies:
03460     - KA-66
03461     Produces_Artifacts: 'Yes'
03462     Audit_Events:
03463     - 'Yes'
03464     Version: 1.0.0
03465     Owner: UKG/USKD Core
03466     Status: Active
03467     Notes: null
03468     Implementation_Mode: HYBRID
03469     Primary_Libraries:
03470     - numpy
03471     - scipy
03472     - networkx
03473     - pydantic
03474     Primary_Library_Versions: numpy>=1.26,<3; scipy>=1.11,<2; networkx>=3.2,<4; \
pydantic>=2.6,<3
03475     Fallback_Libraries:
03476     - pandas
03477     Fallback_Library_Versions: pandas>=2.2,<3
03478     Runtime_Env: python + in-model
03479     Test_Harness: pytest + reasoning-evals
03480     Has_Math: 'No'
03481     Math_Component_Type: null
03482     Math_Object_IDs: null
03483     Math_Scope: null
03484     Variables_Used: null
03485     Units_Normalization: null
03486     Assumptions_Constraints: null
03487 KA-071:
03488     KA_Name: Knowledge Provenance Tracker
03489     Short_Name: K-PROV
03490     Purpose: Maintain evidence trees for stored knowledge
03491     Category: Governance
03492     Primary_Layers:
03493     - L6
03494     - L10
03495     Allowed_Layers:
03496     - L6
03497     - L10
03498     Inputs: Source metadata; commit log
03499     Outputs: Provenance tree; audit links
03500     Reads_Memory: 'Yes'
03501     Writes_Memory: 'Yes'
03502     Can_Invoke_Chaos: 'No'
```

```
03503     Can_Invoke_External_Research: 'No'
03504     Can_Trigger_Recursion: 'No'
03505     Can_Veto: 'No'
03506     Risk_Class: High
03507     Confidence_Impact: Neutral
03508     Entropy_Signal: 'No'
03509     Default_Params: null
03510     Dependencies: null
03511     Produces_Artifacts: 'Yes'
03512     Audit_Events:
03513       - 'Yes'
03514     Version: 1.0.0
03515     Owner: UKG/USKD Core
03516     Status: Active
03517     Notes: null
03518     Implementation_Mode: PYTHON_LIB
03519     Primary_Libraries:
03520       - pydantic
03521       - opentelemetry
03522       - sqlalchemy
03523       - psycpg
03524       - alembic
03525       - redis
03526     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
03527       psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
03528     Fallback_Libraries:
03529       - asyncpg
03530     Fallback_Library_Versions: asyncpg>=0.29,<1
03531     Runtime_Env: python
03532     Test_Harness: pytest + compliance fixtures
03533     Has_Math: 'No'
03534     Math_Component_Type: null
03535     Math_Object_IDs: null
03536     Math_Scope: null
03537     Variables_Used: null
03538     Units_Normalization: null
03539     Assumptions_Constraints: null
03540 KA-072:
03541     KA_Name: Context Window Optimizer
03542     Short_Name: CTX-OPT
03543     Purpose: Select optimal context elements under token budget
03544     Category: Control
03545     Primary_Layers:
03546       - L2
03547       - L9
03548     Allowed_Layers:
03549       - L2
03550       - L9
03551     Inputs: Candidate context; budget
03552     Outputs: Optimized context set
03553     Reads_Memory: 'Yes'
03554     Writes_Memory: 'No'
03555     Can_Invoke_Chaos: 'No'
03556     Can_Invoke_External_Research: 'No'
03557     Can_Trigger_Recursion: 'No'
03558     Can_Veto: 'No'
03559     Risk_Class: Medium
03560     Confidence_Impact: Indirect ↑
03561     Entropy_Signal: 'No'
03562     Default_Params: null
03563     Dependencies: null
03564     Produces_Artifacts: 'Yes'
03565     Audit_Events:
03566       - 'Yes'
03567     Version: 1.0.0
03568     Owner: UKG/USKD Core
```

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03569     Status: Active
03570     Notes: null
03571     Implementation_Mode: HYBRID
03572     Primary_Libraries:
03573     - pydantic
03574     - pyyaml
03575     - orjson
03576     - opentelemetry
03577     - prometheus-client
03578     - networkx
03579     Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \
    opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
03580     prometheus-client>=0.20,<1; networkx>=3.2,<4
03581     Fallback_Libraries:
03582     - redis
03583     Fallback_Library_Versions: redis>=5.0,<6
03584     Runtime_Env: python + in-model
03585     Test_Harness: pytest + golden-trace tests
03586     Has_Math: 'No'
03587     Math_Component_Type: null
03588     Math_Object_IDs: null
03589     Math_Scope: null
03590     Variables_Used: null
03591     Units_Normalization: null
03592     Assumptions_Constraints: null
03593     KA-073:
03594     KA_Name: Intent Clarifier
03595     Short_Name: INTENT
03596     Purpose: Resolve linguistic ambiguity; extract intent
03597     Category: Routing
03598     Primary_Layers:
03599     - L1
03600     Allowed_Layers:
03601     - L1
03602     Inputs: Parsed query; entities
03603     Outputs: Disambiguated intent; intent confidence
03604     Reads_Memory: 'No'
03605     Writes_Memory: 'No'
03606     Can_Invoke_Chaos: 'No'
03607     Can_Invoke_External_Research: 'No'
03608     Can_Trigger_Recursion: 'No'
03609     Can_Veto: 'No'
03610     Risk_Class: Medium
03611     Confidence_Impact: Indirect ↑
03612     Entropy_Signal: Produces
03613     Default_Params: null
03614     Dependencies: null
03615     Produces_Artifacts: 'Yes'
03616     Audit_Events:
03617     - 'Yes'
03618     Version: 1.0.0
03619     Owner: UKG/USKD Core
03620     Status: Active
03621     Notes: null
03622     Implementation_Mode: HYBRID
03623     Primary_Libraries:
03624     - pydantic
03625     - pyyaml
03626     - orjson
03627     - opentelemetry
03628     - prometheus-client
03629     - networkx
03630     Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \
    opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
03631     prometheus-client>=0.20,<1; networkx>=3.2,<4
03632     Fallback_Libraries:
03633     - redis

```

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03634     Fallback_Library_Versions: redis>=5.0,<6
03635     Runtime_Env: python + in-model
03636     Test_Harness: pytest + golden-trace tests
03637     Has_Math: 'No'
03638     Math_Component_Type: null
03639     Math_Object_IDs: null
03640     Math_Scope: null
03641     Variables_Used: null
03642     Units_Normalization: null
03643     Assumptions_Constraints: null
03644 KA-074:
03645     KA_Name: Privacy Preserver
03646     Short_Name: PRIVACY
03647     Purpose: Remove/anonymize sensitive data in outputs/logs
03648     Category: Safety
03649     Primary_Layers:
03650     - L8
03651     - L9
03652     Allowed_Layers:
03653     - L8
03654     - L9
03655     Inputs: Draft output; trace
03656     Outputs: Redacted output; privacy log
03657     Reads_Memory: 'No'
03658     Writes_Memory: 'No'
03659     Can_Invoke_Chaos: 'No'
03660     Can_Invoke_External_Research: 'No'
03661     Can_Trigger_Recursion: 'No'
03662     Can_Veto: 'Yes'
03663     Risk_Class: Critical
03664     Confidence_Impact: Neutral
03665     Entropy_Signal: 'No'
03666     Default_Params: null
03667     Dependencies:
03668     - KA-107
03669     Produces_Artifacts: 'Yes'
03670     Audit_Events:
03671     - 'Yes'
03672     Version: 1.0.0
03673     Owner: UKG/USKD Core
03674     Status: Active
03675     Notes: null
03676     Implementation_Mode: PYTHON_LIB
03677     Primary_Libraries:
03678     - pydantic
03679     - opentelemetry
03680     - presidio
03681     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; presidio-analyzer>=2.2,<3;
03682     presidio-anonymizer>=2.2,<3
03683     Fallback_Libraries:
03684     - asyncpg
03685     Fallback_Library_Versions: asyncpg>=0.29,<1
03686     Runtime_Env: python
03687     Test_Harness: pytest + compliance fixtures
03688     Has_Math: 'No'
03689     Math_Component_Type: null
03690     Math_Object_IDs: null
03691     Math_Scope: null
03692     Variables_Used: null
03693     Units_Normalization: null
03694     Assumptions_Constraints: null
03695 KA-075:
03696     KA_Name: Bias Mitigation Engine
03697     Short_Name: BIAS-MIT
03698     Purpose: Correct detected bias while preserving truth
03699     Category: Ethics
```

```
03700     Primary_Layers:
03701     - L9
03702     Allowed_Layers:
03703     - L9
03704     Inputs: Bias vectors; draft output
03705     Outputs: Balanced output; mitigation log
03706     Reads_Memory: 'Yes'
03707     Writes_Memory: 'No'
03708     Can_Invoke_Chaos: 'No'
03709     Can_Invoke_External_Research: 'No'
03710     Can_Trigger_Recursion: 'No'
03711     Can_Veto: 'No'
03712     Risk_Class: High
03713     Confidence_Impact: Neutral
03714     Entropy_Signal: 'No'
03715     Default_Params: null
03716     Dependencies:
03717     - KA-10
03718     - KA-45
03719     Produces_Artifacts: 'Yes'
03720     Audit_Events:
03721     - 'Yes'
03722     Version: 1.0.0
03723     Owner: UKG/USKD Core
03724     Status: Active
03725     Notes: null
03726     Implementation_Mode: PYTHON_LIB
03727     Primary_Libraries:
03728     - pydantic
03729     - opentelemetry
03730     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
03731     Fallback_Libraries:
03732     - asynccpg
03733     Fallback_Library_Versions: asynccpg>=0.29,<1
03734     Runtime_Env: python
03735     Test_Harness: pytest + compliance fixtures
03736     Has_Math: 'No'
03737     Math_Component_Type: null
03738     Math_Object_IDs: null
03739     Math_Scope: null
03740     Variables_Used: null
03741     Units_Normalization: null
03742     Assumptions_Constraints: null
03743 KA-076:
03744     KA_Name: Knowledge Graph Pruner
03745     Short_Name: PRUNE
03746     Purpose: Remove outdated/low-value/misleading nodes
03747     Category: Lifecycle
03748     Primary_Layers:
03749     - L10
03750     Allowed_Layers:
03751     - L10
03752     Inputs: KB snapshot; importance scores
03753     Outputs: Pruned KB; deletion log
03754     Reads_Memory: 'Yes'
03755     Writes_Memory: 'Yes'
03756     Can_Invoke_Chaos: 'No'
03757     Can_Invoke_External_Research: 'No'
03758     Can_Trigger_Recursion: 'No'
03759     Can_Veto: 'No'
03760     Risk_Class: High
03761     Confidence_Impact: Indirect ↑/↓
03762     Entropy_Signal: 'No'
03763     Default_Params: null
03764     Dependencies:
03765     - KA-77
```

```
03766     - KA-94
03767     Produces_Artifacts: 'Yes'
03768     Audit_Events:
03769     - 'Yes'
03770     Version: 1.0.0
03771     Owner: UKG/USKD Core
03772     Status: Active
03773     Notes: null
03774     Implementation_Mode: PYTHON_LIB
03775     Primary_Libraries:
03776     - pydantic
03777     - opentelemetry
03778     - sqlalchemy
03779     - psycpg
03780     - alembic
03781     - redis
03782     - networkx
03783     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
03784     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6; networkx>=3.2,<4
03785     Fallback_Libraries:
03786     - asyncpg
03787     - neo4j
03788     - chromadb
03789     Fallback_Library_Versions: asyncpg>=0.29,<1; neo4j>=5.18,<6; chromadb>=0.5,<1
03790     Runtime_Env: python
03791     Test_Harness: pytest + graph fixtures
03792     Has_Math: 'No'
03793     Math_Component_Type: null
03794     Math_Object_IDs: null
03795     Math_Scope: null
03796     Variables_Used: null
03797     Units_Normalization: null
03798     Assumptions_Constraints: null
03799     KA-077:
03800     KA_Name: Knowledge Importance Scorer
03801     Short_Name: IMPORT
03802     Purpose: Rank knowledge by relevance/reuse
03803     Category: Lifecycle
03804     Primary_Layers:
03805     - L10
03806     Allowed_Layers:
03807     - L10
03808     Inputs: Usage metrics; citations
03809     Outputs: Importance scores
03810     Reads_Memory: 'Yes'
03811     Writes_Memory: 'Yes'
03812     Can_Invoke_Chaos: 'No'
03813     Can_Invoke_External_Research: 'No'
03814     Can_Trigger_Recursion: 'No'
03815     Can_Veto: 'No'
03816     Risk_Class: Medium
03817     Confidence_Impact: Neutral
03818     Entropy_Signal: 'No'
03819     Default_Params: null
03820     Dependencies: null
03821     Produces_Artifacts: 'Yes'
03822     Audit_Events:
03823     - 'Yes'
03824     Version: 1.0.0
03825     Owner: UKG/USKD Core
03826     Status: Active
03827     Notes: null
03828     Implementation_Mode: PYTHON_LIB
03829     Primary_Libraries:
03830     - pydantic
03831     - opentelemetry
```

```
03832     - sqlalchemy
03833     - psycpg
03834     - alembic
03835     - redis
03836     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
03837     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
03838     Fallback_Libraries:
03839     - asyncpg
03840     Fallback_Library_Versions: asyncpg>=0.29,<1
03841     Runtime_Env: python
03842     Test_Harness: pytest + compliance fixtures
03843     Has_Math: 'No'
03844     Math_Component_Type: null
03845     Math_Object_IDs: null
03846     Math_Scope: null
03847     Variables_Used: null
03848     Units_Normalization: null
03849     Assumptions_Constraints: null
03850 KA-078:
03851     KA_Name: Memory Tier Classifier
03852     Short_Name: TIER
03853     Purpose: Assign short-term vs long-term vs archive
03854     Category: Memory
03855     Primary_Layers:
03856     - L10
03857     Allowed_Layers:
03858     - L10
03859     Inputs: Knowledge candidate; risk/containment
03860     Outputs: Tier assignment
03861     Reads_Memory: 'Yes'
03862     Writes_Memory: 'Yes'
03863     Can_Invoke_Chaos: 'No'
03864     Can_Invoke_External_Research: 'No'
03865     Can_Trigger_Recursion: 'No'
03866     Can_Veto: 'No'
03867     Risk_Class: High
03868     Confidence_Impact: Neutral
03869     Entropy_Signal: 'No'
03870     Default_Params: null
03871     Dependencies:
03872     - KA-109
03873     Produces_Artifacts: 'Yes'
03874     Audit_Events:
03875     - 'Yes'
03876     Version: 1.0.0
03877     Owner: UKG/USKD Core
03878     Status: Active
03879     Notes: null
03880     Implementation_Mode: PYTHON_LIB
03881     Primary_Libraries:
03882     - pydantic
03883     - opentelemetry
03884     - sqlalchemy
03885     - psycpg
03886     - alembic
03887     - redis
03888     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
03889     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
03890     Fallback_Libraries:
03891     - asyncpg
03892     Fallback_Library_Versions: asyncpg>=0.29,<1
03893     Runtime_Env: python
03894     Test_Harness: pytest + compliance fixtures
03895     Has_Math: 'No'
03896     Math_Component_Type: null
```

```
03897     Math_Object_IDs: null
03898     Math_Scope: null
03899     Variables_Used: null
03900     Units_Normalization: null
03901     Assumptions_Constraints: null
03902 KA-079:
03903     KA_Name: Knowledge Promotion Gate
03904     Short_Name: PROMOTE
03905     Purpose: Enforce criteria before long-term commit
03906     Category: Memory
03907     Primary_Layers:
03908     - L10
03909     Allowed_Layers:
03910     - L10
03911     Inputs: Confidence; trust; containment
03912     Outputs: Approve/reject; rationale
03913     Reads_Memory: 'Yes'
03914     Writes_Memory: 'No'
03915     Can_Invoke_Chaos: 'No'
03916     Can_Invoke_External_Research: 'No'
03917     Can_Trigger_Recursion: 'No'
03918     Can_Veto: 'Yes'
03919     Risk_Class: Critical
03920     Confidence_Impact: Direct ↑/↓
03921     Entropy_Signal: 'No'
03922     Default_Params: null
03923     Dependencies:
03924     - KA-14
03925     - KA-62
03926     - KA-109
03927     Produces_Artifacts: 'Yes'
03928     Audit_Events:
03929     - 'Yes'
03930     Version: 1.0.0
03931     Owner: UKG/USKD Core
03932     Status: Active
03933     Notes: null
03934     Implementation_Mode: PYTHON_LIB
03935     Primary_Libraries:
03936     - pydantic
03937     - opentelemetry
03938     - sqlalchemy
03939     - psycpg
03940     - alembic
03941     - redis
03942     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
03943     opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
03944     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
03945     Fallback_Libraries:
03946     - asyncpg
03947     Fallback_Library_Versions: asyncpg>=0.29,<1
03948     Runtime_Env: python
03949     Test_Harness: pytest + compliance fixtures
03950     Has_Math: 'No'
03951     Math_Component_Type: null
03952     Math_Object_IDs: null
03953     Math_Scope: null
03954     Variables_Used: null
03955     Units_Normalization: null
03956     Assumptions_Constraints: null
03957 KA-080:
03958     KA_Name: Simulation Cost Estimator
03959     Short_Name: COST
03960     Purpose: Estimate compute/time cost of deeper simulation
03961     Category: Control
03962     Primary_Layers:
03963     - L1
```



```
03963     - L7
03964     Allowed_Layers:
03965     - L1
03966     - L7
03967     Inputs: Complexity tier; branch count
03968     Outputs: Cost estimate; budget recommendation
03969     Reads_Memory: 'No'
03970     Writes_Memory: 'No'
03971     Can_Invoke_Chaos: 'No'
03972     Can_Invoke_External_Research: 'No'
03973     Can_Trigger_Recursion: 'No'
03974     Can_Veto: 'No'
03975     Risk_Class: Medium
03976     Confidence_Impact: Neutral
03977     Entropy_Signal: 'No'
03978     Default_Params: null
03979     Dependencies: null
03980     Produces_Artifacts: 'Yes'
03981     Audit_Events:
03982     - 'Yes'
03983     Version: 1.0.0
03984     Owner: UKG/USKD Core
03985     Status: Active
03986     Notes: null
03987     Implementation_Mode: HYBRID
03988     Primary_Libraries:
03989     - pydantic
03990     - pyyaml
03991     - orjson
03992     - opentelemetry
03993     - prometheus-client
03994     - networkx
03995     Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \
    opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
03996     prometheus-client>=0.20,<1; networkx>=3.2,<4
03997     Fallback_Libraries:
03998     - redis
03999     Fallback_Library_Versions: redis>=5.0,<6
04000     Runtime_Env: python + in-model
04001     Test_Harness: pytest + golden-trace tests
04002     Has_Math: 'Yes'
04003     Math_Component_Type: Equation
04004     Math_Object_IDs:
04005     - F-COST-01
04006     Math_Scope: FROST
04007     Variables_Used:
04008     - n_branches
04009     - d
04010     - n_tokens
04011     Units_Normalization: seconds/tokens
04012     Assumptions_Constraints: Approximate; calibrate
04013     KA-081:
04014     KA_Name: Simulation Budget Enforcer
04015     Short_Name: BUDGET
04016     Purpose: Enforce compute/recursion limits
04017     Category: Safety
04018     Primary_Layers:
04019     - L7
04020     Allowed_Layers:
04021     - L7
04022     Inputs: Cost estimate; policy caps
04023     Outputs: Allow/restrict; new caps
04024     Reads_Memory: 'No'
04025     Writes_Memory: 'No'
04026     Can_Invoke_Chaos: 'No'
04027     Can_Invoke_External_Research: 'No'
04028     Can_Trigger_Recursion: 'No'
```

```
04029     Can_Veto: 'Yes'
04030     Risk_Class: Critical
04031     Confidence_Impact: Neutral
04032     Entropy_Signal: 'No'
04033     Default_Params: null
04034     Dependencies: null
04035     Produces_Artifacts: 'Yes'
04036     Audit_Events:
04037     - 'Yes'
04038     Version: 1.0.0
04039     Owner: UKG/USKD Core
04040     Status: Active
04041     Notes: null
04042     Implementation_Mode: PYTHON_LIB
04043     Primary_Libraries:
04044     - pydantic
04045     - opentelemetry
04046     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
04047     Fallback_Libraries:
04048     - asyncpg
04049     Fallback_Library_Versions: asyncpg>=0.29,<1
04050     Runtime_Env: python
04051     Test_Harness: pytest + compliance fixtures
04052     Has_Math: 'Yes'
04053     Math_Component_Type: Algorithm
04054     Math_Object_IDs:
04055     - A-BUDGET-01
04056     Math_Scope: FROST
04057     Variables_Used:
04058     - B
04059     - K■
04060     Units_Normalization: budget units
04061     Assumptions_Constraints: Hard caps enforced
04062 KA-082:
04063     KA_Name: Confidence Drift Monitor
04064     Short_Name: DRIFT
04065     Purpose: Detect gradual confidence degradation
04066     Category: Lifecycle
04067     Primary_Layers:
04068     - L10
04069     Allowed_Layers:
04070     - L10
04071     Inputs: Confidence history
04072     Outputs: Drift alerts
04073     Reads_Memory: 'Yes'
04074     Writes_Memory: 'Yes'
04075     Can_Invoke_Chaos: 'No'
04076     Can_Invoke_External_Research: 'No'
04077     Can_Trigger_Recursion: 'No'
04078     Can_Veto: 'No'
04079     Risk_Class: Medium
04080     Confidence_Impact: Direct ↓
04081     Entropy_Signal: Produces
04082     Default_Params: null
04083     Dependencies: null
04084     Produces_Artifacts: 'Yes'
04085     Audit_Events:
04086     - 'Yes'
04087     Version: 1.0.0
04088     Owner: UKG/USKD Core
04089     Status: Active
04090     Notes: null
04091     Implementation_Mode: PYTHON_LIB
04092     Primary_Libraries:
04093     - pydantic
04094     - opentelemetry
```

ka_registry.yaml

```
04095     - sqlalchemy
04096     - psycpg
04097     - alembic
04098     - redis
04099     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
04100         psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
04101     Fallback_Libraries:
04102     - asyncpg
04103     Fallback_Library_Versions: asyncpg>=0.29,<1
04104     Runtime_Env: python
04105     Test_Harness: pytest + compliance fixtures
04106     Has_Math: 'Yes'
04107     Math_Component_Type: Equation
04108     Math_Object_IDs:
04109     - F-DRIFT-01
04110     Math_Scope: System
04111     Variables_Used:
04112     - C_t
04113     - Δt
04114     Units_Normalization: C∈[0,1]
04115     Assumptions_Constraints: Requires history
04116     KA-083:
04117     KA_Name: Knowledge Revalidation Scheduler
04118     Short_Name: REVALIDATE
04119     Purpose: Schedule periodic re-checks of stored knowledge
04120     Category: Lifecycle
04121     Primary_Layers:
04122     - L10
04123     Allowed_Layers:
04124     - L10
04125     Inputs: Drift alerts; time
04126     Outputs: Revalidation tasks
04127     Reads_Memory: 'Yes'
04128     Writes_Memory: 'Yes'
04129     Can_Invoke_Chaos: 'No'
04130     Can_Invoke_External_Research: 'No'
04131     Can_Trigger_Recursion: 'No'
04132     Can_Veto: 'No'
04133     Risk_Class: Medium
04134     Confidence_Impact: Neutral
04135     Entropy_Signal: 'No'
04136     Default_Params: null
04137     Dependencies: null
04138     Produces_Artifacts: 'Yes'
04139     Audit_Events:
04140     - 'Yes'
04141     Version: 1.0.0
04142     Owner: UKG/USKD Core
04143     Status: Active
04144     Notes: null
04145     Implementation_Mode: PYTHON_LIB
04146     Primary_Libraries:
04147     - pydantic
04148     - opentelemetry
04149     - sqlalchemy
04150     - psycpg
04151     - alembic
04152     - redis
04153     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
04154         psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
04155     Fallback_Libraries:
04156     - asyncpg
04157     Fallback_Library_Versions: asyncpg>=0.29,<1
04158     Runtime_Env: python
04159     Test_Harness: pytest + compliance fixtures
```

```
04160     Has_Math: 'Yes'
04161     Math_Component_Type: Algorithm
04162     Math_Object_IDs:
04163     - A-REVAL-01
04164     Math_Scope: System
04165     Variables_Used:
04166     - age
04167     - drift
04168     - crit
04169     Units_Normalization: days
04170     Assumptions_Constraints: Policy-driven
04171 KA-084:
04172   KA_Name: Cross-Instance Consensus Engine
04173   Short_Name: X-CONS
04174   Purpose: Compare answers across UKG instances for consensus
04175   Category: Truth
04176   Primary_Layers:
04177   - L6
04178   - L10
04179   Allowed_Layers:
04180   - L6
04181   - L10
04182   Inputs: Parallel outputs; trust scores
04183   Outputs: Consensus score; discrepancies
04184   Reads_Memory: 'Yes'
04185   Writes_Memory: 'No'
04186   Can_Invoke_Chaos: 'No'
04187   Can_Invoke_External_Research: 'No'
04188   Can_Trigger_Recursion: 'No'
04189   Can_Veto: 'No'
04190   Risk_Class: High
04191   Confidence_Impact: Indirect ↑/↓
04192   Entropy_Signal: Produces
04193   Default_Params: null
04194   Dependencies: null
04195   Produces_Artifacts: 'Yes'
04196   Audit_Events:
04197   - 'Yes'
04198   Version: 1.0.0
04199   Owner: UKG/USKD Core
04200   Status: Active
04201   Notes: null
04202   Implementation_Mode: PYTHON_LIB
04203   Primary_Libraries:
04204   - pydantic
04205   - numpy
04206   - pandas
04207   - rapidfuzz
04208   - opentelemetry
04209   Primary_Library_Versions: pydantic>=2.6,<3; numpy>=1.26,<3; pandas>=2.2,<3; \
    rapidfuzz>=3.6,<4; opentelemetry-api>=1.26,<2;
04210     opentelemetry-sdk>=1.26,<2
04211   Fallback_Libraries:
04212   - scikit-learn
04213   Fallback_Library_Versions: scikit-learn>=1.4,<2
04214   Runtime_Env: python
04215   Test_Harness: pytest + evidence fixtures
04216   Has_Math: 'No'
04217   Math_Component_Type: null
04218   Math_Object_IDs: null
04219   Math_Scope: null
04220   Variables_Used: null
04221   Units_Normalization: null
04222   Assumptions_Constraints: null
04223 KA-085:
04224   KA_Name: Anomaly Detection Engine
04225   Short_Name: ANOM
```

```
04226     Purpose: Detect unusual reasoning/output patterns
04227     Category: Safety
04228     Primary_Layers:
04229     - L6
04230     Allowed_Layers:
04231     - L6
04232     Inputs: Trace stats; outputs
04233     Outputs: Anomaly flags
04234     Reads_Memory: 'Yes'
04235     Writes_Memory: 'No'
04236     Can_Invoke_Chaos: 'No'
04237     Can_Invoke_External_Research: 'No'
04238     Can_Trigger_Recursion: 'No'
04239     Can_Veto: 'No'
04240     Risk_Class: High
04241     Confidence_Impact: Neutral
04242     Entropy_Signal: Produces
04243     Default_Params: null
04244     Dependencies: null
04245     Produces_Artifacts: 'Yes'
04246     Audit_Events:
04247     - 'Yes'
04248     Version: 1.0.0
04249     Owner: UKG/USKD Core
04250     Status: Active
04251     Notes: null
04252     Implementation_Mode: PYTHON_LIB
04253     Primary_Libraries:
04254     - pydantic
04255     - opentelemetry
04256     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
04257     Fallback_Libraries:
04258     - asynccpg
04259     Fallback_Library_Versions: asynccpg>=0.29,<1
04260     Runtime_Env: python
04261     Test_Harness: pytest + compliance fixtures
04262     Has_Math: 'No'
04263     Math_Component_Type: null
04264     Math_Object_IDs: null
04265     Math_Scope: null
04266     Variables_Used: null
04267     Units_Normalization: null
04268     Assumptions_Constraints: null
04269 KA-086:
04270     KA_Name: Knowledge Usage Analytics
04271     Short_Name: ANALYTICS
04272     Purpose: Track knowledge usage across queries
04273     Category: Governance
04274     Primary_Layers:
04275     - L10
04276     Allowed_Layers:
04277     - L10
04278     Inputs: Execution logs
04279     Outputs: Analytics dashboards data
04280     Reads_Memory: 'Yes'
04281     Writes_Memory: 'Yes'
04282     Can_Invoke_Chaos: 'No'
04283     Can_Invoke_External_Research: 'No'
04284     Can_Trigger_Recursion: 'No'
04285     Can_Veto: 'No'
04286     Risk_Class: Low
04287     Confidence_Impact: Neutral
04288     Entropy_Signal: 'No'
04289     Default_Params: null
04290     Dependencies: null
04291     Produces_Artifacts: 'Yes'
```

```

04292     Audit_Events:
04293     - 'Yes'
04294     Version: 1.0.0
04295     Owner: UKG/USKD Core
04296     Status: Active
04297     Notes: null
04298     Implementation_Mode: PYTHON_LIB
04299     Primary_Libraries:
04300     - pydantic
04301     - opentelemetry
04302     - sqlalchemy
04303     - psycpg
04304     - alembic
04305     - redis
04306     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
04307     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
04308     Fallback_Libraries:
04309     - asynpg
04310     Fallback_Library_Versions: asynpg>=0.29,<1
04311     Runtime_Env: python
04312     Test_Harness: pytest + compliance fixtures
04313     Has_Math: 'No'
04314     Math_Component_Type: null
04315     Math_Object_IDs: null
04316     Math_Scope: null
04317     Variables_Used: null
04318     Units_Normalization: null
04319     Assumptions_Constraints: null
04320 KA-087:
04321     KA_Name: Explainability Coverage Checker
04322     Short_Name: EXPLAIN-COV
04323     Purpose: Ensure explanations cover critical reasoning steps
04324     Category: QA
04325     Primary_Layers:
04326     - L9
04327     Allowed_Layers:
04328     - L9
04329     Inputs: Trace DAG; explanation
04330     Outputs: Coverage score; missing points
04331     Reads_Memory: 'Yes'
04332     Writes_Memory: 'No'
04333     Can_Invoke_Chaos: 'No'
04334     Can_Invoke_External_Research: 'No'
04335     Can_Trigger_Recursion: 'No'
04336     Can_Veto: 'No'
04337     Risk_Class: Medium
04338     Confidence_Impact: Indirect ↑
04339     Entropy_Signal: 'No'
04340     Default_Params: null
04341     Dependencies:
04342     - KA-56
04343     Produces_Artifacts: 'Yes'
04344     Audit_Events:
04345     - 'Yes'
04346     Version: 1.0.0
04347     Owner: UKG/USKD Core
04348     Status: Active
04349     Notes: null
04350     Implementation_Mode: IN_MODEL
04351     Primary_Libraries:
04352     - pydantic
04353     Primary_Library_Versions: pydantic>=2.6,<3
04354     Fallback_Libraries: null
04355     Fallback_Library_Versions: null
04356     Runtime_Env: in-model
04357     Test_Harness: pytest + reasoning-evals

```

```
04358     Has_Math: 'No'
04359     Math_Component_Type: null
04360     Math_Object_IDs: null
04361     Math_Scope: null
04362     Variables_Used: null
04363     Units_Normalization: null
04364     Assumptions_Constraints: null
04365 KA-088:
04366     KA_Name: Knowledge Lifecycle Manager
04367     Short_Name: LIFECYCLE
04368     Purpose: Coordinate create→validate→decay→archive
04369     Category: Lifecycle
04370     Primary_Layers:
04371     - L10
04372     Allowed_Layers:
04373     - L10
04374     Inputs: KB state; policies
04375     Outputs: Lifecycle actions
04376     Reads_Memory: 'Yes'
04377     Writes_Memory: 'Yes'
04378     Can_Invoke_Chaos: 'No'
04379     Can_Invoke_External_Research: 'No'
04380     Can_Trigger_Recursion: 'No'
04381     Can_Veto: 'No'
04382     Risk_Class: High
04383     Confidence_Impact: Neutral
04384     Entropy_Signal: 'No'
04385     Default_Params: null
04386     Dependencies:
04387     - KA-52
04388     - KA-53
04389     - KA-76
04390     - KA-83
04391     Produces_Artifacts: 'Yes'
04392     Audit_Events:
04393     - 'Yes'
04394     Version: 1.0.0
04395     Owner: UKG/USKD Core
04396     Status: Active
04397     Notes: null
04398     Implementation_Mode: PYTHON_LIB
04399     Primary_Libraries:
04400     - pydantic
04401     - opentelemetry
04402     - sqlalchemy
04403     - psycpg
04404     - alembic
04405     - redis
04406     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
04407     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
04408     Fallback_Libraries:
04409     - asyncpg
04410     Fallback_Library_Versions: asyncpg>=0.29,<1
04411     Runtime_Env: python
04412     Test_Harness: pytest + compliance fixtures
04413     Has_Math: 'No'
04414     Math_Component_Type: null
04415     Math_Object_IDs: null
04416     Math_Scope: null
04417     Variables_Used: null
04418     Units_Normalization: null
04419     Assumptions_Constraints: null
04420 KA-089:
04421     KA_Name: Policy Evolution Tracker
04422     Short_Name: POLICY-EVO
04423     Purpose: Track changes in laws/policies/standards
```

```

04424     Category: Compliance
04425     Primary_Layers:
04426     - L8
04427     - L10
04428     Allowed_Layers:
04429     - L8
04430     - L10
04431     Inputs: Policy updates; feeds
04432     Outputs: Updated mappings; alerts
04433     Reads_Memory: 'Yes'
04434     Writes_Memory: 'Yes'
04435     Can_Invoke_Chaos: 'No'
04436     Can_Invoke_External_Research: 'No'
04437     Can_Trigger_Recursion: 'No'
04438     Can_Veto: 'No'
04439     Risk_Class: Critical
04440     Confidence_Impact: Indirect ↑/↓
04441     Entropy_Signal: 'No'
04442     Default_Params: null
04443     Dependencies: null
04444     Produces_Artifacts: 'Yes'
04445     Audit_Events:
04446     - 'Yes'
04447     Version: 1.0.0
04448     Owner: UKG/USKD Core
04449     Status: Active
04450     Notes: null
04451     Implementation_Mode: PYTHON_LIB
04452     Primary_Libraries:
04453     - pydantic
04454     - opentelemetry
04455     - pyyaml
04456     - httpx
04457     - tenacity
04458     - sqlalchemy
04459     - psycpg
04460     - alembic
04461     - redis
04462     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
04463     opentelemetry-sdk>=1.26,<2; PyYAML>=6.0,<7;
04463     httpx>=0.27,<1; tenacity>=8.2,<9; sqlalchemy>=2.0,<3; psycpg[binary]>=3.1,<4; \
04463     alembic>=1.13,<2; redis>=5.0,<6
04464     Fallback_Libraries:
04465     - asyncpg
04466     Fallback_Library_Versions: asyncpg>=0.29,<1
04467     Runtime_Env: python
04468     Test_Harness: pytest + compliance fixtures
04469     Has_Math: 'No'
04470     Math_Component_Type: null
04471     Math_Object_IDs: null
04472     Math_Scope: null
04473     Variables_Used: null
04474     Units_Normalization: null
04475     Assumptions_Constraints: null
04476     KA-090:
04477     KA_Name: Compliance Regression Validator
04478     Short_Name: COMP-REG
04479     Purpose: Ensure updates don't violate compliance baselines
04480     Category: Compliance
04481     Primary_Layers:
04482     - L8
04483     - L10
04484     Allowed_Layers:
04485     - L8
04486     - L10
04487     Inputs: Updated KB; controls
04488     Outputs: Pass/fail; diffs

```



```
04489     Reads_Memory: 'Yes'
04490     Writes_Memory: 'No'
04491     Can_Invoke_Chaos: 'No'
04492     Can_Invoke_External_Research: 'No'
04493     Can_Trigger_Recursion: 'No'
04494     Can_Veto: 'Yes'
04495     Risk_Class: Critical
04496     Confidence_Impact: Direct ↑/↓
04497     Entropy_Signal: 'No'
04498     Default_Params: null
04499     Dependencies:
04500     - KA-16
04501     - KA-89
04502     Produces_Artifacts: 'Yes'
04503     Audit_Events:
04504     - 'Yes'
04505     Version: 1.0.0
04506     Owner: UKG/USKD Core
04507     Status: Active
04508     Notes: null
04509     Implementation_Mode: PYTHON_LIB
04510     Primary_Libraries:
04511     - pydantic
04512     - opentelemetry
04513     - pyyaml
04514     - httpx
04515     - tenacity
04516     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
04517     opentelemetry-sdk>=1.26,<2; PyYAML>=6.0,<7;
04518     httpx>=0.27,<1; tenacity>=8.2,<9
04518     Fallback_Libraries:
04519     - asyncpg
04520     Fallback_Library_Versions: asyncpg>=0.29,<1
04521     Runtime_Env: python
04522     Test_Harness: pytest + compliance fixtures
04523     Has_Math: 'No'
04524     Math_Component_Type: null
04525     Math_Object_IDs: null
04526     Math_Scope: null
04527     Variables_Used: null
04528     Units_Normalization: null
04529     Assumptions_Constraints: null
04530 KA-091:
04531     KA_Name: Scenario Outcome Archivist
04532     Short_Name: ARCHIVE
04533     Purpose: Store major simulation outcomes for reuse
04534     Category: Memory
04535     Primary_Layers:
04536     - L10
04537     Allowed_Layers:
04538     - L10
04539     Inputs: Final outputs; traces
04540     Outputs: Scenario archive entries
04541     Reads_Memory: 'Yes'
04542     Writes_Memory: 'Yes'
04543     Can_Invoke_Chaos: 'No'
04544     Can_Invoke_External_Research: 'No'
04545     Can_Trigger_Recursion: 'No'
04546     Can_Veto: 'No'
04547     Risk_Class: Medium
04548     Confidence_Impact: Neutral
04549     Entropy_Signal: 'No'
04550     Default_Params: null
04551     Dependencies: null
04552     Produces_Artifacts: 'Yes'
04553     Audit_Events:
04554     - 'Yes'
```

```
04555     Version: 1.0.0
04556     Owner: UKG/USKD Core
04557     Status: Active
04558     Notes: null
04559     Implementation_Mode: PYTHON_LIB
04560     Primary_Libraries:
04561     - pydantic
04562     - opentelemetry
04563     - sqlalchemy
04564     - psycpg
04565     - alembic
04566     - redis
04567     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
04568     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
04569     Fallback_Libraries:
04570     - asyncpg
04571     Fallback_Library_Versions: asyncpg>=0.29,<1
04572     Runtime_Env: python
04573     Test_Harness: pytest + compliance fixtures
04574     Has_Math: 'No'
04575     Math_Component_Type: null
04576     Math_Object_IDs: null
04577     Math_Scope: null
04578     Variables_Used: null
04579     Units_Normalization: null
04580     Assumptions_Constraints: null
04581 KA-092:
04582     KA_Name: Knowledge Dependency Auditor
04583     Short_Name: IMPACT
04584     Purpose: Audit downstream dependencies of changes
04585     Category: Governance
04586     Primary_Layers:
04587     - L10
04588     Allowed_Layers:
04589     - L10
04590     Inputs: Changed nodes; dependency graph
04591     Outputs: Impact report
04592     Reads_Memory: 'Yes'
04593     Writes_Memory: 'No'
04594     Can_Invoke_Chaos: 'No'
04595     Can_Invoke_External_Research: 'No'
04596     Can_Trigger_Recursion: 'No'
04597     Can_Veto: 'No'
04598     Risk_Class: High
04599     Confidence_Impact: Neutral
04600     Entropy_Signal: 'No'
04601     Default_Params: null
04602     Dependencies:
04603     - KA-25
04604     Produces_Artifacts: 'Yes'
04605     Audit_Events:
04606     - 'Yes'
04607     Version: 1.0.0
04608     Owner: UKG/USKD Core
04609     Status: Active
04610     Notes: null
04611     Implementation_Mode: PYTHON_LIB
04612     Primary_Libraries:
04613     - pydantic
04614     - opentelemetry
04615     - sqlalchemy
04616     - psycpg
04617     - alembic
04618     - redis
04619     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
```

```

04620     psychpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
04621     Fallback_Libraries:
04622     - asyncpg
04623     Fallback_Library_Versions: asyncpg>=0.29,<1
04624     Runtime_Env: python
04625     Test_Harness: pytest + compliance fixtures
04626     Has_Math: 'No'
04627     Math_Component_Type: null
04628     Math_Object_IDs: null
04629     Math_Scope: null
04630     Variables_Used: null
04631     Units_Normalization: null
04632     Assumptions_Constraints: null
04633     KA-093:
04634     KA_Name: Trust Decay Engine
04635     Short_Name: TRUST-DECAY
04636     Purpose: Reduce trust for unused/stale knowledge
04637     Category: Lifecycle
04638     Primary_Layers:
04639     - L10
04640     Allowed_Layers:
04641     - L10
04642     Inputs: Usage+time
04643     Outputs: Updated trust scores
04644     Reads_Memory: 'Yes'
04645     Writes_Memory: 'Yes'
04646     Can_Invoke_Chaos: 'No'
04647     Can_Invoke_External_Research: 'No'
04648     Can_Trigger_Recursion: 'No'
04649     Can_Veto: 'No'
04650     Risk_Class: Medium
04651     Confidence_Impact: Indirect ↓
04652     Entropy_Signal: 'No'
04653     Default_Params: null
04654     Dependencies: null
04655     Produces_Artifacts: 'Yes'
04656     Audit_Events:
04657     - 'Yes'
04658     Version: 1.0.0
04659     Owner: UKG/USKD Core
04660     Status: Active
04661     Notes: null
04662     Implementation_Mode: PYTHON_LIB
04663     Primary_Libraries:
04664     - pydantic
04665     - opentelemetry
04666     - sqlalchemy
04667     - psychpg
04668     - alembic
04669     - redis
04670     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
04671     psychpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
04672     Fallback_Libraries:
04673     - asyncpg
04674     Fallback_Library_Versions: asyncpg>=0.29,<1
04675     Runtime_Env: python
04676     Test_Harness: pytest + compliance fixtures
04677     Has_Math: 'Yes'
04678     Math_Component_Type: Equation
04679     Math_Object_IDs:
04680     - F-DECAY-01
04681     Math_Scope: System
04682     Variables_Used:
04683     - T0
04684     - λ
04685     - t

```

```
04686     - u
04687     Units_Normalization: T∈[0,1]
04688     Assumptions_Constraints: Exponential decay default
04689 KA-094:
04690     KA_Name: Knowledge Quarantine Engine
04691     Short_Name: QUAR
04692     Purpose: Isolate suspicious/disputed knowledge
04693     Category: Safety
04694     Primary_Layers:
04695     - L10
04696     Allowed_Layers:
04697     - L10
04698     Inputs: Anomalies; conflicts
04699     Outputs: Quarantined nodes; quarantine log
04700     Reads_Memory: 'Yes'
04701     Writes_Memory: 'Yes'
04702     Can_Invoke_Chaos: 'No'
04703     Can_Invoke_External_Research: 'No'
04704     Can_Trigger_Recursion: 'No'
04705     Can_Veto: 'Yes'
04706     Risk_Class: Critical
04707     Confidence_Impact: Neutral
04708     Entropy_Signal: 'No'
04709     Default_Params: null
04710     Dependencies:
04711     - KA-85
04712     - KA-26
04713     Produces_Artifacts: 'Yes'
04714     Audit_Events:
04715     - 'Yes'
04716     Version: 1.0.0
04717     Owner: UKG/USKD Core
04718     Status: Active
04719     Notes: null
04720     Implementation_Mode: PYTHON_LIB
04721     Primary_Libraries:
04722     - pydantic
04723     - opentelemetry
04724     - sqlalchemy
04725     - psycpg
04726     - alembic
04727     - redis
04728     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
04729     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
04730     Fallback_Libraries:
04731     - asyncpg
04732     Fallback_Library_Versions: asyncpg>=0.29,<1
04733     Runtime_Env: python
04734     Test_Harness: pytest + compliance fixtures
04735     Has_Math: 'No'
04736     Math_Component_Type: null
04737     Math_Object_IDs: null
04738     Math_Scope: null
04739     Variables_Used: null
04740     Units_Normalization: null
04741     Assumptions_Constraints: null
04742 KA-095:
04743     KA_Name: Human-in-the-Loop Escalation
04744     Short_Name: HITL
04745     Purpose: Escalate uncertain/high-risk cases to humans
04746     Category: Governance
04747     Primary_Layers:
04748     - L9
04749     - L10
04750     Allowed_Layers:
04751     - L9
```

```
04752     - L10
04753     Inputs: Low confidence; high risk
04754     Outputs: Review packet; questions
04755     Reads_Memory: 'Yes'
04756     Writes_Memory: 'No'
04757     Can_Invoke_Chaos: 'No'
04758     Can_Invoke_External_Research: 'No'
04759     Can_Trigger_Recursion: 'No'
04760     Can_Veto: 'No'
04761     Risk_Class: Critical
04762     Confidence_Impact: Neutral
04763     Entropy_Signal: Consumes
04764     Default_Params: null
04765     Dependencies: null
04766     Produces_Artifacts: 'Yes'
04767     Audit_Events:
04768     - 'Yes'
04769     Version: 1.0.0
04770     Owner: UKG/USKD Core
04771     Status: Active
04772     Notes: null
04773     Implementation_Mode: PYTHON_LIB
04774     Primary_Libraries:
04775     - pydantic
04776     - opentelemetry
04777     - sqlalchemy
04778     - psycpg
04779     - alembic
04780     - redis
04781     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
04782     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
04783     Fallback_Libraries:
04784     - asyncpg
04785     Fallback_Library_Versions: asyncpg>=0.29,<1
04786     Runtime_Env: python
04787     Test_Harness: pytest + compliance fixtures
04788     Has_Math: 'No'
04789     Math_Component_Type: null
04790     Math_Object_IDs: null
04791     Math_Scope: null
04792     Variables_Used: null
04793     Units_Normalization: null
04794     Assumptions_Constraints: null
04795 KA-096:
04796     KA_Name: Knowledge Release Manager
04797     Short_Name: RELEASE
04798     Purpose: Stage and control when new knowledge becomes active
04799     Category: Governance
04800     Primary_Layers:
04801     - L10
04802     Allowed_Layers:
04803     - L10
04804     Inputs: Approved updates
04805     Outputs: Release state; rollout plan
04806     Reads_Memory: 'Yes'
04807     Writes_Memory: 'Yes'
04808     Can_Invoke_Chaos: 'No'
04809     Can_Invoke_External_Research: 'No'
04810     Can_Trigger_Recursion: 'No'
04811     Can_Veto: 'No'
04812     Risk_Class: High
04813     Confidence_Impact: Neutral
04814     Entropy_Signal: 'No'
04815     Default_Params: null
04816     Dependencies: null
04817     Produces_Artifacts: 'Yes'
```

```
04818     Audit_Events:
04819     - 'Yes'
04820     Version: 1.0.0
04821     Owner: UKG/USKD Core
04822     Status: Active
04823     Notes: null
04824     Implementation_Mode: PYTHON_LIB
04825     Primary_Libraries:
04826     - pydantic
04827     - opentelemetry
04828     - sqlalchemy
04829     - psycpg
04830     - alembic
04831     - redis
04832     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
04833     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
04834     Fallback_Libraries:
04835     - asynpg
04836     Fallback_Library_Versions: asynpg>=0.29,<1
04837     Runtime_Env: python
04838     Test_Harness: pytest + compliance fixtures
04839     Has_Math: 'No'
04840     Math_Component_Type: null
04841     Math_Object_IDs: null
04842     Math_Scope: null
04843     Variables_Used: null
04844     Units_Normalization: null
04845     Assumptions_Constraints: null
04846 KA-097:
04847     KA_Name: System Performance Optimizer
04848     Short_Name: OPTIMIZE
04849     Purpose: Tune overall KA/layer performance and defaults
04850     Category: Learning
04851     Primary_Layers:
04852     - L10
04853     Allowed_Layers:
04854     - L10
04855     Inputs: Metrics; perf profiles
04856     Outputs: Optimization actions
04857     Reads_Memory: 'Yes'
04858     Writes_Memory: 'Yes'
04859     Can_Invoke_Chaos: 'No'
04860     Can_Invoke_External_Research: 'No'
04861     Can_Trigger_Recursion: 'No'
04862     Can_Veto: 'No'
04863     Risk_Class: Medium
04864     Confidence_Impact: Neutral
04865     Entropy_Signal: 'No'
04866     Default_Params: null
04867     Dependencies: null
04868     Produces_Artifacts: 'Yes'
04869     Audit_Events:
04870     - 'Yes'
04871     Version: 1.0.0
04872     Owner: UKG/USKD Core
04873     Status: Active
04874     Notes: null
04875     Implementation_Mode: PYTHON_LIB
04876     Primary_Libraries:
04877     - pydantic
04878     - opentelemetry
04879     - sqlalchemy
04880     - psycpg
04881     - alembic
04882     - redis
04883     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
```

```
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
04884      psycopg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
04885      Fallback_Libraries:
04886      - asyncpg
04887      Fallback_Library_Versions: asyncpg>=0.29,<1
04888      Runtime_Env: python
04889      Test_Harness: pytest + compliance fixtures
04890      Has_Math: 'No'
04891      Math_Component_Type: null
04892      Math_Object_IDs: null
04893      Math_Scope: null
04894      Variables_Used: null
04895      Units_Normalization: null
04896      Assumptions_Constraints: null
04897  KA-098:
04898    KA_Name: Self-Evaluation & Benchmarking
04899    Short_Name: BENCH
04900    Purpose: Measure accuracy vs benchmarks/test suites
04901    Category: QA
04902    Primary_Layers:
04903    - L10
04904    Allowed_Layers:
04905    - L10
04906    Inputs: Test suites; outputs
04907    Outputs: Benchmark scores; gaps
04908    Reads_Memory: 'Yes'
04909    Writes_Memory: 'No'
04910    Can_Invoke_Chaos: 'No'
04911    Can_Invoke_External_Research: 'No'
04912    Can_Trigger_Recursion: 'No'
04913    Can_Veto: 'No'
04914    Risk_Class: High
04915    Confidence_Impact: Direct ↑/↓
04916    Entropy_Signal: 'No'
04917    Default_Params: null
04918    Dependencies: null
04919    Produces_Artifacts: 'Yes'
04920    Audit_Events:
04921    - 'Yes'
04922    Version: 1.0.0
04923    Owner: UKG/USKD Core
04924    Status: Active
04925    Notes: null
04926    Implementation_Mode: IN_MODEL
04927    Primary_Libraries: null
04928    Primary_Library_Versions: null
04929    Fallback_Libraries: null
04930    Fallback_Library_Versions: null
04931    Runtime_Env: in-model
04932    Test_Harness: pytest + reasoning-evals
04933    Has_Math: 'No'
04934    Math_Component_Type: null
04935    Math_Object_IDs: null
04936    Math_Scope: null
04937    Variables_Used: null
04938    Units_Normalization: null
04939    Assumptions_Constraints: null
04940  KA-099:
04941    KA_Name: System Integrity Auditor
04942    Short_Name: SYS-INTEGRITY
04943    Purpose: Audit full system consistency and health
04944    Category: Safety
04945    Primary_Layers:
04946    - L10
04947    Allowed_Layers:
04948    - L10
04949    Inputs: Full snapshot
```

```
04950      Outputs: Integrity report; actions
04951      Reads_Memory: 'Yes'
04952      Writes_Memory: 'No'
04953      Can_Invoke_Chaos: 'No'
04954      Can_Invoke_External_Research: 'No'
04955      Can_Trigger_Recursion: 'No'
04956      Can_Veto: 'Yes'
04957      Risk_Class: Critical
04958      Confidence_Impact: Direct ↑/↓
04959      Entropy_Signal: Produces
04960      Default_Params: null
04961      Dependencies:
04962      - KA-50
04963      - KA-65
04964      Produces_Artifacts: 'Yes'
04965      Audit_Events:
04966      - 'Yes'
04967      Version: 1.0.0
04968      Owner: UKG/USKD Core
04969      Status: Active
04970      Notes: null
04971      Implementation_Mode: PYTHON_LIB
04972      Primary_Libraries:
04973      - pydantic
04974      - opentelemetry
04975      Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
04976      Fallback_Libraries:
04977      - asynpg
04978      Fallback_Library_Versions: asynpg>=0.29,<1
04979      Runtime_Env: python
04980      Test_Harness: pytest + compliance fixtures
04981      Has_Math: 'No'
04982      Math_Component_Type: null
04983      Math_Object_IDs: null
04984      Math_Scope: null
04985      Variables_Used: null
04986      Units_Normalization: null
04987      Assumptions_Constraints: null
04988  KA-100:
04989      KA_Name: Autonomous System Evolution Controller
04990      Short_Name: EVOLVE
04991      Purpose: Govern safe self-improvement with bounded rules
04992      Category: Governance
04993      Primary_Layers:
04994      - L10
04995      Allowed_Layers:
04996      - L10
04997      Inputs: Learning signals; approvals
04998      Outputs: Approved evolution actions
04999      Reads_Memory: 'Yes'
05000      Writes_Memory: 'Yes'
05001      Can_Invoke_Chaos: 'No'
05002      Can_Invoke_External_Research: 'No'
05003      Can_Trigger_Recursion: 'No'
05004      Can_Veto: 'No'
05005      Risk_Class: Critical
05006      Confidence_Impact: Neutral
05007      Entropy_Signal: Consumes
05008      Default_Params: null
05009      Dependencies:
05010      - KA-107
05011      - KA-108
05012      - KA-111
05013      Produces_Artifacts: 'Yes'
05014      Audit_Events:
05015      - 'Yes'
```



```
05016      Version: 1.0.0
05017      Owner: UKG/USKD Core
05018      Status: Active
05019      Notes: null
05020      Implementation_Mode: PYTHON_LIB
05021      Primary_Libraries:
05022      - pydantic
05023      - opentelemetry
05024      - sqlalchemy
05025      - psycpg
05026      - alembic
05027      - redis
05028      Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
05029      psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
05030      Fallback_Libraries:
05031      - asyncpg
05032      Fallback_Library_Versions: asyncpg>=0.29,<1
05033      Runtime_Env: python
05034      Test_Harness: pytest + compliance fixtures
05035      Has_Math: 'No'
05036      Math_Component_Type: null
05037      Math_Object_IDs: null
05038      Math_Scope: null
05039      Variables_Used: null
05040      Units_Normalization: null
05041      Assumptions_Constraints: null
05042      KA-101:
05043      KA_Name: Chaos Injection Governor
05044      Short_Name: CHAOS-GOV
05045      Purpose: Policy control for chaos injection magnitude and scope
05046      Category: Control
05047      Primary_Layers:
05048      - L3
05049      - L4
05050      - L5
05051      - L6
05052      - L7
05053      Allowed_Layers:
05054      - L3
05055      - L4
05056      - L5
05057      - L6
05058      - L7
05059      Inputs: Entropy; risk; mode
05060      Outputs: Chaos bounds; allowed KAs
05061      Reads_Memory: 'Yes'
05062      Writes_Memory: 'No'
05063      Can_Invoke_Chaos: 'No'
05064      Can_Invoke_External_Research: 'No'
05065      Can_Trigger_Recursion: 'No'
05066      Can_Veto: 'Yes'
05067      Risk_Class: Critical
05068      Confidence_Impact: Neutral
05069      Entropy_Signal: Consumes
05070      Default_Params: null
05071      Dependencies:
05072      - KA-102
05073      - KA-107
05074      Produces_Artifacts: 'Yes'
05075      Audit_Events:
05076      - 'Yes'
05077      Version: 1.0.0
05078      Owner: UKG/USKD Core
05079      Status: Active
05080      Notes: Must disable in compliance/memory layers.
05081      Implementation_Mode: HYBRID
```

```
05082     Primary_Libraries:
05083     - pydantic
05084     - pyyaml
05085     - orjson
05086     - opentelemetry
05087     - prometheus-client
05088     - networkx
05089     Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \
    opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
05090     prometheus-client>=0.20,<1; networkx>=3.2,<4
05091     Fallback_Libraries:
05092     - redis
05093     Fallback_Library_Versions: redis>=5.0,<6
05094     Runtime_Env: python + in-model
05095     Test_Harness: pytest + golden-trace tests
05096     Has_Math: 'Yes'
05097     Math_Component_Type: Algorithm
05098     Math_Object_IDs:
05099     - A-CHAOS-01
05100     Math_Scope: FROST
05101     Variables_Used:
05102     - H*
05103     - r
05104     - εmax
05105     Units_Normalization: ε bounded
05106     Assumptions_Constraints: Chaos disabled in L8-L10
05107     KA-102:
05108     KA_Name: Global Entropy Quantifier
05109     Short_Name: ENTROPY
05110     Purpose: Compute normalized entropy score for simulation
05111     Category: Control
05112     Primary_Layers:
05113     - L6
05114     Allowed_Layers:
05115     - L6
05116     Inputs: Branch variance; persona variance; evidence variance
05117     Outputs: Entropy score [0..1]; components
05118     Reads_Memory: 'Yes'
05119     Writes_Memory: 'No'
05120     Can_Invoke_Chaos: 'No'
05121     Can_Invoke_External_Research: 'No'
05122     Can_Trigger_Recursion: 'No'
05123     Can_Veto: 'No'
05124     Risk_Class: High
05125     Confidence_Impact: Neutral
05126     Entropy_Signal: Produces
05127     Default_Params: null
05128     Dependencies:
05129     - KA-2
05130     - KA-12
05131     - KA-26
05132     Produces_Artifacts: 'Yes'
05133     Audit_Events:
05134     - 'Yes'
05135     Version: 1.0.0
05136     Owner: UKG/USKD Core
05137     Status: Active
05138     Notes: null
05139     Implementation_Mode: HYBRID
05140     Primary_Libraries:
05141     - pydantic
05142     - pyyaml
05143     - orjson
05144     - opentelemetry
05145     - prometheus-client
05146     - networkx
05147     Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \
```

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    opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
05148     prometheus-client>=0.20,<1; networkx>=3.2,<4
05149     Fallback_Libraries:
05150     - redis
05151     Fallback_Library_Versions: redis>=5.0,<6
05152     Runtime_Env: python + in-model
05153     Test_Harness: pytest + golden-trace tests
05154     Has_Math: 'Yes'
05155     Math_Component_Type: Equation
05156     Math_Object_IDs:
05157     - F-ENT-01
05158     - F-ENT-02
05159     Math_Scope: TruthEngine/FROST
05160     Variables_Used:
05161     - p_k
05162     - Var
05163     - D
05164     Units_Normalization:  $H \in [0,1]$ 
05165     Assumptions_Constraints: Normalize entropy
05166 KA-103:
05167     KA_Name: Simulation State Rollback Manager
05168     Short_Name: ROLLBACK
05169     Purpose: Checkpoint and rollback simulation state
05170     Category: Safety
05171     Primary_Layers:
05172     - L7
05173     Allowed_Layers:
05174     - L7
05175     Inputs: Checkpoint state; failure flags
05176     Outputs: Restored state; rollback log
05177     Reads_Memory: 'Yes'
05178     Writes_Memory: 'Yes'
05179     Can_Invoke_Chaos: 'No'
05180     Can_Invoke_External_Research: 'No'
05181     Can_Trigger_Recursion: 'No'
05182     Can_Veto: 'Yes'
05183     Risk_Class: Critical
05184     Confidence_Impact: Neutral
05185     Entropy_Signal: 'No'
05186     Default_Params: null
05187     Dependencies:
05188     - KA-32
05189     - KA-81
05190     Produces_Artifacts: 'Yes'
05191     Audit_Events:
05192     - 'Yes'
05193     Version: 1.0.0
05194     Owner: UKG/USKD Core
05195     Status: Active
05196     Notes: null
05197     Implementation_Mode: PYTHON_LIB
05198     Primary_Libraries:
05199     - pydantic
05200     - opentelemetry
05201     - sqlalchemy
05202     - psycpg
05203     - alembic
05204     - redis
05205     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
05206     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
05207     Fallback_Libraries:
05208     - asyncpg
05209     Fallback_Library_Versions: asyncpg>=0.29,<1
05210     Runtime_Env: python
05211     Test_Harness: pytest + compliance fixtures
05212     Has_Math: 'Yes'

```

```
05213     Math_Component_Type: Algorithm
05214     Math_Object_IDs:
05215     - A-ROLLBACK-01
05216     Math_Scope: FROST
05217     Variables_Used:
05218     - checkpoint
05219     Units_Normalization: null
05220     Assumptions_Constraints: Deterministic checkpoints
05221 KA-104:
05222     KA_Name: Truth vs Utility Arbiter
05223     Short_Name: TUA
05224     Purpose: Balance correctness vs operational safety/utility
05225     Category: Safety
05226     Primary_Layers:
05227     - L8
05228     Allowed_Layers:
05229     - L8
05230     Inputs: Truth scores; risk profile; audience
05231     Outputs: Adjusted framing; veto if harmful
05232     Reads_Memory: 'Yes'
05233     Writes_Memory: 'No'
05234     Can_Invoke_Chaos: 'No'
05235     Can_Invoke_External_Research: 'No'
05236     Can_Trigger_Recursion: 'No'
05237     Can_Veto: 'Yes'
05238     Risk_Class: Critical
05239     Confidence_Impact: Neutral
05240     Entropy_Signal: 'No'
05241     Default_Params: null
05242     Dependencies:
05243     - KA-22
05244     - KA-24
05245     Produces_Artifacts: 'Yes'
05246     Audit_Events:
05247     - 'Yes'
05248     Version: 1.0.0
05249     Owner: UKG/USKD Core
05250     Status: Active
05251     Notes: null
05252     Implementation_Mode: PYTHON_LIB
05253     Primary_Libraries:
05254     - pydantic
05255     - opentelemetry
05256     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
05257     Fallback_Libraries:
05258     - asynpg
05259     Fallback_Library_Versions: asynpg>=0.29,<1
05260     Runtime_Env: python
05261     Test_Harness: pytest + compliance fixtures
05262     Has_Math: 'No'
05263     Math_Component_Type: null
05264     Math_Object_IDs: null
05265     Math_Scope: null
05266     Variables_Used: null
05267     Units_Normalization: null
05268     Assumptions_Constraints: null
05269 KA-105:
05270     KA_Name: Conceptual Obsolescence Monitor
05271     Short_Name: OBSOLETE
05272     Purpose: Detect paradigm-level obsolescence (not just staleness)
05273     Category: Lifecycle
05274     Primary_Layers:
05275     - L10
05276     Allowed_Layers:
05277     - L10
05278     Inputs: Contradiction trends; policy shifts; citations
```

```
05279      Outputs: Obsolescence flags; revalidation requests
05280      Reads_Memory: 'Yes'
05281      Writes_Memory: 'Yes'
05282      Can_Invoke_Chaos: 'No'
05283      Can_Invoke_External_Research: 'No'
05284      Can_Trigger_Recursion: 'No'
05285      Can_Veto: 'No'
05286      Risk_Class: High
05287      Confidence_Impact: Indirect ↓
05288      Entropy_Signal: Produces
05289      Default_Params: null
05290      Dependencies:
05291      - KA-82
05292      - KA-83
05293      Produces_Artifacts: 'Yes'
05294      Audit_Events:
05295      - 'Yes'
05296      Version: 1.0.0
05297      Owner: UKG/USKD Core
05298      Status: Active
05299      Notes: null
05300      Implementation_Mode: PYTHON_LIB
05301      Primary_Libraries:
05302      - pydantic
05303      - opentelemetry
05304      - sqlalchemy
05305      - psycpg
05306      - alembic
05307      - redis
05308      Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
05309      psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
05310      Fallback_Libraries:
05311      - asyncpg
05312      Fallback_Library_Versions: asyncpg>=0.29,<1
05313      Runtime_Env: python
05314      Test_Harness: pytest + compliance fixtures
05315      Has_Math: 'No'
05316      Math_Component_Type: null
05317      Math_Object_IDs: null
05318      Math_Scope: null
05319      Variables_Used: null
05320      Units_Normalization: null
05321      Assumptions_Constraints: null
05322      KA-106:
05323      KA_Name: Human Override Reason Capture
05324      Short_Name: OVERRIDE
05325      Purpose: Capture human correction rationale in structured form
05326      Category: Governance
05327      Primary_Layers:
05328      - L10
05329      Allowed_Layers:
05330      - L10
05331      Inputs: Human decision; rationale
05332      Outputs: Structured correction; training signal
05333      Reads_Memory: 'Yes'
05334      Writes_Memory: 'Yes'
05335      Can_Invoke_Chaos: 'No'
05336      Can_Invoke_External_Research: 'No'
05337      Can_Trigger_Recursion: 'No'
05338      Can_Veto: 'No'
05339      Risk_Class: High
05340      Confidence_Impact: Neutral
05341      Entropy_Signal: 'No'
05342      Default_Params: null
05343      Dependencies:
05344      - KA-95
```

```
05345     Produces_Artifacts: 'Yes'
05346     Audit_Events:
05347     - 'Yes'
05348     Version: 1.0.0
05349     Owner: UKG/USKD Core
05350     Status: Active
05351     Notes: null
05352     Implementation_Mode: PYTHON_LIB
05353     Primary_Libraries:
05354     - pydantic
05355     - opentelemetry
05356     - sqlalchemy
05357     - psycpg
05358     - alembic
05359     - redis
05360     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
05361     psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
05362     Fallback_Libraries:
05363     - asyncpg
05364     Fallback_Library_Versions: asyncpg>=0.29,<1
05365     Runtime_Env: python
05366     Test_Harness: pytest + compliance fixtures
05367     Has_Math: 'No'
05368     Math_Component_Type: null
05369     Math_Object_IDs: null
05370     Math_Scope: null
05371     Variables_Used: null
05372     Units_Normalization: null
05373     Assumptions_Constraints: null
05374 KA-107:
05375     KA_Name: Reasoning Boundary Enforcer
05376     Short_Name: BOUNDARY
05377     Purpose: Enforce hard boundaries on capabilities and layer entry
05378     Category: Safety
05379     Primary_Layers:
05380     - L1
05381     - L7
05382     Allowed_Layers:
05383     - L1
05384     - L7
05385     Inputs: Policy; query class; planned pipeline
05386     Outputs: Allowed capabilities; blocks
05387     Reads_Memory: 'No'
05388     Writes_Memory: 'No'
05389     Can_Invoke_Chaos: 'No'
05390     Can_Invoke_External_Research: 'No'
05391     Can_Trigger_Recursion: 'No'
05392     Can_Veto: 'Yes'
05393     Risk_Class: Critical
05394     Confidence_Impact: Neutral
05395     Entropy_Signal: 'No'
05396     Default_Params: null
05397     Dependencies:
05398     - KA-4
05399     - KA-5
05400     - KA-113
05401     Produces_Artifacts: 'Yes'
05402     Audit_Events:
05403     - 'Yes'
05404     Version: 1.0.0
05405     Owner: UKG/USKD Core
05406     Status: Active
05407     Notes: null
05408     Implementation_Mode: PYTHON_LIB
05409     Primary_Libraries:
05410     - pydantic
```

```
05411     - opentelemetry
05412     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
05413       opentelemetry-sdk>=1.26,<2
05413     Fallback_Libraries:
05414     - asyncpg
05415     Fallback_Library_Versions: asyncpg>=0.29,<1
05416     Runtime_Env: python
05417     Test_Harness: pytest + compliance fixtures
05418     Has_Math: 'No'
05419     Math_Component_Type: null
05420     Math_Object_IDs: null
05421     Math_Scope: null
05422     Variables_Used: null
05423     Units_Normalization: null
05424     Assumptions_Constraints: null
05425 KA-108:
05426     KA_Name: Capability Escalation Detector
05427     Short_Name: CAP-ESC
05428     Purpose: Detect unsafe capability drift/escalation patterns
05429     Category: Safety
05430     Primary_Layers:
05431     - L6
05432     - L10
05433     Allowed_Layers:
05434     - L6
05435     - L10
05436     Inputs: KA interaction patterns; emergence flags
05437     Outputs: Escalation alerts; containment actions
05438     Reads_Memory: 'Yes'
05439     Writes_Memory: 'No'
05440     Can_Invoke_Chaos: 'No'
05441     Can_Invoke_External_Research: 'No'
05442     Can_Trigger_Recursion: 'No'
05443     Can_Veto: 'Yes'
05444     Risk_Class: Critical
05445     Confidence_Impact: Neutral
05446     Entropy_Signal: Produces
05447     Default_Params: null
05448     Dependencies:
05449     - KA-21
05450     - KA-112
05451     Produces_Artifacts: 'Yes'
05452     Audit_Events:
05453     - 'Yes'
05454     Version: 1.0.0
05455     Owner: UKG/USKD Core
05456     Status: Active
05457     Notes: null
05458     Implementation_Mode: PYTHON_LIB
05459     Primary_Libraries:
05460     - pydantic
05461     - opentelemetry
05462     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
05463       opentelemetry-sdk>=1.26,<2
05463     Fallback_Libraries:
05464     - asyncpg
05465     Fallback_Library_Versions: asyncpg>=0.29,<1
05466     Runtime_Env: python
05467     Test_Harness: pytest + compliance fixtures
05468     Has_Math: 'No'
05469     Math_Component_Type: null
05470     Math_Object_IDs: null
05471     Math_Scope: null
05472     Variables_Used: null
05473     Units_Normalization: null
05474     Assumptions_Constraints: null
05475 KA-109:
```

```
05476     KA_Name: Knowledge Containment Classifier
05477     Short_Name: CONTAIN
05478     Purpose: Classify knowledge persistence safety (public/restricted/never)
05479     Category: Governance
05480     Primary_Layers:
05481     - L10
05482     Allowed_Layers:
05483     - L10
05484     Inputs: Knowledge candidate; risk signals
05485     Outputs: Containment class; persistence rules
05486     Reads_Memory: 'Yes'
05487     Writes_Memory: 'Yes'
05488     Can_Invoke_Chaos: 'No'
05489     Can_Invoke_External_Research: 'No'
05490     Can_Trigger_Recursion: 'No'
05491     Can_Veto: 'Yes'
05492     Risk_Class: Critical
05493     Confidence_Impact: Neutral
05494     Entropy_Signal: 'No'
05495     Default_Params: null
05496     Dependencies:
05497     - KA-24
05498     - KA-74
05499     Produces_Artifacts: 'Yes'
05500     Audit_Events:
05501     - 'Yes'
05502     Version: 1.0.0
05503     Owner: UKG/USKD Core
05504     Status: Active
05505     Notes: null
05506     Implementation_Mode: PYTHON_LIB
05507     Primary_Libraries:
05508     - pydantic
05509     - opentelemetry
05510     - presidio
05511     - sqlalchemy
05512     - psycpg
05513     - alembic
05514     - redis
05515     Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
05516     opentelemetry-sdk>=1.26,<2; presidio-analyzer>=2.2,<3;
05517     presidio-anonymizer>=2.2,<3; sqlalchemy>=2.0,<3; psycpg[binary]>=3.1,<4; \
05518     alembic>=1.13,<2; redis>=5.0,<6
05519     Fallback_Libraries:
05520     - asyncpg
05521     Fallback_Library_Versions: asyncpg>=0.29,<1
05522     Runtime_Env: python
05523     Test_Harness: pytest + compliance fixtures
05524     Has_Math: 'No'
05525     Math_Component_Type: null
05526     Math_Object_IDs: null
05527     Math_Scope: null
05528     Variables_Used: null
05529     Units_Normalization: null
05530     Assumptions_Constraints: null
05531     KA-110:
05532     KA_Name: Cross-Domain Coupling Risk Analyzer
05533     Short_Name: COUPLE-RISK
05534     Purpose: Detect risky cross-domain coupling pathways
05535     Category: Safety
05536     Primary_Layers:
05537     - L6
05538     - L7
05539     Allowed_Layers:
05540     - L6
05541     - L7
05542     Inputs: Domain tags; planned KAs; knowledge links
```



```
05541      Outputs: Coupling risk score; blocked combos
05542      Reads_Memory: 'Yes'
05543      Writes_Memory: 'No'
05544      Can_Invoke_Chaos: 'No'
05545      Can_Invoke_External_Research: 'No'
05546      Can_Trigger_Recursion: 'No'
05547      Can_Veto: 'Yes'
05548      Risk_Class: Critical
05549      Confidence_Impact: Neutral
05550      Entropy_Signal: Produces
05551      Default_Params: null
05552      Dependencies:
05553      - KA-5
05554      - KA-107
05555      Produces_Artifacts: 'Yes'
05556      Audit_Events:
05557      - 'Yes'
05558      Version: 1.0.0
05559      Owner: UKG/USKD Core
05560      Status: Active
05561      Notes: null
05562      Implementation_Mode: PYTHON_LIB
05563      Primary_Libraries:
05564      - pydantic
05565      - opentelemetry
05566      Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
05567      Fallback_Libraries:
05568      - asynpg
05569      Fallback_Library_Versions: asynpg>=0.29,<1
05570      Runtime_Env: python
05571      Test_Harness: pytest + compliance fixtures
05572      Has_Math: 'No'
05573      Math_Component_Type: null
05574      Math_Object_IDs: null
05575      Math_Scope: null
05576      Variables_Used: null
05577      Units_Normalization: null
05578      Assumptions_Constraints: null
05579  KA-111:
05580      KA_Name: Long-Horizon Goal Drift Monitor
05581      Short_Name: GOAL-DRIFT
05582      Purpose: Detect latent goal persistence across runs
05583      Category: Safety
05584      Primary_Layers:
05585      - L10
05586      Allowed_Layers:
05587      - L10
05588      Inputs: Evolution actions; traces
05589      Outputs: Goal-drift alerts; constraints
05590      Reads_Memory: 'Yes'
05591      Writes_Memory: 'No'
05592      Can_Invoke_Chaos: 'No'
05593      Can_Invoke_External_Research: 'No'
05594      Can_Trigger_Recursion: 'No'
05595      Can_Veto: 'Yes'
05596      Risk_Class: Critical
05597      Confidence_Impact: Neutral
05598      Entropy_Signal: Produces
05599      Default_Params: null
05600      Dependencies:
05601      - KA-100
05602      - KA-112
05603      Produces_Artifacts: 'Yes'
05604      Audit_Events:
05605      - 'Yes'
05606      Version: 1.0.0
```

```
05607      Owner: UKG/USKD Core
05608      Status: Active
05609      Notes: null
05610      Implementation_Mode: PYTHON_LIB
05611      Primary_Libraries:
05612      - pydantic
05613      - opentelemetry
05614      Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2
05615      Fallback_Libraries:
05616      - asynpg
05617      Fallback_Library_Versions: asynpg>=0.29,<1
05618      Runtime_Env: python
05619      Test_Harness: pytest + compliance fixtures
05620      Has_Math: 'No'
05621      Math_Component_Type: null
05622      Math_Object_IDs: null
05623      Math_Scope: null
05624      Variables_Used: null
05625      Units_Normalization: null
05626      Assumptions_Constraints: null
05627  KA-112:
05628      KA_Name: System Self-Introspection Auditor
05629      Short_Name: INTROSPECT
05630      Purpose: 'Audit system-level behavior: chaos usage, overrides, drift'
05631      Category: Governance
05632      Primary_Layers:
05633      - L10
05634      Allowed_Layers:
05635      - L10
05636      Inputs: System metrics; logs
05637      Outputs: Audit report; governance actions
05638      Reads_Memory: 'Yes'
05639      Writes_Memory: 'Yes'
05640      Can_Invoke_Chaos: 'No'
05641      Can_Invoke_External_Research: 'No'
05642      Can_Trigger_Recursion: 'No'
05643      Can_Veto: 'No'
05644      Risk_Class: High
05645      Confidence_Impact: Neutral
05646      Entropy_Signal: 'No'
05647      Default_Params: null
05648      Dependencies: null
05649      Produces_Artifacts: 'Yes'
05650      Audit_Events:
05651      - 'Yes'
05652      Version: 1.0.0
05653      Owner: UKG/USKD Core
05654      Status: Active
05655      Notes: null
05656      Implementation_Mode: PYTHON_LIB
05657      Primary_Libraries:
05658      - pydantic
05659      - opentelemetry
05660      - sqlalchemy
05661      - psycpg
05662      - alembic
05663      - redis
05664      Primary_Library_Versions: pydantic>=2.6,<3; opentelemetry-api>=1.26,<2; \
    opentelemetry-sdk>=1.26,<2; sqlalchemy>=2.0,<3;
05665      psycpg[binary]>=3.1,<4; alembic>=1.13,<2; redis>=5.0,<6
05666      Fallback_Libraries:
05667      - asynpg
05668      Fallback_Library_Versions: asynpg>=0.29,<1
05669      Runtime_Env: python
05670      Test_Harness: pytest + compliance fixtures
05671      Has_Math: 'No'
```

```
05672     Math_Component_Type: null
05673     Math_Object_IDs: null
05674     Math_Scope: null
05675     Variables_Used: null
05676     Units_Normalization: null
05677     Assumptions_Constraints: null
05678 KA-113:
05679     KA_Name: Query Analysis & Complexity Router
05680     Short_Name: QROUTER
05681     Purpose: Select how much of the system runs based on difficulty/stakes
05682     Category: Routing
05683     Primary_Layers:
05684     - L1
05685     Allowed_Layers:
05686     - L1
05687     Inputs: Normalized query; org mode; policy
05688     Outputs: Complexity tier; layer depth; budgets
05689     Reads_Memory: 'No'
05690     Writes_Memory: 'No'
05691     Can_Invoke_Chaos: 'No'
05692     Can_Invoke_External_Research: 'No'
05693     Can_Trigger_Recursion: 'No'
05694     Can_Veto: 'No'
05695     Risk_Class: Critical
05696     Confidence_Impact: Neutral
05697     Entropy_Signal: Produces
05698     Default_Params: null
05699     Dependencies:
05700     - KA-5
05701     - KA-4
05702     - KA-80
05703     Produces_Artifacts: 'Yes'
05704     Audit_Events:
05705     - 'Yes'
05706     Version: 1.0.0
05707     Owner: UKG/USKD Core
05708     Status: Active
05709     Notes: null
05710     Implementation_Mode: HYBRID
05711     Primary_Libraries:
05712     - pydantic
05713     - pyyaml
05714     - orjson
05715     - opentelemetry
05716     - prometheus-client
05717     - networkx
05718     Primary_Library_Versions: pydantic>=2.6,<3; PyYAML>=6.0,<7; orjson>=3.10,<4; \
05719     opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2;
05720     prometheus-client>=0.20,<1; networkx>=3.2,<4
05721     Fallback_Libraries:
05722     - redis
05723     Fallback_Library_Versions: redis>=5.0,<6
05724     Runtime_Env: python + in-model
05725     Test_Harness: pytest + golden-trace tests
05726     Has_Math: 'Yes'
05727     Math_Component_Type: Algorithm
05728     Math_Object_IDs:
05729     - A-ROUTER-01
05730     Math_Scope: System
05731     Variables_Used:
05732     - tier
05733     - r
05734     - H*
05735     - B
05736     Units_Normalization: tiers
05737     Assumptions_Constraints: Policy + classifier required
05738 KA-114:
```

```
05738     KA_Name: External Deep Research Invocation
05739     Short_Name: DEEP-EXT
05740     Purpose: Invoke bounded Deep Research provider and return evidence packet
05741     Category: Capability
05742     Primary_Layers:
05743     - L3
05744     Allowed_Layers:
05745     - L3
05746     Inputs: Sub-question; scope; timebox; policy
05747     Outputs: Research packet; source list; citations
05748     Reads_Memory: 'Yes'
05749     Writes_Memory: 'No'
05750     Can_Invoke_Chaos: 'No'
05751     Can_Invoke_External_Research: 'Yes'
05752     Can_Trigger_Recursion: 'No'
05753     Can_Veto: 'No'
05754     Risk_Class: Critical
05755     Confidence_Impact: Neutral
05756     Entropy_Signal: 'No'
05757     Default_Params: null
05758     Dependencies:
05759     - KA-113
05760     - KA-31
05761     - KA-107
05762     - KA-24
05763     Produces_Artifacts: 'Yes'
05764     Audit_Events:
05765     - 'Yes'
05766     Version: 1.0.0
05767     Owner: UKG/USKD Core
05768     Status: Active
05769     Notes: Never writes memory directly; must pass validation & trust scoring first.
05770     Implementation_Mode: SERVICE
05771     Primary_Libraries:
05772     - httpx
05773     - tenacity
05774     - pydantic
05775     - opentelemetry
05776     Primary_Library_Versions: httpx>=0.27,<1; tenacity>=8.2,<9; pydantic>=2.6,<3; \
    opentelemetry-api>=1.26,<2; opentelemetry-sdk>=1.26,<2
05777     Fallback_Libraries:
05778     - requests
05779     Fallback_Library_Versions: requests>=2.32,<3
05780     Runtime_Env: python + managed-service
05781     Test_Harness: contract tests + pytest
05782     Has_Math: 'No'
05783     Math_Component_Type: null
05784     Math_Object_IDs: null
05785     Math_Scope: null
05786     Variables_Used: null
05787     Units_Normalization: null
05788     Assumptions_Constraints: null
05789 3) lookup_indexes:
05790     by_category:
05791     Reasoning:
05792     - KA-001
05793     - KA-002
05794     - KA-006
05795     - KA-011
05796     - KA-015
05797     - KA-035
05798     - KA-038
05799     - KA-055
05800     - KA-066
05801     - KA-067
05802     - KA-070
05803     Analysis:
```

05804 - KA-003
05805 - KA-021
05806 - KA-025
05807 - KA-037
05808 - KA-042
05809 Safety:
05810 - KA-004
05811 - KA-022
05812 - KA-024
05813 - KA-034
05814 - KA-050
05815 - KA-061
05816 - KA-065
05817 - KA-074
05818 - KA-081
05819 - KA-085
05820 - KA-094
05821 - KA-099
05822 - KA-103
05823 - KA-104
05824 - KA-107
05825 - KA-108
05826 - KA-110
05827 - KA-111
05828 Routing:
05829 - KA-005
05830 - KA-031
05831 - KA-073
05832 - KA-113
05833 Control:
05834 - KA-007
05835 - KA-020
05836 - KA-032
05837 - KA-059
05838 - KA-060
05839 - KA-068
05840 - KA-072
05841 - KA-080
05842 - KA-101
05843 - KA-102
05844 QA:
05845 - KA-008
05846 - KA-087
05847 - KA-098
05848 Truth:
05849 - KA-009
05850 - KA-014
05851 - KA-018
05852 - KA-026
05853 - KA-041
05854 - KA-048
05855 - KA-062
05856 - KA-084
05857 Ethics:
05858 - KA-010
05859 - KA-027
05860 - KA-045
05861 - KA-075
05862 Persona:
05863 - KA-012
05864 - KA-013
05865 - KA-028
05866 Compliance:
05867 - KA-016
05868 - KA-089
05869 - KA-090
05870 Context:

```
05871      - KA-017
05872      - KA-029
05873      - KA-040
05874      - KA-054
05875      Synthesis:
05876      - KA-019
05877      - KA-056
05878      Lifecycle:
05879      - KA-023
05880      - KA-039
05881      - KA-049
05882      - KA-052
05883      - KA-053
05884      - KA-076
05885      - KA-077
05886      - KA-082
05887      - KA-083
05888      - KA-088
05889      - KA-093
05890      - KA-105
05891      Coordination:
05892      - KA-030
05893      Reserved:
05894      - KA-033
05895      Optimization:
05896      - KA-036
05897      Governance:
05898      - KA-043
05899      - KA-071
05900      - KA-086
05901      - KA-092
05902      - KA-095
05903      - KA-096
05904      - KA-100
05905      - KA-106
05906      - KA-109
05907      - KA-112
05908      Creativity:
05909      - KA-044
05910      Memory:
05911      - KA-046
05912      - KA-078
05913      - KA-079
05914      - KA-091
05915      Meta:
05916      - KA-047
05917      Learning:
05918      - KA-051
05919      - KA-063
05920      - KA-064
05921      - KA-097
05922      UX:
05923      - KA-057
05924      - KA-058
05925      - KA-069
05926      Capability:
05927      - KA-114
05928      by_primary_layer:
05929      L1:
05930      - KA-001
05931      - KA-004
05932      - KA-005
05933      - KA-031
05934      - KA-032
05935      - KA-058
05936      - KA-059
05937      - KA-061
```

05938 - KA-073
05939 - KA-080
05940 - KA-107
05941 - KA-113
05942 L2:
05943 - KA-001
05944 - KA-003
05945 - KA-006
05946 - KA-017
05947 - KA-029
05948 - KA-032
05949 - KA-035
05950 - KA-068
05951 - KA-072
05952 L3:
05953 - KA-002
05954 - KA-011
05955 - KA-015
05956 - KA-025
05957 - KA-029
05958 - KA-032
05959 - KA-034
05960 - KA-038
05961 - KA-040
05962 - KA-054
05963 - KA-055
05964 - KA-066
05965 - KA-101
05966 - KA-114
05967 L7:
05968 - KA-002
05969 - KA-006
05970 - KA-007
05971 - KA-021
05972 - KA-032
05973 - KA-034
05974 - KA-035
05975 - KA-036
05976 - KA-037
05977 - KA-044
05978 - KA-047
05979 - KA-060
05980 - KA-066
05981 - KA-067
05982 - KA-070
05983 - KA-080
05984 - KA-081
05985 - KA-101
05986 - KA-103
05987 - KA-107
05988 - KA-110
05989 L9:
05990 - KA-008
05991 - KA-010
05992 - KA-019
05993 - KA-032
05994 - KA-038
05995 - KA-055
05996 - KA-056
05997 - KA-057
05998 - KA-058
05999 - KA-069
06000 - KA-072
06001 - KA-074
06002 - KA-075
06003 - KA-087
06004 - KA-095

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06005      L6:
06006      - KA-009
06007      - KA-014
06008      - KA-018
06009      - KA-026
06010      - KA-032
06011      - KA-039
06012      - KA-040
06013      - KA-041
06014      - KA-042
06015      - KA-048
06016      - KA-062
06017      - KA-071
06018      - KA-084
06019      - KA-085
06020      - KA-101
06021      - KA-102
06022      - KA-108
06023      - KA-110
06024      L8:
06025      - KA-010
06026      - KA-016
06027      - KA-022
06028      - KA-024
06029      - KA-027
06030      - KA-032
06031      - KA-045
06032      - KA-074
06033      - KA-089
06034      - KA-090
06035      - KA-104
06036      L4:
06037      - KA-012
06038      - KA-013
06039      - KA-016
06040      - KA-028
06041      - KA-032
06042      - KA-068
06043      - KA-101
06044      L10:
06045      - KA-014
06046      - KA-020
06047      - KA-021
06048      - KA-023
06049      - KA-032
06050      - KA-039
06051      - KA-043
06052      - KA-046
06053      - KA-049
06054      - KA-050
06055      - KA-051
06056      - KA-052
06057      - KA-053
06058      - KA-054
06059      - KA-062
06060      - KA-063
06061      - KA-064
06062      - KA-065
06063      - KA-071
06064      - KA-076
06065      - KA-077
06066      - KA-078
06067      - KA-079
06068      - KA-082
06069      - KA-083
06070      - KA-084
06071      - KA-086
```



```
06072      - KA-088
06073      - KA-089
06074      - KA-090
06075      - KA-091
06076      - KA-092
06077      - KA-093
06078      - KA-094
06079      - KA-095
06080      - KA-096
06081      - KA-097
06082      - KA-098
06083      - KA-099
06084      - KA-100
06085      - KA-105
06086      - KA-106
06087      - KA-108
06088      - KA-109
06089      - KA-111
06090      - KA-112
06091      L5:
06092      - KA-030
06093      - KA-032
06094      - KA-037
06095      - KA-101
```

master_workflow.yaml

Source: ukg_yaml_v2/master_workflow.yaml

Content follows on next page.

```

00001      # =====
00002      # master_workflow.yaml
00003      # Generated: 2025-12-21T22:20:42.681800
00004      # Format: Tree / numbered node style + YAML
00005      # =====
00006      # ASCII TREE (high-level)
00007      # master_workflow
00008      # ■■ 1) subsystems
00009      # ■■ 2) initialization
00010      # ■■ 3) execution_flow (1..5)
00011      # ■■ 4) triggers
00012      # ■■ 5) outputs
00013      #
00014      # -----
00015      master_workflow:
00016        "1) metadata":
00017          name: "UKG Master Workflow"
00018          version: "2.0"
00019          status: "production"
00020          description: "Top-level orchestrator for 17-axis coordinate parsing, quad-persona \
orchestration, 10-layer simulation, and 12-step refinement."
00021          required_confidence_default: 0.995
00022          max_refinement_cycles_default: 3
00023
00024        "2) subsystems":
00025          "2.1) axis_schema_config": "17_axis_coordinate_schema.yaml"
00026          "2.2) simulation_stack_config": "10_layer_simulation_stack.yaml"
00027          "2.3) refinement_engine_config": "12_step_refinement_engine.yaml"
00028          "2.4) persona_templates": "quad_persona_templates.yaml"
00029          "2.5) knowledge_agent_registry": "ukg_knowledge_agents_registry.yaml"
00030          "2.6) ka_registry": "ka_registry.yaml"
00031          "2.7) math_formula_library": "math_formula_library.yaml"
00032          "2.8) truth_engine_config": "truth_engine.yaml"
00033
00034        "3) initialization":
00035          "3.1) load_all_subsystems": true
00036          "3.2) warm_caches":
00037            enabled: true
00038            preload:
00039              - "pillar_taxonomy"
00040              - "industry_code_crosswalks"
00041              - "regulatory_index"
00042              - "persona_templates"
00043          "3.3) security_baseline":
00044            invoke_ka:
00045              - "KA-004" # Input Validation & Normalization
00046              - "KA-061" # Adversarial Input Shield
00047              - "KA-107" # Reasoning Boundary Enforcer
00048
00049        "4) execution_flow":
00050          "4.1) step_1":
00051            operation: "Parse Input → 17-axis coordinates"
00052            invoke:
00053              config: "17_axis_coordinate_schema.yaml"
00054              ka_pipeline: ["KA-004", "KA-005", "KA-113"]
00055              outputs: ["parsed_query", "axis_coordinates", "complexity_tier", "stakes_profile"]
00056
00057          "4.2) step_2":
00058            operation: "Activate Quad Personas + optional stakeholders"
00059            invoke:
00060              config: "quad_persona_templates.yaml"
00061              ka_pipeline: ["KA-008", "KA-009", "KA-010", "KA-011"]
00062              outputs: ["persona_instances", "persona_weighting"]
00063
00064          "4.3) step_3":
00065            operation: "Run 10-layer simulation stack"
00066            invoke:

```

```
00067         config: "10_layer_simulation_stack.yaml"
00068         outputs: ["draft_solution", "reasoning_trace", "local_working_subgraph", \
00069                  "evidence_pack"]
00070     "4.4) step_4":
00071         operation: "Run 12-step refinement engine"
00072         invoke:
00073             config: "12_step_refinement_engine.yaml"
00074             outputs: ["validated_solution", "final_confidence", "audit_bundle"]
00075
00076     "4.5) step_5":
00077         operation: "Run Truth Engine (truth scoring + certification)"
00078         invoke:
00079             config: "truth_engine.yaml"
00080             ka_pipeline: ["KA-014", "KA-026", "KA-102"]
00081             outputs: ["truth_report", "final_confidence", "attestations", "veto_signals"]
00082
00083     "4.6) step_6":
00084         operation: "Output Finalization"
00085         invoke:
00086             module: "final_output_module"
00087             ka_pipeline: ["KA-014", "KA-026", "KA-071"]
00088         outputs:
00089             - "answer"
00090             - "confidence"
00091             - "provenance"
00092             - "compliance_attestation"
00093             - "security_attestation"
00094             - "metadata"
00095     "5) triggers":
00096         - condition: "persona_conflict_detected"
00097           action: "persona_arbitration"
00098           invoke_ka: ["KA-026", "KA-012"]
00099           details: "Initiate structured debate; resolve by evidence or explicit trade-offs."
00100
00101         - condition: "compliance_violation_detected"
00102           action: "containment_rollback"
00103           invoke_ka: ["KA-066", "KA-107"]
00104           details: "Rollback to safe state; reformulate within constraints; log full audit \
00105 event."
00106
00107         - condition: "refinement_loop_incomplete"
00108           action: "iteration_feedback"
00109           invoke_ka: ["KA-003", "KA-102"]
00110           details: "Feed gaps/entropy signals back to simulation layers; repeat refinement."
00111
00112         - condition: "threshold_not_met_after_max"
00113           action: "manual_review_alert"
00114           invoke_ka: ["KA-022", "KA-026"]
00115           details: "Escalate with full reasoning trace and unresolved gaps."
00116
00117         - condition: "all_clear"
00118           action: "finalize_output"
00119           details: "All checks passed; produce final answer with provenance + attestations."
00120     "6) outputs":
00121         schema:
00122             answer: "string"
00123             confidence: "float [0,1]"
00124             provenance:
00125                 reasoning_trace: "list"
00126                 sources: "list"
00127                 persona_consensus: "object"
00128             metadata:
00129                 processing_time_ms: "int"
00130                 complexity_tier: "string"
00131                 layers_executed: "list[int]"
```

00132 refinement_cycles: "int"

math_formula_library.yaml

Source: ukg_yaml_v2/math_formula_library.yaml

Content follows on next page.

math_formula_library.yaml

```
00001      # =====
00002      # math_formula_library.yaml
00003      # Generated: 2025-12-21T22:20:43.013965
00004      # Format: Tree / numbered node style + YAML
00005      # =====
00006      # ASCII TREE (high-level)
00007      # math_formula_library
00008 # ■■ 1 metadata
00009 # ■■ 2 objects (F-*)
00010 # ■■ 3 lookup_indexes
00011 #
00012      # -----
00013 math_formula_library:
00014   1) metadata:
00015     name: UKG Mathematical Formula Library
00016     version: '2.0'
00017     count: 16
00018     source: UKG_KA_1_to_114_Enterprise_Registry_with_Libraries_and_Formulas.xlsx / \
Formula_Library
00019   2) objects:
00020     F-CONF-01:
00021       Type: Equation
00022       Name: Confidence aggregation
00023       LaTeX:  $C = \sigma \left( \alpha \sum_i w_i e_i - \beta \cdot \right.$ 
 $\left. \mathrm{conflicts} - \gamma \cdot H \right)$ 
00024       Plain_English: Aggregate evidence strength, penalize conflicts and entropy, squash \
to [0,1].
00025       Inputs: w_i, e_i, conflicts, H
00026       Outputs: C
00027       Valid_Ranges:  $C \in [0,1]$ ;  $\sum w_i = 1$ 
00028       Used_By_KA_IDs:
00029         - KA-014
00030       Version: 1.0.0
00031       Notes: Run after evidence validation & contradiction scoring.
00032     F-CONF-02:
00033       Type: Equation
00034       Name: Confidence threshold gate
00035       LaTeX:  $\mathrm{PASS} = \mathbb{1}[C \geq C_{\min}]$ 
00036       Plain_English: Pass if confidence meets threshold.
00037       Inputs: C, C_min
00038       Outputs: PASS
00039       Valid_Ranges: C_min per policy (e.g., 0.995 for persistence)
00040       Used_By_KA_IDs:
00041         - KA-014
00042         - KA-079
00043       Version: 1.0.0
00044       Notes: Separate thresholds for response vs memory commit.
00045     F-ENT-01:
00046       Type: Equation
00047       Name: Branch entropy
00048       LaTeX:  $H = - \sum_k p_k \log p_k$ 
00049       Plain_English: Uncertainty across competing branches.
00050       Inputs: p_k
00051       Outputs: H
00052       Valid_Ranges: Normalize via  $H / \log K$  to [0,1]
00053       Used_By_KA_IDs:
00054         - KA-102
00055       Version: 1.0.0
00056       Notes: Use K=number of active branches.
00057     F-ENT-02:
00058       Type: Equation
00059       Name: Composite entropy
00060       LaTeX:  $H^* = \lambda_1 H_{\text{branch}} + \lambda_2 \mathrm{Var}(C_{\text{persona}}) + \lambda_3 D_{\text{evidence}}$ 
00061       Plain_English: Combine branch entropy, persona disagreement and evidence \
disagreement.
00062       Inputs: H_branch, Var(C_persona), D_evidence
```

```
00063       Outputs: H*
00064       Valid_Ranges: H*∈[0,1] after normalization
00065       Used_By_KA_IDs:
00066       - KA-102
00067       - KA-014
00068       Version: 1.0.0
00069       Notes: Weights λ set by policy/mode.
00070 F-IMPUTE-01:
00071   Type: Equation
00072   Name: Bayesian imputation (Gaussian example)
00073   LaTeX:  $p(x \mid D) \propto p(D \mid x) p(x) \prod x \sim \mathcal{N}(\mu, \sigma^2)$ 
00074   Plain_English: Compute posterior for missing x given prior and observed data.
00075   Inputs: prior p(x), data D
00076   Outputs: posterior p(x|D)
00077   Valid_Ranges: σ>0; model-specific
00078   Used_By_KA_IDs:
00079   - KA-035
00080   Version: 1.0.0
00081   Notes: Store uncertainty along with imputed value.
00082 A-PARETO-01:
00083   Type: Algorithm
00084   Name: Pareto front selection
00085   LaTeX:  $\{s \in S : \nexists s' \in S, \forall i \ f_i(s') \leq f_i(s) \ \wedge \ \exists j \ f_j(s') < f_j(s)\}$ 
00086   Plain_English: Keep solutions not dominated across objectives.
00087   Inputs: S, objectives f_i
00088   Outputs: Pareto set
00089   Valid_Ranges: Objectives normalized or comparable
00090   Used_By_KA_IDs:
00091   - KA-036
00092   Version: 1.0.0
00093   Notes: Used for trade-off recommendations.
00094 F-TRUST-01:
00095   Type: Equation
00096   Name: Trust score (weighted provenance)
00097   LaTeX:  $T = \frac{\sum_i w_i p_i}{\sum_i w_i}$ 
00098   Plain_English: Weighted average of provenance scores.
00099   Inputs: w_i, p_i
00100   Outputs: T
00101   Valid_Ranges: T∈[0,1]
00102   Used_By_KA_IDs:
00103   - KA-062
00104   - KA-018
00105   Version: 1.0.0
00106   Notes: Weights incorporate authority and recency.
00107 F-COST-01:
00108   Type: Equation
00109   Name: Simulation cost estimate
00110   LaTeX:  $\hat{K} = k_0 + k_1 n_{\text{branches}} + k_2 d + k_3 n_{\text{tokens}}$ 
00111   Plain_English: Estimate simulation cost from branches, depth and token volume.
00112   Inputs: n_branches, depth d, n_tokens
00113   Outputs: K-hat
00114   Valid_Ranges: 'Units: tokens/sec/$ (chosen)
00115   Used_By_KA_IDs:
00116   - KA-080
00117   Version: 1.0.0
00118   Notes: Calibrate constants with telemetry.
00119 A-BUDGET-01:
00120   Type: Algorithm
00121   Name: Budget enforcement
00122   LaTeX:  $\text{If } \hat{K} > B \text{ then prune/reduce depth until } \hat{K} \leq B$ 
00123   Plain_English: Enforce hard budget by pruning and depth reduction.
00124   Inputs: K-hat, budget B
00125   Outputs: caps/pruning
00126   Valid_Ranges: B in chosen units
00127   Used_By_KA_IDs:
00128   - KA-081
```



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00129     - KA-060
00130     Version: 1.0.0
00131     Notes: Hard stop when minimum viable reasoning reached.
00132   F-DRIFT-01:
00133     Type: Equation
00134     Name: Confidence drift rate
00135     LaTeX:  $\Delta C / \Delta t = (C_t - C_{t-\Delta t}) / \Delta t$ 
00136     Plain_English: Measures change in confidence over time.
00137     Inputs: C_t, C_{t-Δt}, Δt
00138     Outputs: drift
00139     Valid_Ranges: Δt>0
00140     Used_By_KA_IDs:
00141     - KA-082
00142     Version: 1.0.0
00143     Notes: Use rolling windows.
00144   F-DECAY-01:
00145     Type: Equation
00146     Name: Trust/belief decay
00147     LaTeX:  $T(t) = T_0 e^{-\lambda t} \cdot g(u)$ 
00148     Plain_English: Trust decays with time, modulated by usage.
00149     Inputs: T0, λ, age t, usage u
00150     Outputs: T(t)
00151     Valid_Ranges: T∈[0,1]
00152     Used_By_KA_IDs:
00153     - KA-093
00154     - KA-023
00155     Version: 1.0.0
00156     Notes: g(u) increases trust for frequent use.
00157   A-CHAOS-01:
00158     Type: Algorithm
00159     Name: Chaos injection policy
00160     LaTeX:  $\epsilon = \min(\epsilon_{\max}, a \cdot H^* \cdot (1-r))$ 
00161     Plain_English: Set perturbation magnitude from entropy and risk.
00162     Inputs: H*, risk r
00163     Outputs: epsilon
00164     Valid_Ranges: epsilon∈[0,εmax]
00165     Used_By_KA_IDs:
00166     - KA-101
00167     Version: 1.0.0
00168     Notes: Disable when risk high or in L8-L10.
00169   A-ROLLBACK-01:
00170     Type: Algorithm
00171     Name: Checkpoint rollback
00172     LaTeX:  $\text{'\text{On fail: state} \rightarrow \mathrm{checkpoint}(t^*)'}$ 
00173     Plain_English: Restore last stable checkpoint.
00174     Inputs: checkpoint id
00175     Outputs: state
00176     Valid_Ranges: null
00177     Used_By_KA_IDs:
00178     - KA-103
00179     Version: 1.0.0
00180     Notes: Requires deterministic checkpointing.
00181   A-LOAD-01:
00182     Type: Algorithm
00183     Name: Load balancing (benefit-cost pruning)
00184     LaTeX:  $\text{'\text{Keep } i \text{ if } b_i/c_i \geq \tau'}$ 
00185     Plain_English: Prune branches with low benefit per cost.
00186     Inputs: b_i, c_i, τ
00187     Outputs: kept set
00188     Valid_Ranges: b_i, c_i>0
00189     Used_By_KA_IDs:
00190     - KA-060
00191     Version: 1.0.0
00192     Notes: Benefit estimated as expected confidence gain.
00193   A-REVAL-01:
00194     Type: Algorithm
00195     Name: Revalidation scheduling
```

```
00196 LaTeX: \text{Schedule if }(\mathrm{age}>\theta_a)\lor(\mathrm{drift}>\theta_d)\lor(\mathrm{a}
athrm{crit}>\theta_c)
00197 Plain_English: Revalidate based on age, drift, or criticality.
00198 Inputs: age, drift, crit
00199 Outputs: tasks
00200 Valid_Ranges: thresholds by policy
00201 Used_By_KA_IDs:
00202 - KA-083
00203 - KA-105
00204 Version: 1.0.0
00205 Notes: Supports policy-driven cadences.
00206 A-ROUTER-01:
00207 Type: Algorithm
00208 Name: Complexity routing policy
00209 LaTeX: \text{Depth}=f(\mathrm{tier}, r, H^*, B)
00210 Plain_English: Select layer depth based on tier, risk, entropy and budget.
00211 Inputs: tier, r, H*, B
00212 Outputs: depth
00213 Valid_Ranges: Depth in {L1..L10}
00214 Used_By_KA_IDs:
00215 - KA-113
00216 Version: 1.0.0
00217 Notes: Produces budgets used by KA-31/KA-81.
00218 3) lookup_indexes:
00219 by_ka_id:
00220 KA-014:
00221 - F-CONF-01
00222 - F-CONF-02
00223 - F-ENT-02
00224 KA-079:
00225 - F-CONF-02
00226 KA-102:
00227 - F-ENT-01
00228 - F-ENT-02
00229 KA-035:
00230 - F-IMPUTE-01
00231 KA-036:
00232 - A-PARETO-01
00233 KA-062:
00234 - F-TRUST-01
00235 KA-018:
00236 - F-TRUST-01
00237 KA-080:
00238 - F-COST-01
00239 KA-081:
00240 - A-BUDGET-01
00241 KA-060:
00242 - A-BUDGET-01
00243 - A-LOAD-01
00244 KA-082:
00245 - F-DRIFT-01
00246 KA-093:
00247 - F-DECAY-01
00248 KA-023:
00249 - F-DECAY-01
00250 KA-101:
00251 - A-CHAOS-01
00252 KA-103:
00253 - A-ROLLBACK-01
00254 KA-083:
00255 - A-REVAL-01
00256 KA-105:
00257 - A-REVAL-01
00258 KA-113:
00259 - A-ROUTER-01
```

quad_persona_templates.yaml

Source: ukg_yaml_v2/quad_persona_templates.yaml
Content follows on next page.

```

00001      # =====
00002      # quad_persona_templates.yaml
00003      # Generated: 2025-12-21T22:20:42.686542
00004      # Format: Tree / numbered node style + YAML
00005      # =====
00006      # ASCII TREE (high-level)
00007      # personas
00008      # ■■ P8 Knowledge (Scholar)
00009      # ■■ P9 Sector (Practitioner)
00010      # ■■ P10 Regulatory (Octopus)
00011      # ■■ P11 Compliance (Spiderweb)
00012      #
00013      # -----
00014      personas:
00015          "1) schema":
00016              description: "7-part persona schema (plus optional extensions) used for instantiation."
00017              required_fields:
00018                  - "a) role"
00019                  - "b) education"
00020                  - "c) certifications"
00021                  - "d) skills"
00022                  - "e) traits"
00023                  - "f) job_training"
00024                  - "g) related_roles"
00025              optional_fields:
00026                  - "career_path"
00027                  - "years_experience"
00028                  - "domain_specialties"
00029                  - "jurisdiction_scope"
00030                  - "risk_tolerance_profile"
00031
00032          "2) templates":
00033              "2.1) P8_Knowledge_Scholar":
00034                  axis_id: 8
00035                  title: "Knowledge Expert (Scholar)"
00036                  objective: "Theoretical rigor, correctness, and evidence-backed reasoning."
00037                  defaults:
00038                      "a) role": "Domain Scholar / Researcher"
00039                      "b) education": ["PhD or equivalent in relevant pillar/branch"]
00040                      "c) certifications": []
00041                      "d) skills": ["theoretical analysis", "literature synthesis", "formal reasoning"]
00042                      "e) traits": ["precision", "skepticism", "clarity", "methodical"]
00043                      "f) job_training": ["research labs", "peer-review practice", "methods training"]
00044                      "g) related_roles": ["Professor", "Research Scientist", "Principal Investigator"]
00045                  guardrails:
00046                      must:
00047                          - "cite evidence nodes for non-obvious claims"
00048                          - "flag uncertainty explicitly"
00049                      must_not:
00050                          - "overstate confidence"
00051
00052              "2.2) P9_Sector_Practitioner":
00053                  axis_id: 9
00054                  title: "Sector Expert (Practitioner)"
00055                  objective: "Operational feasibility, best practices, and real-world constraints."
00056                  defaults:
00057                      "a) role": "Industry Operator / Implementation Lead"
00058                      "b) education": ["BS/MS in relevant field or equivalent experience"]
00059                      "c) certifications": ["sector-specific (optional)"]
00060                      "d) skills": ["implementation planning", "workflow optimization", "stakeholder \
management"]
00061                      "e) traits": ["pragmatic", "risk-aware", "time/budget sensitive"]
00062                      "f) job_training": ["on-call operations", "incident response", "process \
improvement"]
00063                      "g) related_roles": ["COO", "Program Manager", "Operations Director", "Site Lead"]
00064                  guardrails:
00065                      must:

```

```

00066         - "identify resource constraints"
00067         - "propose phased rollout options"
00068
00069     "2.3) P10_Regulatory_Octopus":
00070         axis_id: 10
00071         title: "Regulatory Expert (Octopus)"
00072         objective: "External law, regulations, standards, enforcement precedents."
00073         defaults:
00074             "a) role": "Regulatory Counsel / DPO / Standards Specialist"
00075             "b) education": ["JD/LLM or equivalent legal/regulatory training (as applicable)"]
00076             "c) certifications": ["CIPP/E", "CIPP/US", "ISO lead auditor (optional)"]
00077             "d) skills": ["statutory interpretation", "standards mapping", "audit readiness"]
00078             "e) traits": ["conservative", "detail-oriented", "risk-minimizing"]
00079             "f) job_training": ["regulator engagement", "compliance audits", "legal review \
cycles"]
00080             "g) related_roles": ["Privacy Counsel", "Regulatory Affairs Director"]
00081         guardrails:
00082             must:
00083                 - "separate jurisdiction-specific requirements"
00084                 - "flag when legal advice requires human counsel"
00085
00086     "2.4) P11_Compliance_Spiderweb":
00087         axis_id: 11
00088         title: "Compliance Expert (Spiderweb)"
00089         objective: "Internal controls, ethics, policy mesh, reputational risk."
00090         defaults:
00091             "a) role": "Chief Compliance Officer / Risk Officer"
00092             "b) education": ["MBA/Risk/Compliance training or equivalent"]
00093             "c) certifications": ["CCEP", "CHC", "CISA (optional)"]
00094             "d) skills": ["control design", "policy enforcement", "ethics review", "risk \
management"]
00095             "e) traits": ["balanced", "systems thinker", "fairness-aware"]
00096             "f) job_training": ["internal audit", "incident management", "policy governance"]
00097             "g) related_roles": ["Internal Auditor", "Ethics Officer", "GRC Lead"]
00098         guardrails:
00099             must:
00100                 - "run bias/ethics checks for people-impacting outputs"
00101                 - "identify conflicts among policies/standards"
00102     weights:
00103         default:
00104             P8_Knowledge_Scholar: 0.28
00105             P9_Sector_Practitioner: 0.26
00106             P10_Regulatory_Octopus: 0.24
00107             P11_Compliance_Spiderweb: 0.22
00108     adaptive_rules:
00109         - if: "stakes_profile == 'high_regulatory'"
00110             adjust:
00111                 P10_Regulatory_Octopus: "+0.10"
00112                 P9_Sector_Practitioner: "-0.05"
00113                 P8_Knowledge_Scholar: "-0.03"
00114                 P11_Compliance_Spiderweb: "-0.02"

```

truth_engine.yaml

Source: ukg_yaml_v2/truth_engine.yaml

Content follows on next page.

```

00001 # =====
00002 # truth_engine.yaml
00003 # Generated: 2025-12-21T00:00:00Z
00004 # Format: Tree / numbered node style + YAML
00005 # =====
00006 # ASCII TREE (high-level)
00007 # truth_engine
00008 # ■■ 1) metadata
00009 # ■■ 2) contracts (inputs/outputs)
00010 # ■■ 3) truth_primitives (claims/evidence/provenance)
00011 # ■■ 4) scoring_model (truth score → confidence)
00012 # ■■ 5) contradiction_resolution
00013 # ■■ 6) governance (thresholds, vetoes, escalation)
00014 # ■■ 7) storage (truth_ledger, audit_bundle)
00015 # ■■ 8) integration_hooks (simulation/refinement/api)
00016 # ■■ 9) observability (metrics, alerts)
00017 # -----
00018
00019 truth_engine:
00020   "1) metadata":
00021     name: "UKG Truth Engine"
00022     version: "2.0"
00023     status: "production"
00024     purpose: "Convert evidence + reasoning traces into auditable truth scores, calibrated \
confidence, and signed attestations."
00025     aligns_to_axes:
00026       - "Axis 14: Certainty and Evidence"
00027       - "Axis 13: Temporal Context (validity windows)"
00028       - "Axis 12: Location Context (jurisdictional truth constraints)"
00029     required_confidence_by_stakes:
00030       low: 0.85
00031       medium: 0.95
00032       high: 0.995
00033       safety_critical: 0.999
00034
00035   "2) contracts":
00036     "2.1) inputs":
00037       evidence_pack:
00038         description: "Normalized evidence objects (sources, excerpts, measurements, rules)
with provenance."
00039         schema:
00040           evidence_id: "string"
00041           type: "SOURCE | FACT | RULE | MEASUREMENT | POLICY | CASE_LAW | STANDARD"
00042           payload_ref: "uri | doc_id | coordinate"
00043           time_validity: "ISO_8601_interval"
00044           jurisdiction: "location_id"
00045           reliability:
00046             source_authority: "0..1"
00047             corroboration_count: "int"
00048             freshness_score: "0..1"
00049           links:
00050             supports_claims: "list[claim_id]"
00051             contradicts_claims: "list[claim_id]"
00052       reasoning_trace:
00053         description: "Stepwise trace from simulation + refinement (AoT/ToT graphs, \
decisions, trade-offs)."
00054         schema:
00055           step_id: "string"
00056           layer: "L1..L10"
00057           operation: "string"
00058           inputs: "list[ref]"
00059           outputs: "list[ref]"
00060           assumptions: "list[string]"
00061       query_context:
00062         description: "17-axis coordinates + stakes profile + user constraints."
00063         schema:
00064           axis_coordinates: "object"

```

truth_engine.yaml

```
00065         stakes_profile: "low|medium|high|safety_critical"
00066         required_confidence: "float [0,1]"
00067         audience: "engineer|executive|regulator|general"
00068
00069     "2.2) outputs":
00070         truth_report:
00071             description: "Truth-scored claim set with provenance, conflicts, and final \
confidence certification."
00072             schema:
00073                 claim_set: "list[claim]"
00074                 overall_confidence: "float [0,1]"
00075                 pass_threshold: "bool"
00076                 attestations:
00077                     compliance: "object"
00078                     security: "object"
00079                     provenance: "object"
00080                 audit_bundle_ref: "uri"
00081         veto_signals:
00082             description: "Blocking signals when hard constraints fail."
00083             values:
00084                 - "INSUFFICIENT_EVIDENCE"
00085                 - "UNRESOLVED_CONTRADICTION"
00086                 - "REGULATORY_CONFLICT"
00087                 - "TEMPORAL_INVALIDITY"
00088                 - "JURISDICTION_MISMATCH"
00089
00090     "3) truth_primitives":
00091         "3.1) claim_object":
00092             description: "Atomic assertion produced by the system. Everything is decomposed to \
claims."
00093             schema:
00094                 claim_id: "string"
00095                 text: "string"
00096                 coordinate: "optional coordinate tuple"
00097                 claim_type: "FACT | INFERENCE | RECOMMENDATION | CONSTRAINT | WARNING"
00098                 temporal_scope: "ISO_8601_interval"
00099                 jurisdiction_scope: "location_id"
00100                 supports: "list[evidence_id]"
00101                 contradicts: "list[claim_id]"
00102                 dependencies: "list[claim_id]"
00103                 scores:
00104                     truth_score: "0..1"
00105                     uncertainty: "0..1"
00106                     risk_weight: "0..1"
00107                     bias_risk: "0..1"
00108                 status: "ACCEPTED | REJECTED | PENDING | ESCALATED"
00109
00110         "3.2) evidence_object":
00111             normalization_pipeline:
00112                 - "De-duplicate by semantic fingerprint"
00113                 - "Extract authoritative metadata (publisher, standard body, regulator)"
00114                 - "Bind to validity interval + jurisdiction"
00115                 - "Assign reliability components (authority, corroboration, freshness)"
00116             identity:
00117                 fingerprint:
00118                     algorithm: "sha256(canonical_payload)"
00119                     collision_policy: "treat_as_same_evidence_when_hash_equal"
00120
00121         "3.3) provenance_graph":
00122             standard: "W3C PROV compatible (prov:Entity, prov:Activity, prov:Agent)"
00123             required_edges:
00124                 - "claim ←generatedBy– reasoning_step"
00125                 - "reasoning_step ←used– evidence"
00126                 - "evidence ←derivedFrom– source"
00127             tamper_resistance:
00128                 hash_chain:
00129                     enabled: true
```



```
00130         algorithm: "sha256"
00131         link_field: "prev_hash"
00132         signatures:
00133             enabled: true
00134             algorithm: "ed25519"
00135             signing_scope: "audit_bundle + truth_report summary"
00136
00137     "4) scoring_model":
00138         "4.1) stage_pipeline":
00139             # NOTE: KA numbers are resolved via ka_registry.yaml
00140             "4.1.1) claim_extraction":
00141                 invoke_ka: ["KA-001", "KA-003"]
00142                 outputs: ["claim_set_raw"]
00143             "4.1.2) evidence_binding":
00144                 invoke_ka: ["KA-020", "KA-041"]
00145                 outputs: ["claim_evidence_graph"]
00146             "4.1.3) local_truth_scoring":
00147                 invoke_ka: ["KA-014", "KA-102"]
00148                 outputs: ["claim_truth_scores", "entropy_signal"]
00149             "4.1.4) calibration":
00150                 invoke_ka: ["KA-014"]
00151                 method: "isotonic + per-domain calibration curves"
00152                 outputs: ["calibrated_confidence"]
00153             "4.1.5) stake_adjustment":
00154                 rule: "overall_confidence = min(calibrated_confidence, 1 - risk_penalty)"
00155                 risk_penalty_inputs: ["stakes_profile", "risk_weight", "bias_risk", \
00156                     "unresolved_conflicts"]
00157             "4.1.6) certification":
00158                 invoke_ka: ["KA-026"]
00159                 outputs: ["pass_threshold", "cert_bundle"]
00160
00161         "4.2) truth_score_components":
00162             formula_reference_ids:
00163                 - "F-CONF-01" # From ka_registry math objects
00164             components:
00165                 authority_weight: "source_authority"
00166                 corroboration_weight: "log(1 + corroboration_count)"
00167                 freshness_weight: "freshness_score"
00168                 chain_strength_weight: "function(reasoning_depth, dependency_count)"
00169                 conflict_penalty: "function(conflict_graph_density)"
00170             aggregation:
00171                 per_claim:
00172                     method: "weighted_sum → sigmoid"
00173                 overall:
00174                     method: "min( weakest_link , consensus_weighted_mean )"
00175                     weakest_link_policy: "use claims tagged FACT or CONSTRAINT as hard floor"
00176
00177     "5) contradiction_resolution":
00178         "5.1) detection":
00179             types:
00180                 - "direct negation"
00181                 - "parameter mismatch (units/definitions)"
00182                 - "temporal mismatch (valid then vs valid now)"
00183                 - "jurisdiction mismatch (EU vs US)"
00184                 - "scope mismatch (population/scale differences)"
00185             invoke_ka: ["KA-003", "KA-102"]
00186
00187         "5.2) resolution_strategies":
00188             "5.2.1) normalize_and_retry":
00189                 description: "Attempt reconciliation via unit normalization, definition alignment,
00190                     and scope matching."
00191                 invoke_ka: ["KA-020"]
00192             "5.2.2) evidence_escalation":
00193                 description: "Request additional authoritative evidence or trigger deeper research
00194                     agents."
00195                 invoke_ka: ["KA-113"]
00196                 policy:
```

```

00194         allowed_when: "stakes_profile != safety_critical OR human_gate_present"
00195     "5.2.3) forked_truth":
00196         description: "Maintain multiple truth branches when reality differs by \
time/jurisdiction; surface both."
00197         output_format: "truth_variants[]" with applicability conditions"
00198     "5.2.4) hard_veto":
00199         description: "If unresolved contradictions hit hard-claim set, veto output."
00200         emits: "UNRESOLVED_CONTRADICTION"
00201
00202     "6) governance":
00203         "6.1) hard_constraints":
00204             - id: "TEMPORAL_VALIDITY_REQUIRED"
00205               rule: "all FACT claims must have time_validity covering query temporal context"
00206               veto: "TEMPORAL_INVALIDITY"
00207             - id: "JURISDICTION_MATCH_REQUIRED"
00208               rule: "REGULATORY claims must match query location/jurisdiction"
00209               veto: "JURISDICTION_MISMATCH"
00210             - id: "MIN_EVIDENCE_FOR_HIGH_STAKES"
00211               rule: "high+ stakes require corroboration_count >= 2 for FACT claims"
00212               veto: "INSUFFICIENT_EVIDENCE"
00213
00214         "6.2) escalation":
00215             triggers:
00216                 - "REGULATORY_CONFLICT"
00217                 - "safety_critical AND confidence < required_confidence"
00218             action: "manual_review_alert"
00219             payload: "audit_bundle + conflict_summary + recommended_questions"
00220
00221         "6.3) persona_influence":
00222             description: "Truth engine consumes persona consensus weights but can override via \
evidence."
00223             rules:
00224                 - "if evidence contradicts persona consensus, lower consensus weight"
00225                 - "for Regulatory/Compliance claims, prefer Octopus/Spiderweb personas unless \
evidence is stronger"
00226
00227     "7) storage":
00228         "7.1) truth_ledger":
00229             description: "Append-only store for truth reports (hash-chained) enabling audit \
replay."
00230             backend_options:
00231                 - "postgres + immutable table"
00232                 - "object storage + manifest"
00233                 - "ledger DB (optional)"
00234             retention:
00235                 default_days: 365
00236                 safety_critical_days: 2555    # ~7 years
00237             indexing:
00238                 by_coordinate: true
00239                 by_claim_id: true
00240                 by_source_id: true
00241
00242         "7.2) audit_bundle":
00243             contents:
00244                 - "claim_set"
00245                 - "evidence_pack (normalized)"
00246                 - "provenance_graph"
00247                 - "conflict_graph"
00248                 - "calibration_artifacts"
00249                 - "attestations + signatures"
00250             serialization:
00251                 format: "jsonl + gz"
00252                 encryption: "AES-256-GCM at rest"
00253
00254     "8) integration_hooks":
00255         "8.1) master_workflow_hook":
00256             placement: "after 12-step refinement, before output finalization"

```

```
00257     provides: ["final_confidence", "truth_report", "attestations"]
00258 "8.2) refinement_engine_hook":
00259     placement: "Step 12"
00260     behavior: "Truth Engine must certify or force iteration/escalation"
00261 "8.3) api_endpoints":
00262     - method: "POST"
00263       path: "/truth/score"
00264       returns: "truth_report"
00265     - method: "POST"
00266       path: "/truth/verify"
00267       returns: "verification_result"
00268     - method: "GET"
00269       path: "/truth/ledger/{query_id}"
00270       returns: "audit_bundle_ref"
00271
00272 "9) observability":
00273     metrics:
00274     - "truth_engine_latency_ms"
00275     - "claims_scored_count"
00276     - "contradictions_detected_count"
00277     - "veto_rate"
00278     - "calibration_error_ece"
00279     - "confidence_distribution_by_domain"
00280     alerts:
00281     - name: "High veto rate"
00282       condition: "veto_rate > 0.10 over 1h"
00283       action: "page_oncall + open incident"
00284     - name: "Calibration drift"
00285       condition: "ece > 0.05 over 24h"
00286       action: "trigger recalibration job"
```

ukg_knowledge_agents_registry.yaml

Source: ukg_yaml_v2/ukg_knowledge_agents_registry.yaml
Content follows on next page.

```
00001      # =====
00002      # ukg_knowledge_agents_registry.yaml
00003      # Generated: 2025-12-21T22:20:42.687041
00004      # Format: Tree / numbered node style + YAML
00005      # =====
00006      # ASCII TREE (high-level)
00007      # agents
00008 # ■■ L3_research_agents
00009 # ■■ domain_agents (examples)
00010 # ■■ governance
00011 #
00012      # -----
00013 agents:
00014   "1) metadata":
00015     name: "UKG Knowledge Agents Registry"
00016     version: "2.0"
00017     description: "Specialized research agents spawned during Simulation Layer 3."
00018
00019   "2) L3_research_agents":
00020     - agent_id: "AG-PRIVACY-01"
00021       name: "Data Protection Specialist"
00022       focus: ["GDPR", "HIPAA", "data minimization", "lawful basis"]
00023       invoke_ka: ["KA-067", "KA-066", "KA-003"]
00024       outputs: ["reg_requirement_summary", "risk_flags", "control_recommendations"]
00025
00026     - agent_id: "AG-SEC-01"
00027       name: "Security Architect"
00028       focus: ["threat modeling", "encryption", "access control", "zero trust"]
00029       invoke_ka: ["KA-061", "KA-022"]
00030       outputs: ["threat_model", "security_controls", "attack_surface_notes"]
00031
00032     - agent_id: "AG-DATA-01"
00033       name: "Data Scientist / Analytics"
00034       focus: ["model selection", "validation", "bias checks", "privacy-preserving ML"]
00035       invoke_ka: ["KA-011", "KA-010", "KA-018"]
00036       outputs: ["model_plan", "validation_plan", "fairness_notes"]
00037
00038     - agent_id: "AG-OPS-01"
00039       name: "Operations Implementer"
00040       focus: ["workflows", "rollout planning", "change management", "SLA/SLO"]
00041       invoke_ka: ["KA-022", "KA-012"]
00042       outputs: ["implementation_plan", "resource_estimates", "operational_constraints"]
00043
00044     - agent_id: "AG-REG-01"
00045       name: "Standards & Audit Specialist"
00046       focus: ["SOC2", "ISO27001", "NIST CSF", "evidence for audits"]
00047       invoke_ka: ["KA-066", "KA-040"]
00048       outputs: ["control_mapping", "audit_artifacts_list", "evidence_requirements"]
00049
00050   "3) governance":
00051     spawn_limits:
00052       max_agents_default: 7
00053       max_agents_high_stakes: 10
00054     safety:
00055       require_ka_before_spawn: ["KA-004", "KA-061", "KA-107"]
```

README.md

Source: ukg_yaml_v2/README.md

Content follows on next page.

```
00001 # UKG YAML Pack (v2.0)
00002
00003 This pack contains the executable-style configuration YAMLs for:
00004 - 17-axis coordinate schema
00005 - Quad-persona templates
00006 - 10-layer simulation stack
00007 - 12-step refinement engine
00008 - Knowledge Agents registry
00009 - KA Registry (KA-001..KA-114)
00010 - Mathematical Formula Library
00011 - Truth Engine (evidence → truth score → calibrated confidence + attestations)
00012
00013 All YAMLs use a **tree + numbered node** style at the top for human scanability, followed \
    by structured YAML for programmatic consumption.
```